

Course Overview :

India is a developing nation and Agriculture is unquestionably its foundation. The country is leading in the production of food, fodder, fibre, fuel, fruit, flower, fish, and timber contributing raw materials to several large and small-scale industries. In a nutshell, Agriculture contributes a lion's share to the international trade. The mammoth population of the country is exclusively dependent on Agriculture, either directly or indirectly. Being India's largest private enterprise, it shares 18.8% of the country's GDP (PRS India, 2021). We find a number of triumph tales that transformed the nation from the perception of a "begging bowl" to one that is not only "self-sufficient" in food- grains but also a leading exporter of some Agricultural produces in the trans-boundary markets. In this laborious long voyage, the country has added a number of gems to her crown by impressive revolutions in the Agricultural sector -Green, White, Blue and Yellow; of which the "Green Revolution is considered as the most astounding one at the global level thumping the population growth rate. Despite this glorious progress during the last few decades we can not ignore the grey part of the story as well. Although we have made commendable progress in food supply, 21.9% of our population now lives below the poverty line, and it is really ironical that a large portion of our society, mostly women and children goes to bed every night hungry (Asian Development Bank, 2020). The growth achieved in Agricultural sector has been attributed to the concerted efforts of skilled human resource developed through Agricultural Education System (AES).

Success of a curriculum is rated not only based on development of technologies and knowledge but also on demand and marketability of its product, i.e. student. Agricultural education must ensure employment of these 'products' to make it more relevant to the society and the nation.

There is a great demand and opportunities for Agriculture graduates in government departments and non-governmental organizations, banking & insurance sectors, retailing industry, multi-national companies.

It is estimated that, more than 16000 scientific manpower would be required to cater to the needs of R&D in the country. Meagre production of Agriculture graduates vis-a-vis Masters and Doctorates would not suffice this demand. So, the country along with its States would have to produce ample skilled manpower holding diploma, certificate or any other supporting degrees. This would satisfy the prescribed norm of one Agriculture extension person for every 1,000 population as against current availability of one per 10,000. At present day, there is a huge gap between the demand and supply of manpower in this sector. This data clearly show that the importance of AGRICULTURE in the school level is escalating day by day.

That means sincere efforts are required to attract a greater number of students towards Higher Agricultural Education. There is a vast scope for young graduates to undertake Agriculture as their profession which is directly or indirectly contributing to the economic and social sectors.

Agriculture as a subject at Higher Secondary level can play a crucial role in this context. There is a vast scope for young graduates to undertake Agriculture as their profession which is directly or indirectly contributing to the economic and social domains.

The new syllabus may help to reduce the gap between demand and supply of skilled manpower in this sector.

This course and lessons are designed in such a manner that the students will have both knowledge and skills of Agriculture.

Course Objectives :

The course is a planned sequence of instructions consisting of units meant for developing the knowledge as well as skills regarding this course.

- To introduce the students with the new subject of Agriculture. They can understand the importance of Agriculture and other associated occupation related to Agriculture in Indian economy.
- To expose the relationship of crop growth with weather to the students. They will get a clear conception about different Agro climatic regions of West Bengal.
- To familiarize the students with soil and physical, chemical and biological properties of soil.
- To develop familiarity with the basic concept of Genetics and Plant Breeding in Agriculture. They will get basic idea about different selection methods like hybridisation, different types of modern crop varieties.
- To introduce the students with varied cultivation practices including handling of different manures, fertilisers, pesticides, fungicides and herbicides.
- To expose the students to different package of practices (POP) of important Agricultural and horticultural crop along with crop protection. They will get a comprehensive idea about IPM, INM, IDM and IWM.
- To help the students to comprehend the facts and importance of livestock management.
- To develop knowledge about different farming systems, cropping pattern, dry land Agriculture as well as different important schemes and policies related to Agriculture.
- The major objective of practical course is to expose students with first-hand knowledge about different aspects. They will be able to run important machines, solve practical mathematical problems, handle semi-sophisticated tools like pH meter, barometer etc.
- To make the students acquainted with the health and environmental management with special reference to Agricultural systems management.
- This course may enlighten rather steer the students in mining their future occupations / jobs through escalating their skills and knowledge. There remains wide opportunity of selecting entrepreneurship too.

CLASS - XI
SUBJECT: AGRICULTURE (AGRI)
SEMESTER – I

FULL MARKS : 35

CONTACT HOURS: 100 Hours

COURSE CODE: THEORY

UNIT NO.	TOPICS	CONTACT HOURS	MARKS
I : An Introduction to Agriculture	▪ Agriculture and Its Different Branches, Relationship of Agriculture with other Disciplines ▪ Importance of Agriculture in Indian Economy, Green Revolution ▪ Allied Agriculture (Lac Culture, Apiculture, Sericulture, Pisciculture)–a brief concept, Subsidiary occupation related to Agriculture	12	09
II : Weather, Climate and Crop Seasons	▪ Weather and Climate– A Brief Concept ▪ Factors Affecting Weather and Crop Growth (Temperature, Sunshine, Cloud, Rainfall, Humidity, Fog, Frost, Wind and Storm) ▪ Cropping Seasons ▪ Broad Classification of Crops According to Use ▪ Agro-Climatic Regions of West Bengal ▪ Climate Change, Global Warming, and Its Effect on Agriculture	25	10
III : Soil and Physical Properties of Soil	▪ Definition of Soil, Components of Soil, Formation of Soil Weathering (Physical, Chemical and Biological), Factors of Soil Formation ▪ Soil Texture and Soil Structure ▪ Density of Soil (Bulk Density, Particle Density, Porosity and their Inter-Relationship) ▪ Soil Profile, Soils of India (Name, Area, Basic Properties) ▪ Soil Water, Types of Soil Water and their Characteristics, Soil Moisture Conservation, Calculation of Soil Moisture	25	10
IV : Genetics and Plant breeding in Agriculture – An Introduction	▪ Mendel's Law of Inheritance ▪ Concept of Gene, Allele, Mutation ▪ Role of Genetics in Plant Breeding ▪ Selection Methods of Plant Breeding (Bulk selection, Pure line selection, Mass selection and Pedigree selection) ▪ Hybridization and Tissue Culture Technique–Basic Concept ▪ Crop Varieties- Indigenous, HYV, Hybrid, G.M, and Transgenic– only definition with examples	18	06
	Contact Hours	80	
	Practical Work	20	
	Total Hours	100	

CLASS - XI
SUBJECT: AGRICULTURE (AGRI)
SEMESTER – II

FULL MARKS : 35

CONTACT HOURS: 100 HOURS

COURSE CODE : THEORY

UNIT NO.	TOPICS	CONTACT HOURS	MARKS
V : Cultivation Practices	▪ Tillage (Definition, Objectives, Types) ▪ Seeds— Categories of Seeds, Seed Germination, Seed Treatment ▪ Land Preparation and Seed Bed Preparation, Sowing, Transplanting and intercultural Operations, Harvesting, Winnowing and Storage ▪ Different Kinds of Farm Implements and Their Uses	10	06
VI : Crop Production	▪ Package and Practices of Jute, Potato, Onion, Maize, Pulse (Gram), Oil seed(Mustard and Groundnut), Millet (Jowar, Bajra) ▪ Fodder crop(Berseem), Fruits (Papaya, Guava)	25	11
VII : Irrigation and Drainage	▪ Irrigation—Concepts, Sources, Types of Irrigation with Special Emphasis on Micro-Irrigation, Fertigation ▪ Water requirement of crops, Irrigation Requirements, Irrigation Efficiency and Scheduling of Irrigation. ▪ Drainage—concept, types and importance.	15	09
VIII: Farming System	▪ Farming System—Concept, Objectives, Components/ Enterprises ▪ Mixed Farming, Subsistence Farming, Specialized Farming, Crop Rotation, Mixed Cropping, Inter-Cropping, Relay Cropping, Paira Cropping, Multiple Cropping, Multi-Storied Cropping, Cropping Intensity	20	09
	Contact Hours	70	
	Practical Work	10	
	Home Assignment, Remedial and Tutorial Class	20	
	Total Hours	100	

CLASS : XI

SUBJECT: AGRICULTURE (AGRI)

COURSE CODE: PRACTICAL

GROUP: A (20 Marks)

1. Measurement and record of temperature, rainfall, air pressure.
2. Determination of soil texture by feel method.
3. Study of a soil profile.
4. Acquaintance with common farm implements and their operation (Country plough, mould board plough, seed drill, sprayer and duster).
5. Determination of Test Weight, Germination Percentage.
6. Preparation of seed bed, sowing of seeds.
7. Seed treatment.
8. Calculation of the seed requirement for a particular crop related to theory syllabus.

GROUP : B (10 Marks)

1. Project work related to theory course
2. Practical notebook
3. Viva voce

CLASS - XII

SUBJECT: AGRICULTURE (AGRI)

SEMESTER – III

FULL MARKS : 35

CONTACT HOURS : 100 Hours

COURSE CODE : THEORY

UNIT NO.	TOPICS	CONTACT HOURS	MARKS
I : Growth of Crops	• Concept of Growth and Development • C3,C4,CAM Plants –in brief • Yield Components and Yield Calculation	12	06
II : Soil Chemical and Biological Properties	• Different Plant Nutrients and Their Availability to Plants • Mobility of Plant Nutrients in Soil and Plant • Soil Fertility and Productivity • Soil Colloid • Soil Reaction • Problem of Soil (Acid soil, Saline soil, Alkaline soil) • Soil MicroOrganism-Important Bacteria, Algae, Fungi, Actinomycetes (in brief) • Soil Organic Matter, Mineralization and Emmobilization • Soil Erosion and Its Control	25	10
III: Manures and Fertilizers	• Role of Manures and Fertilizers in Crop Production • Important Manures and Fertilizers– Compost, Farm Yard Manure (FYM), Green Manure, Vermicompost, Oil Cake, Bone meal, Ammonium Sulphate, Urea, Calcium, Ammonium Nitrate, Super-Phosphate, Potassium Sulphate, Potassium Chloride • Mixed Fertilizers - Their Properties and Uses • Bio-Fertilizers • Application of Fertilizers • Law of diminishing return in fertiliser use • INM (Integrated Nutrient Management)	25	10
IV: Livestock Management	• Importance of Livestock in Agriculture and Industry, White Revolution in India • Important Breeds of Cows, Buffaloes, Goats and Poultry in India	18	09

UNIT NO.	TOPICS	CONTACT HOURS	MARKS
	- Signs of Sick Animals, Symptoms of Common Diseases in Cattle and Poultry- Rinderpest, Black Quarter, Foot-and-Mouth, Mastitis, Hemorrhagic Septicemia, Foul Pox, Ranikhet and Bird Flu– Their Control - Artificial Insemination– Brief Concept		
	Contact Hours	80	
	Practical Work	20	
	Total Hours	100	

CLASS - XII

SUBJECT: AGRICULTURE (AGRI)

SEMESTER – IV

FULL MARKS : 35

CONTACT HOURS : 100 Hours

COURSE CODE : THEORY

UNIT NO.	TOPICS	CONTACT HOURS	MARKS
V : Crop Production	Package and Practices – Land Preparation, Time of Sowing, Suitable Varieties, Cultivation Procedures, Manures and Fertilizers Application, Irrigation and Drainage, Intercultural Operations and Yield of Rice, Wheat, Sugarcane, Brinjal, Tomato, Mango and Banana (In tabular form-Cabbage, Cauliflower, Bottle Gourd, Bitter Gourd)	20	11
VI : Crop Protection	- Concept of Pests and Pesticides (with Special Reference to Target Pests and Mode of Action – Basic Idea) - Concept of IPM, IDM and IWM	20	10

UNIT NO.	TOPICS	CONTACT HOURS	MARKS
	- Control of Different Insects, Diseases and Weeds of Different Crops (Rice, Wheat, Jute, Potato, Sugarcane, Brinjal, Mango, Banana, Mustard, Gram) - Control of Rodents - Handling of Pesticides		
VII : Food Processing and Preservation	- Principles and Methods of Fruits and Vegetables preservation - Preparation of Jelly, Jam, Ketchup, Chips and Their Packing	15	07
VIII : Dry land AGNM culture and Important Policies	- Special Features of Dryland Farming / Rainfed Agriculture - Different Important Schemes: Crop Insurance, Kishan Credit Card	15	07
	Contact Hours	70	35
	Practical Work	10	
	Home Assignment, Remedial and Tutorial Class	20	
	Total Hours	100	

CLASS: XII

SUBJECT : AGRICULTURE (AGRI)

COURSE CODE : PRACTICAL

GROUP : A (20 Marks)

1. Collection of Soil sample and preparation of the soil samples for laboratory experiment.
2. Determination of soil pH by pH meter.
3. Identification of different manures and fertilisers.
4. Preparation of compost pit.
5. Identification of crop seeds (related to syllabus only), different farm weeds, important insect pests and diseases and their damages.
6. Mathematical calculation for the formulation of spraying chemicals.
7. Calculation of yield of different crops by studying yield attributes of crop related to theory portion.
8. Mathematical calculation for the requirement of fertiliser in the crop field.
9. Determination of cropping intensity.
10. Cost of cultivation of different crops related to theory portion.
11. Students practice air layering, budding, inarching and Gooti making.

GROUP : B (10 Marks)

1. Project work related to theory course.
2. Practical notebook or laboratory notebook.
3. Viva voce.

CLASS - XI**Course Overviews and Objectives :****SEMESTER – I****Unit-I****Arabic Prose (النثر):**

Content: أبو الحسن علي الحسيني الندوي by من يضع الحجر , كامل الكيلاني by أبو صير و أبوقير , سورة الفيل

Objectives:

- Develop a comprehensive understanding of Surah Al-Fil from the Quran, exploring its linguistic and thematic aspects.
- Enhance linguistic skills and vocabulary through the analysis of modern prose.
- Foster critical thinking through the evaluation of themes, characters, and literary techniques introduced in the selected prose works.

Unit-II**Arabic Poetry (النظم):**

Content: محمود درويش by البنت , أحمد شوقي بك by الأم , أمية بن أبي الصلت by في صفة الله تعالى

Objectives:

- Connect poetic expressions with the cultural and historical contexts, understanding how poets reflect and influence the societies.
- Develop expressive and creative language skills through the study of various poetic styles.
- Gain insights into societal values, emotions, and historical contexts reflected in Arabic poetry.

Unit-III**History of Arabic Literature (تاريخ الأدب العربي):**

Content: History of Arabic literature from pre Islamic period to Islamic period. Poetry with special references to Antara Bin Shaddad, Imraul Qais, Zuhair bin Abi Sulma, Hassan bin Thabit, Khansa.

Objectives:

- Understand the historical development of Arabic literature, providing insight into cultural and intellectual evolution.
- Appreciate the literary contributions of Antara Bin Shaddad, Imraul Qais, Zuhair bin Abi Sulma, Hassan bin Thabit, Khansa.
- Develop historical awareness by exploring literary responses to socio-political events.

Unit-IV

Grammar (النحو والصرف):

Content: Basic Arabic grammar like huruf, kalimah, harakat, muftada, khabar, mudaf etc.

Objectives:

- Achieve knowledge about basic grammatical rules, enhancing language proficiency.
- Improve communication skills by understanding and applying basic grammatical concepts.
- Foster accuracy in expression and grammatical correctness for effective communication.

Unit-V

Translation (الترجمة):

Content: Translation of simple sentences from English / Vernacular language to Arabic or from Arabic to English / Vernacular language.

Objectives:

- Develop translation skills by applying grammar concepts.
- Translating sentences from Arabic to English and vice versa.
- Gain cultural sensitivity and understanding through translation.
- Apply grammatical knowledge to ensure accurate and nuanced translations.

SEMESTER – II

Unit-I

Arabic Prose (النثر):

Content: أبو الحسن by من يضع الحجر , كامل الكيلاني by محمد ﷺ في البداية , أحمد شهاب الدين القليوبي by نواذر علي الحسني الندوي

Objectives:

- Analyze various Arabic stories by Ahmed Shihabuddin Al Qalyubi, Samee Atif Al Zain and Muhammad Atiya Al Ibrashi.
- Enhance linguistic skills and vocabulary through the analysis of the selected stories.
- Acquire moral values through the analysis of these stories.

Unit-II

Arabic Poetry (النظم):

Content: عيسى الناعوري by أخي الانسان , كامل كيلاني by الوقت , حسان بن ثابت by أغر عليه للنبوة خاتم

Objectives:

- Cultivate an appreciation for Arabic poetry, recognizing its cultural and emotional nuances.
- Introducing Arabic poetry to the students so that they can be benefited from thoughts of the poets of different periods and develops their knowledge about basic forms and structure of Arabic poetry.

Unit-III

History of Arabic Literature (تاريخ الأدب العربي):

Content: History of Arabic literature (Umayyad Period). Poetry with special references to Akhtal, Farazdaq, Jarir, Umar Bin Abi Rabiah

Objectives:

- Understand the historical development of Arabic literature, providing insight into cultural and intellectual evolution.
- Appreciate the literary contributions of Akhtal, Farazdaq, Jarir, Umar Bin Abi Rabiah
- Providing information about cultural & social aspects of Arabic literature in Umayyad period.

Unit-IV

Grammar (النحو والصرف):

Content: Basic Arabic grammar like Ismul isarah, Ismul mousool, harful jar, Fa'l etc.

Objectives:

- Teaching the students basic forms of Arabic structure like Ismul isarah, Prepositions, Conjunctions etc.
- Improve communication skills by understanding and applying grammatical concepts.
- Achieve grammatical correctness for effective communication.

Unit-V

Translation (الترجمة):

Content: Translation of simple sentences from English / Vernacular language to Arabic or from Arabic to English / Vernacular language.

Objectives:

- To translate the basic sentences from Arabic to English and English to Arabic.
- Gain communication skills through translation.
- Apply grammatical knowledge to ensure accurate and nuanced translations.

PROJECT :

Content: Project on various topics like Origin and Development Arabic Language, Muslim Festivals, A prominent Educationist of Bengal, Muslim Architecture in India.

Objectives:

- Develop research and presentation skills through a project on one of the provided topics.
- Fostering independent inquiry and critical thinking.
- Delving into historical figures and architectural elements.

Assessment:

Assessments include written analyses of prose and poetry, translation exercises, grammar assessments, and research projects on historical figures in Arabic literature.

Overall Goals:

- Equip students with comprehensive Arabic language skills.
- Foster cultural appreciation, critical thinking, and expressive communication.
- Provide a historical and literary context for a deeper understanding of Arabic culture and literature.

Arabic Syllabus for Class XI

FULL MARKS -100

Prose:	20 Marks
Poetry:	20 Marks
History of Arabic Literature:	15 Marks
Grammar:	15 Marks
Translation:	10 Marks
Project:	20 Marks

CLASS - XI**SUBJECT : ARABIC (ARBC)****SEMESTER – I****FULL MARKS: 40****CONTACT HOURS: 100 Hours****COURSE CODE: THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – I	النثر-Prose Text Book: Arabic Selection for H.S, 2024 A.D 1. سورة الفيل القرآن الكريم 2. أبو صير وأبوقير كامل الكيلاني 3. من يضع الحجر أبو الحسن علي الحسني الندوي	20	10
Unit - II	النظم-Poetry 1. في صفة الله تعالى أمية بن أبي الصلت 2. الأم أحمد شوقي بك 3. البنات محمود درويش	20	10
Unit - III	تاريخ الأدب العربي-History of Arabic Literature (Pre- Islamic & Islamic Period) • Imraul Qais • Zuhair bin Abi Sulma • Antara Bin Shaddad • Hassan bin Thabit • Khansa	20	10
Unit - IV	النحو والصرف-Grammar 1. الحروف الشمسية و الحروف القمرية 2. حروف العلة 3. الحركات 4. حالة الإعراب 5. الكلمة وأقسامها 6. المعرفة والنكرة 7. المذكر والمؤنث 8. الاسم: المفرد والمثنى والجمع 9. المبتدأ والخبر 10. الموصوف والصفة 11. المضاف والمضاف إليه.	20	5
Unit - V	الترجمة-Translation Based on the above-mentioned Grammar	20	5

CLASS - XI**SUBJECT : ARABIC (ARBC)****SEMESTER – II****FULL MARKS: 40****CONTACT HOURS: 80 HOURS****COURSE CODE: THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - I	النثر - Prose Text Book: Arabic Selection for H.S, 2024 A.D 1. النوادر (الحكايات 1- 3) أحمد شهاب الدين القليوبي 2. مجد في البادية سميح عاطف الزين 3. الملك و الراعي مجد عطية الإبريشي	15	10
Unit - II	النظم - Poetry 1. في مدح الرسول حسان بن ثابت الانصاري 2. الوقت مجموعة القصص العربية لكامل الكيلاني 3. أخي الإنسان (8 أبيات) عيسى الناعوري	15	10
Unit - III	تاريخ الأدب العربي - History of Arabic Literature (Umayyad Period) • Akhtal • Farazdaq • Jarir • Umar Bin Abi Rabiah	15	5
Unit - IV	النحو والصرف - Grammar 1. اسم الإشارة 2. الأسماء الموصولة 3. حروف الجر 4. الضمائر 5. أداة الاستفهام 6. الفعل اللازم و المتعدي 7. الفعل الماضي و صيغه 8. الفعل المضارع و صيغه	15	10
Unit - V	الترجمة - Translation Based on the above-mentioned Grammar (A passage or few sentences to be translated from Arabic to English or vernacular language / English or Bengali to Arabic)	20	5

CLASS - XI

PROJECT

SUBJECT : ARABIC (ARBC)

FULL MARKS: 20

CONTACT HOURS: 20 HOURS

Sub Topics :

1	Six important books of Hadith (Any one)
2	Role of Fairs in promotion of Arabic Language and Literature in the pre- Islamic period
3	Influence of Quran and Hadith on Arabic Language and Literature
4	Origin and Development Arabic Language
5	Muslim Festival (Any one)
6	A prominent Educationist of Bengal
7	Muslim Architecture in India
8	Book review: (Any Arabic Book)
9	A prominent Arabic Scholar of India

OR

Communicative Arabic for Class XI

Conversation & Related Vocabularies:

- Introduction, Greeting & Gratitude
- Body parts
- Dresses
- Colours & Directions
- Days & months
- Flowers & Fruits
- Interrogative Words
- In the hospital
- In the bank
- In the embassy
- At the airport

SUGGESTED READING:

1. Al- Qirayatur Rashida by Abul Hasan Ali An-Nadwi
2. Al-Qasasul Adabiyyah by Atiah al- Ibrishi
3. Kitab- An-Nawadir by Shihabuddin al-Kalyubi
4. Majmuah Al-Qasas al-Arabiyyah by Kamil Al-Kilani
5. Ahlul Kahaf by Taufik al- Hakim
6. Mukhtarat Min Siyiril Arabi al- Hadith by Dr.Tajuddin Mannani &
Dr. Ahamad K.M.Ain. Ahmad Zubair
7. H.S.Arabic Selection by WBCHSE
8. Tarikh al-Adab Al-Arabi by Ahmad Hussain Zaiyyat
9. Tarikh Al-Adab Al-Arabi by Jurji Jaidan
10. A literary History of Arabs by R.A. Nicholson
11. Teach Yourself by S.A. Rahman
12. An-Nahwal Wadih by Al- Jarim & Mustafa Amin
13. Spoken Arabic Made Easy by Amanulla Vadakkangara
14. Al-Arabiyyah Lil Hayat, Vol I, Published by King Saud University, Riyadh
15. Madinah Arabic Reader by Dr. V. Abdur Rahim

CLASS - XII

Course Overviews and Objectives :

SEMESTER – I

Arabic Prose (النثر):

Content: أبو الحسن علي الحسيني الندوي by شهامة اليتيم , عبد الملك بن هشام by فتح مكة , سورة الأعلى

Objectives:

- Understand and analyze Sura Al-A'la focusing on linguistic and thematic elements from the Quran.
- Explore the historical narrative of conquest Makkah.
- Foster critical thinking by interpreting diverse prose genres.

Unit-II

Arabic Poetry (النظم):

Content: أحمد شوقي by غاندي , كامل كيلاني by الطالب النشيط , حاتم الطائي by مهلا نوار!

Objectives:

- Analyze classical and modern poems, exploring the various expressions of emotions, social issues and personal reflections.
- Develop expressive and creative language skills through the study of various poetic forms.
- Gain insights into societal values, emotions, and historical contexts reflected in Arabic poetry.

Unit-III

History of Arabic Literature (تاريخ الأدب العربي):

Content: History of Arabic literature Abbasid period. Arabic prose & poetry with special references to Abul Atahia, Mutanabbi, Abu Nuwas, Ibnul Muqaffa, Jahez, Hariri,

Objectives:

- Providing information about cultural & social aspects of Arabic literature in Abbasid period.
- Appreciate the literary contributions of Abul Atahia, Mutanabbi, Abu Nuwas, and others.
- Develop historical awareness by exploring literary responses to socio-political events.

Unit-IV

Grammar (النحو والصرف):

Content: Basic Arabic grammar like Jumla Ismiah, Jumla Faliyah, M'urab, Mabni etc.

Objectives:

- Teaching the student basic forms of Arabic structure like Morab, Mabni, Jumla Ismiah, Jumla Fa'liyah etc.
- Improve communication skills by understanding and applying grammatical concepts.
- Improve grammatical correctness for effective communication.

Unit-V

Translation (الترجمة):

Content: Translation of simple sentences from English / Vernacular language to Arabic or from Arabic to English / Vernacular language.

Objectives:

- Transfer meaning accurately between Arabic and the target language.
- Improve translation skills to get job opportunities in various fields.
- Apply grammatical knowledge to ensure accurate and nuanced translations.

SEMESTER – IV

Unit-I

Arabic Prose (النثر):

Content: توفيق الحكيم by أهل الكهف , مجد عطية الإبريشي by سيدنا علي بن أبي طالب , الأحاديث المنتخبة

Objectives:

- Study selected hadiths and gaining insights into the sayings and actions of Prophet Muhammad.
- Enhance linguistic skills and vocabulary through the analysis of complex prose.
- Develop critical thinking by interpreting various texts.

Unit-II

Arabic Poetry (النظم):

Content: ابن زيدون by الأمّ المحب , سعيد عقل by أمي , نجيب محفوظ by في وصف القمر

Objectives:

- Analyze the selected poems, exploring poets' reflections on nature and societal issues.
- Improve language skills through the study of various poetic forms.
- Gain insights into societal values, emotions, and historical contexts reflected in Arabic poetry.

Unit-III

History of Arabic Literature (تاريخ الأدب العربي):

Content: History of Arabic literature modern period. Arabic prose & poetry with special references to Ahmad Shawqi, Haiz Ibrahim, Najib Mahfooz, Manfaluti, Abul Hasan Ali Al Nadwi

Objectives:

- Understand the historical development of Arabic literature, providing insight into cultural and intellectual evolution.
- Appreciate the literary contributions of Ahmad Shawqi, Haiz Ibrahim, Najib Mahfooz, Manfaluti, Abul Hasan Ali Al Nadwi.
- Develop historical awareness by exploring literary responses to socio-political events.

Unit-IV

Grammar (النحو والصرف):

Content: Basic Arabic grammar like Munsarif, Mafool, Hal, Tamiz etc.

Objectives:

- Achieve grammatical knowledge, enhancing language proficiency.
- Improve communication skills by understanding and applying grammatical concepts.
- Improve precision in expression and grammatical correctness for effective communication.

Unit-V

Translation (الترجمة):

Content: Translation of simple sentences from English / Vernacular language to Arabic or from Arabic to English / Vernacular language.

Objectives:

- Translate accurately between Arabic and the target language.
- Improve translation skills for getting job in different fields.
- Apply advanced grammatical knowledge to ensure accurate and nuanced translations.

PROJECT

Content: Project on various topics like The Four Caliphs of Islam, Arabic word uses in Bengali Language, A prominent Bengali Muslim Scholar, A Pioneer of Muslim women Education in Bengal, Three Grand Mosques etc.

Objectives:

- Develop research and presentation skills through a project on one of the provided topics.
- Fostering independent inquiry and critical thinking.
- Delving into historical figures and architectural elements.

Assessment:

Assessments include written analyses of prose and poetry, translation exercises, grammar assessments, and research projects on historical figures in Arabic literature.

Overall Goals:

- Equip students with comprehensive Arabic language skills.
- Foster cultural appreciation, critical thinking, and expressive communication.
- Provide a historical and literary context for a deeper understanding of Arabic culture and literature.

Arabic Syllabus for Class XII

FULL MARKS -100

Prose:	20 Marks
Poetry:	20 Marks
History of Arabic Literature:	15 Marks
Grammar:	15 Marks
Translation:	10 Marks
Project:	20 Marks

CLASS - XII

SUBJECT : ARABIC (ARBC)

SEMESTER - III

FULL MARKS: 40

CONTACT HOURS: 100 Hours

COURSE CODE: THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – I	النثر - Prose Text Book: Arabic Selection for H.S, 2024 A.D 1. سورة الأعلى القرآن الكريم 2. فتح مكة عبد الملك بن هشام . من السيرة النبوية 3. شهامة اليتيم أبو الحسن علي الحسيني الندوي	20	10
Unit - II	النظم - Poetry 1. وقال حاتم الطائي حاتم الطائي 2. الطالب النشيط مجموعة القصص العربية لكامل الكيلاني 3. غاندى (أبيات 1-12) أحمد شوقي بك	20	10
Unit - III	تاريخ الأدب العربي - History of Arabic Literature (Abbasid Period) • Abul Atahia • Mutanabbi • Abu Nuwas • Ibnul Muqaffa • Jahez • Hariri	20	10
Unit - IV	النحو والصرف - Grammar 1. الجملة الاسمية 2. الجملة الفعلية 3. المعرب والمبني 4. الجمع المذكر السالم والجمع المؤنث السالم 5. الجمع المكسر 6. الفعل الأمر وصيغته 7. الفعل النهي وصيغته 8. نواصب الفعل المضارع 9. جوازم الفعل المضارع	20	5
Unit - V	الترجمة - Translation Based on the above-mentioned Grammar	20	5

CLASS - XII

SUBJECT : ARABIC (ARBC)

SEMESTER - IV

FULL MARKS: 40

CONTACT HOURS: 80 HOURS

COURSE CODE: THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - I	النثر - Prose Text Book: Arabic Selection for H.S, 2024 A.D 1. الأحاديث المنتخبة من مشكاة المصابيح 2. سيدنا علي بن أبي طالب مجده عطية الإبريشي 3. اهل الكهف (جزء من الفصل الاول) توفيق الحكيم	15	10
Unit - II	النظم - Poetry 1. في وصف القمر نجيب محفوظ 2. أمي سعيد عقل 3. الام المحب ابن زيدون	15	10
Unit - III	تاريخ الأدب العربي - History of Arabic Literature (Modern Period) - Ahmad Shawqi - Hafiz Ibrahim - Najib Mahfooz - Mustafa lutfi al- Manfaluti, - Abul Hasan Ali Al Nadwi	15	5
Unit - IV	النحو والصرف - Grammar 1. منصرف و غير منصرف 2. المفاعيل الخمسة 2. الحروف العطف 3. حروف المشبهة بالفعل 4. الأفعال الناقصة 5. ما و لا المشبهتان بليس 6. لا النافية للجنس 7. الحال 8. التمييز 9. الاستثناء	15	10
Unit - V	الترجمة - Translation Based on the above-mentioned Grammar (A passage or few sentences to be translated from Arabic to English or vernacular language / English or Bengali to Arabic)	20	5

CLASS - XII
PROJECT
SUBJECT : ARABIC (ARBC)

FULL MARKS: 20

CONTACT HOURS: 20 HOURS

Sub Topics :

1	The Four Caliphs of Islam (Any One)
2	Arabic word uses in Bengali Language
3	A prominent Bengali Muslim Scholar
4	A Pioneer of Muslim Women Education in Bengal
5	Three Grand Mosques
6	Masjid al Nabawi
7	Masjid al Haraam
8	Masjid al Aqsa
9	A famous Muslim Institution in India
10	Sunderban
11	Bengali Festival (Any One)
12	An important River of West Bengal
14	Book Review: (Any Arabic Book)

OR

Communicative Arabic for Class XII

Conversation & Related Vocabularies:

- Numbers
- Professions
- Diseases
- Conversation in daily life
- In the school
- At the market
- At the street
- At home

SUGGESTED READING:

1. Al- Qirayatur Rashida by Abul Hasan Ali An-Nadwi
2. Al-Qasasul Adabiyyah by Atiah al- Ibrishi
3. Kitab- An-Nawadir by Shihabuddin al-Kalyubi
4. Majmuah Al-Qasas al-Arabiyah by Kamil Al-Kilani
5. Ahlul Kahaf by Taufik al- Hakim
6. Mukhtarat Min Siyiril Arabi al- Hadith by Dr.Tajuddin Mannani &
Dr. Ahamad K.M.Ain. Ahmad Zubair
7. H.S.Arabic Selection by WBCHSE
8. Tarikh al-Adab Al-Arabi by Ahmad Hussain Zaiyyat
9. Tarikh Al-Adab Al-Arabi by Jurji Jaidan
10. A literary History of Arabs by R.A. Nicholson
11. Teach Yourself by S.A. Rahman
12. An-Nahwal Wadih by Al- Jarim & Mustafa Amin
13. Spoken Arabic Made Easy by Amanulla Vadakkangara
14. Al-Arabiyyah Lil Hayat, Vol I, Published by King Saud University, Riyadh
15. Madinah Arabic Reader by Dr. V. Abdur Rahim

SUBJECT : BIOLOGICAL SCIENCE**CLASS - XI****SUBJECT : BIOLOGICAL SCIENCE (BIOS)****SEMESTER – I****FULL MARKS : 35****CONTACT HOURS : 55 Hours****COURSE CODE: THEORY**

UNIT No.	TOPICS	Contact Hours	Marks
UNIT I (DIVERSITY OF LIVING ORGANISM)	<u>Chapter-1: The Living World</u> Biodiversity; need for classification; three domains of life; Taxonomy and Systematics; concept of species; and taxonomical hierarchy; binomial nomenclature; Tools for study of Biodiversity; Museums; Zoological and Botanical Gardens; Herbaria (Definition: World's largest herbarium, name of the herbarium in Bengal, Importance of herbarium)	19	8
	<u>Chapter-2: Biological Classification</u> Five Kingdoms of Classification; Salient features and classification of Monera; Protista and Fungi into major groups; Lichens, Viruses, Viroids and Prions.		
	<u>Chapter-3: Plant Kingdom</u> Classification of Plants into major Groups, Salient and distinguishing features and a few examples of Algae, Bryophyta, Pteridophyta, Gymnosperm.		
	<u>Chapter-4: Animal Kingdom</u> Salient features and classification of animals, non-chordates up to phyla level and chordates up to class level.		
UNIT II STRUCTURAL ORGANIZATIONS IN PLANTS AND ANIMALS)	<u>Chapter-5: Morphology of Flowering Plants</u> Morphology of different parts of flowering plants: root, stem, leaf, inflorescence, flower, fruit, and seed. Description of families : Malvaceae, Solanaceae, Brassicaceae, Compositae, Leguminosae (Dicots), Poaceae, Liliaceae (Monocots).	16	12
	<u>Chapter-6: Anatomy of Flowering Plants</u> Plant tissue systems including Mechanical tissue systems, anatomy and functions of tissue systems in dicots and monocots.		
	<u>Chapter-7: Structural Organization in Animals</u> Animal Tissue Systems: epithelial, connective, muscular and nervous systems (structure, organization and function); morphology, anatomy and functions of different systems; digestive, circulatory, respiratory, nervous, and reproductive systems of frog.		

UNIT No.	TOPICS	Contact Hours	Marks
UNIT III (CELL STRUCTURE AND FUNCTIONS)	<u>Chapter-8: Cell- The Unit of Life</u> Cell theory and cell as the basic unit of life; structure of prokaryotic and eukaryotic cells; Plant cell and Animal cell; cell envelope; cell membrane, cell wall; cell organelles — structure and function; endo-membrane system, nucleus, endoplasmic reticulum, golgi bodies, lysosomes, vacuoles, mitochondria, ribosomes, plastids, microbodies, cytoskeleton, cilia, flagella, centrioles (ultra-structure and function).	20	15
	<u>Chapter-9: Biomolecules</u> Chemical constituents of living cells: biomolecules; structure and function of proteins; carbohydrates; lipids; and nucleic acids; Enzyme — types; properties; enzyme action.		
	<u>Chapter-10: Cell Cycle and Cell Division</u> Cell cycle; mitosis; meiosis; and their significance.		

CLASS - XI

SUBJECT : BIOLOGICAL SCIENCE (BIOS)

SEMESTER – II

FULL MARKS : 35

CONTACT HOURS : 97 HOURS

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	Marks
UNIT IV (PLANT PHYSIOLOGY)	<u>Chapter-11: Photosynthesis in Higher Plants</u> Photosynthesis as a means of autotrophic nutrition; site of photosynthesis, pigments involved in photosynthesis (structure of chlorophyll; empirical formula of chlorophyll a, b, c, d, e, bacteriochlorophyll, carotene and xanthophyll); photochemical and biosynthetic phases of photosynthesis; cyclic and non-cyclic photophosphorylation; chemiosmotic hypothesis, photorespiration, C3 and C4 pathways, CAM Cycle (schematic pathway only), factors affecting photosynthesis.	34	17
	<u>Chapter-12: Respiration in Plants</u> Exchange of gases; cellular respiration — glycolysis, fermentation (anaerobic), TCA cycle and electron transport system (aerobic); energy relations — number of ATP molecules generated; amphibolic pathways; respiratory quotient.		
	<u>Chapter-13: Plant Growth and Development</u> Seed germination; phases of plant growth and plant growth rate; conditions of growth; differentiation, dedifferentiation and redifferentiation; sequence of developmental processes in a plant cell; plant growth regulators — auxin, gibberellin, cytokinin, ethylene, ABA, Photoperiodism — Definition and different types.		

UNIT No.	TOPICS	CONTACT HOURS	Marks
UNIT V (HUMAN PHYSIOLOGY)	<u>Chapter – 14: Digestion and Absorption</u> Introduction; Structure of human alimentary canal (drawing, labelling and function of different parts including dental arrangement and digestive glands); Role of digestive enzymes and the GI hormone in digestion; Peristalsis; Digestion, absorption and assimilation of protein, carbohydrate and fat; egestion; Nutritional and digestive disorders — PEM (protein energy malnutrition) indigestion, constipation, vomiting, jaundice, diarrhoea.	63	18
	<u>Chapter-15: Breathing and Exchange of Gases</u> Respiratory organs in animals (name only); Respiratory system in humans; mechanism of breathing and its regulation in humans - exchange of gases, transport of gases and regulation of respiration, respiratory volume; disorders related to respiration — asthma, emphysema, occupational respiratory disorders.		
	<u>Chapter-16: Body Fluids and Circulation</u> Composition of blood, blood groups, coagulation of blood; composition of lymph and its function; human circulatory system - Structure of human heart and blood vessels; cardiac cycle, cardiac output, ECG; double circulation; regulation of cardiac activity; disorders of circulatory system — hypertension, coronary artery disease, angina pectoris, heart failure.		
	<u>Chapter-17: Excretory Products and their Elimination</u> Modes of excretion — ammonotelism, ureotelism, uricotelism; human excretory system — structure and function; urine formation, osmoregulation; counter-current mechanism; regulation of kidney function — renin-angiotensin system, atrial natriuretic factor, ADH and diabetes insipidus; role of other organs in excretion; disorders — uremia, renal failure, renal calculi, nephritis; dialysis and artificial kidney, kidney transplant.		
	<u>Chapter-18: Locomotion and Movement</u> Types of movement - ciliary, flagellar, muscular; skeletal muscle, contractile proteins and muscle contraction; skeletal system and its functions; joints; disorders of muscular and skeletal systems - myasthenia gravis, tetany, muscular dystrophy, arthritis, osteoporosis, gout.		
	<u>Chapter-19: Neural Control and Coordination</u> Mechanism of neural control and co-ordination; Neuron and nerves; Nervous system in humans - central nervous system; peripheral nervous system and visceral nervous system; Brain and its major parts- cerebral cortex, thalamus, hypothalamus and limbic system; mid-brain, pons, medulla, cerebellum and spinal cord (function only); Modes of distribution and function of P.N.S. and autonomic nervous system; Generation and conduction of nerve impulse; reflex action and reflex arc; Sense organs – Sensory perception, outline structure and function of eye and ear; Disorders — Parkinson's and Alzheimer's diseases.		
	<u>Chapter-20: Chemical Coordination and Integration</u> Endocrine glands and hormones; human endocrine system — hypothalamus, pituitary, pineal, thyroid, parathyroid, adrenal, pancreas, gonads; mechanism of hormone action (protein and steroid hormones); role of hormones as messengers and regulators, hypo- and hyperactivity and related disorders; dwarfism, acromegaly, cretinism, goitre, exophthalmic goitre, diabetes, Addison's disease.		

CLASS : XI

SUBJECT : BIOLOGICAL SCIENCE (BIOS)

COURSE CODE : PRACTICAL

FULL MARKS : 30

CONTACT HOURS : 30 HOURS

Time allowed : 3 hours.

Max. Marks : 30

EVALUATION SCHEME	MARKS
One major experiment Part A.(experiment no-1,3,7)	6
One minor experiment Part A.(experiment no-6,8,9,10,11)	5
Slide preparation Part A.(experiment no-2,4,5) (any one)	3
Spotting. Part – B (three)	6(2x3)
Practical record + Viva voce	5(3+2)
Investigatory project + Viva voce	5(3+2)
TOTAL	30

A. List of Experiments

1. Study and describe locally available common flowering plants from family Malvaceae, Solanaceae, Brassicaceae, Asteraceae, Leguminosae including dissection and display of floral whorls, Anther and Ovary to show number of chambers (Placentation). (Floral formula and floral diagrams.), Type of root. (Tap and adventitious.); Type of stem. (Herbaceous and woody); Leaf (Arrangement, shape, venation, simple and compound)
2. Preparation and study of TS of dicot and monocot roots and stems. (Primary.)
3. Study of osmosis by Potato Osmometer.
4. Study of plasmolysis in epidermal peels (e.g..Rheo/lily or fleshy scale leaves of onion bulb)
5. Study of distribution of stomata on the upper and lower surfaces of leaves.
6. Comparative study of the rates of transpiration in the upper and lower surfaces of leaves.
7. Test for the presence of sugar. starch, proteins and fats in suitable plant and animal materials.

8. Test for presence of urea in urine.
9. Test for presence of sugar in urine.
10. Test for presence of Albumin in urine.
11. Test for presence of Bile salts in urine.

B. Study and observe the following (Spotting)

1. Parts of a compound microscope.
2. Specimens./Slides./Models. Identify with reasons.—Bacteria, Spirogyra, Rhizopus, mushroom, yeast, liverwort, moss, fern, pine cone: male and female, one monocotyledonous plant, one dicotyledonous plant, one lichen. Different types of inflorescence. (Racemose and Cymose)
3. Virtual specimens/Slides/Models. Identifying features of Amoeba, Hydra, Liver Fluke, Ascaris, Leech, Earthworm, Prawn, Silkworm, Honeybee, Snail, Starfish, Shark, Rohu, Frog, Lizard, Pigeon and Rabbit. Human blood, and Toad blood
4. Mitosis in onion root tip cells and animal cells (Grasshopper) from permanent slides.
5. Human skeleton and different types of joints with the help of Virtual image/Models only.

[Note: ***18 Hours** reserved for Remedial classes, Tutorials and Home Assignments.]

CLASS - XII

SUBJECT : BIOLOGICAL SCIENCE (BIOS)

SEMESTER – III

FULL MARKS : 35

CONTACT HOURS : 90 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	Marks
UNIT VI REPRODUCTION	<u>Chapter 1: Sexual Reproduction in flowering plants</u> Flower structure; Development of male and female Gametophytes; Pollination — Types, Agencies and examples; Out breeding devices; Pollen pistil interaction; Double fertilization; Post-fertilization events-development of endosperm and embryo, development of seed and formation of fruit; Special modes — Apomixis; Parthenocarpy; Polyembryony; Seed dispersal and fruit formation and their significance.	33	15
	<u>Chapter 2: Human reproduction</u> Male and female reproductive systems; Anatomy and Histology of testis and ovary, Gametogenesis-Spermatogenesis and Oogenesis; Menstrual cycle; Fertilization, embryo development up to blastocyst, formation and implantation; Pregnancy and placenta formation and function; Parturition: mechanism and neuroendocrine system involved in this mechanism, Lactation.		
	<u>Chapter 3: Reproductive health</u> Need for reproductive health and prevention of sexually transmitted diseases (STDs); Birth control-need and methods, Contraception and medical termination of pregnancy(MTP); Amniocentesis; Infertility and Assisted reproductive technologies-IVF, ZIFT, GIFT (elementary idea for general awareness).		
UNIT-VII (GENETICS AND EVOLUTION)	<u>Chapter 4: Principles of Inheritance and variation.</u> Heredity and Variation: Mendelian inheritance, Deviations from Mendelism-Incomplete dominance, Co-dominance, Multiple alleles and inheritance of blood groups, Pleiotropy; Elementary idea of polygenic inheritance; Chromosome theory of inheritance; Chromosomes and genes, sex determination in humans, birds and honeybees; Linkage and crossing over; Sex linked inheritance-hemophilia, colour blindness; Mendelian disorders in humans-Thalassemia; Pedigree Analysis; chromosomal disorders in humans; Down's syndrome, Turner's syndrome and Klinefelter's syndrome.	57	20

UNIT No.	TOPICS	CONTACT HOURS	Marks
	<u>Chapter 5: Molecular basis of inheritance</u> Search for genetic material and DNA as genetic material(experiments on bacterial transformation by F. Griffith; Avery, MacLeod and McCarty; Experiment by Hershey and Chase; Structure of DNA and RNA, DNA packaging, DNA replication; Central Dogma; Transcription Genetic Code, Translation, gene expression and regulation-lac operon; Genome, Human and Rice genome projects; DNA fingerprinting.		
	<u>Chapter 6: Evolution</u> Origin of life; Biological evolution and evidences for biological evolution (Palaeontology, Comparative Anatomy, Embryology and molecular evidence); Darwin's contribution, modern synthetic theory; Mechanism of evolution – Variation (Mutation and Recombination) and Natural selection with examples, types of natural selection; Gene flow and genetic drift; Hardy-Weinberg Principle; Adaptive radiation; Human evolution.		

CLASS - XII

SUBJECT : BIOLOGICAL SCIENCE (BIOS)

SEMESTER – IV

FULL MARKS : 35

CONTACT HOURS : 58 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT-VIII (BIOLOGY AND HUMAN WELFARE)	<u>Chapter 7: Human health and diseases</u> Basic concept of immunology.-Immune system, Antigen, Antibody, antigen-antibody reaction, Types of immunity - vaccine and vaccinations; Pathogens: Parasites causing human diseases: Malaria, Filariasis, Chikungunya, Dengue, Ascariasis, Typhoid, Pneumonia, Common Cold, Amoebiasis, Ringworm, SARS (COVID), Allergy and Autoimmune disorders – Symptoms of disease, Name of causative agents, Mode of transmission. Preventive measures. Cancer, HIV and AIDS — Symptoms of diseases; Causative agents, Mode of transmissions, Preventive measures. Adolescence: Drug and alcohol abuse.	20	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	<u>Chapter 8.Improvement In food production</u> Plant breeding, Tissue culture, Single cell Protein.		
	<u>Chapter 9: Microbes in Human Welfare</u> Microbes in food processing, Industrial production, Sewage treatment, Energy generation, Microbes as bio-control agents and biofertilizers, Antibiotics: Production and judicious use.		
UNIT –IX (BIOTECHNO- LOGY AND ITS APPLICATION)	<u>Chapter 10: Biotechnology and its applications</u> Principle, Process of genetic engineering. (Recombinant DNA technology), Application of biotechnology in health and agriculture, Human Insulin and vaccine production, Stem cell therapy, Gene therapy; Genetically modified organisms: Bt Crops; Transgenic animals. Biosafety issues, Biopiracy and patents.	15	12
UNIT- X (ECOLOGY AND ENVIRON- MENT)	<u>Chapter 11: Organisms and Populations</u> Meaning of Environment. Habitat and niche, Population interactions – Mutualism; Competition; Predation; Parasitism. Population attributes – Growth, birth rate and death rate, age distribution.	23	13
	<u>Chapter 12: Ecosystem.</u> Ecosystem and its pattern; Components of ecosystem. Productivity and Decomposition. Energy flow, Pyramids of number, Biomass and energy, Ecological succession.		
	<u>Chapter 13: Biodiversity and its conservation</u> Biodiversity — concept, Patterns, Importance; Loss of biodiversity; Biodiversity conservation; Hotspots, Endangered organisms, Extinction, Red Data book. Sacred Groves, Biosphere reserves, National Parks, Wildlife Sanctuaries and Ramsar sites.		
	<u>Chapter 14: Environmental issues</u> Solid waste management; Radioactive waste management; Success stories addressing environmental issues-Chipko movement, Dasholi Gram Swarajya Mandal movement (DGSM) Silent Valley movement, Amrita Devi Bishnoi (Jaipur) Movement.		

CLASS : XII

SUBJECT : BIOLOGICAL SCIENCE (BIOS)

COURSE CODE : PRACTICAL

FULL MARKS : 30

CONTACT HOURS : 30 HOURS

Time allowed : 3 hours.

Max. Marks : 30

EVALUATION SCHEME.	MARKS
One major experiment. 5.	6
One minor experiment. 2 and 3.	5
Slide preparation. 1 and 4. (Any one)	3
Spotting. (three)	6(2x3)
Practical record + Viva voce	5(3+2)
Investigatory project + Viva voce	5(3+2)
TOTAL	30

A. List of experiments.

1. Prepare a temporary mount to observe pollen germination.
2. Study the plant population density by quadrat method.
3. Study the plant population frequency by quadrat method.
4. Prepare a temporary mount on onion root tip to study mitosis.
5. Isolate DNA from available plant material, such as Spinach, Green pea seeds, Papaya or any other suitable materials.

B. Study and observe the following. (Spotting)

1. Flowers adapted to pollination by different agencies (Wind, Insects, Birds, etc.).
2. Pollen germination on stigma through a permanent slide.

3. Identification of stages of gamete development i.e., T.S. of Mammalian Testis and TS of Mammalian Ovary through permanent slides.
4. Meiosis in Onion Bud cell or Grasshopper testis through permanent slides.
5. TS of Blastula through permanent slides (Mammalian.)
6. Prepare pedigree charts of any one of the genetic traits, such as rolling of tongue, blood groups, ear lobes, Widow's peak and colour blindness.
7. Common disease causing organisms like *Ascaris*, *Entamoeba*, *Plasmodium*, any fungus causing ringworm through Permanent slides/Models or virtual images or specimens, Comment on symptoms of diseases that they cause
8. Models, specimen showing symbiotic association in root nodules of leguminous plants, *Cuscuta* on host, Lichens.
9. Flash card models showing examples of homologous and analogous organs.

[Note: ***22 Hours** reserved for Remedial classes, Tutorials and Home Assignments.]

BNGA**SUBJECT : BENGALI – A****CLASS - XI****SUBJECT : BENGALI – A (BNGA)****SEMESTER – I****FULL MARKS : 40****CONTACT HOURS : 90 Hours****COURSE CODE : THEORY
(MCQ Type Questions)**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	10	08
	প্রবন্ধ	09	05
	কবিতা	12	07
	আন্তর্জাতিক গল্প ও ভারতীয় কবিতা	12	05
	ভাষা	22	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	25	05

CLASS - XI**SUBJECT : BENGALI – A (BNGA)****SEMESTER – II****FULL MARKS : 40****CONTACT HOURS : 60 Hours****COURSE CODE : THEORY
(SAQ and LAQ Type Questions)**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	10	05
	কবিতা	09	05
	নাটক	06	05
	পূর্ণাঙ্গ সহায়ক গ্রন্থ	12	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	13	05
	প্রবন্ধ রচনা	10	10

PROJECT**CLASS - XI****FULL MARKS : 20****CONTACT HOURS : 30 Hours**

একাদশ শ্রেণি

বিষয় : বাংলা – ক

সেমিস্টার – I

পূর্ণমান – ৪০

গল্প –

পুঁইমাচা – বিভূতিভূষণ বন্দ্যোপাধ্যায়

প্রবন্ধ –

বিড়াল – বঙ্কিমচন্দ্র চট্টোপাধ্যায়

কবিতা –

১. ঈশ্বরচন্দ্র বিদ্যাসাগর – মাইকেল মধুসূদন দত্ত

২. সাম্যবাদী – কাজী নজরুল ইসলাম

আন্তর্জাতিক গল্প –

বিশাল ডানাওয়ালা এক থুথুরে বুড়ো – গ্যাব্রিয়াল গারসিয়া মার্কিজ (অনুবাদ – মানবেন্দ্র বন্দ্যোপাধ্যায়)

ভারতীয় কবিতা –

চারণ কবি – ভারভারা রাও (অনুবাদ – শঙ্খ ঘোষ)

ভাষা –

- বিশ্বের ভাষা ও ভাষা পরিবার
- বাংলা ভাষার উদ্ভব ও ক্রমবিকাশ
- ভারতে প্রচলিত ভাষা পরিবার
- বাংলা ভাষার বৈচিত্র্য

বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস –

পর্ব ১ – প্রাচীন বাংলা : সমাজ ও সাহিত্য

পর্ব ২ – মধ্যযুগের বাংলা সমাজ ও সাহিত্যের প্রধান ধারা : শ্রীকৃষ্ণকীর্তন, বৈষ্ণব পদাবলি, চৈতন্য ও চৈতন্য জীবনী, মঙ্গলকাব্য, অনুবাদ, ইসলামীয় ধারা, শাক্ত পদাবলি ।

একাদশ শ্রেণি
বিষয় : বাংলা – ক
সেমিস্টার – II
পূর্ণমান – ৪০

গল্প –

১. ছুটি – রবীন্দ্রনাথ ঠাকুর
২. তেলেনাপোতা আবিষ্কার – প্রেমেন্দ্র মিত্র

কবিতা –

১. ভাব সন্মিলন – বিদ্যাপতি
২. লালনশাহ ফকিরের গান – লালনশাহ
৩. নুন – জয় গোস্বামী

নাটক –

আগুন – বিজন ভট্টাচার্য

পূর্ণাঙ্গ সহায়ক গ্রন্থ –

পঞ্চতন্ত্র – সৈয়দ মুজতবা আলি

- বই কেনা
- আজব শহর কলকাতা
- পঁচিশে বৈশাখ
- আড্ডা

বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস –

পর্ব ৩ –

- আধুনিক বাংলা সাহিত্য : যুগের আধুনিকতা ও ঔপনিবেশিক বাংলার সামাজিক ও ধর্মীয় আন্দোলন, শিক্ষা- সংস্কার, গদ্যের উদ্ভব, কবিতা, উপন্যাস ও ছোটগল্প, নাটক, যাত্রা ও নাট্যমঞ্চ, প্রবন্ধ (সত্তর দশক পর্যন্ত)।
- লৌকিক সাহিত্য : ছড়া, খাঁখা, প্রবাদ, কথা
- কবিতায় মহাকাব্য, আখ্যানকাব্য, গীতিকবিতার ধারা
- কবিতা : মধুসূদন দত্ত, রবীন্দ্রনাথ ঠাকুর, কাজী নজরুল ইসলাম, জীবনানন্দ দাশ ও পরবর্তী ধারা।
- কথাসাহিত্য : বঙ্কিমচন্দ্র চট্টোপাধ্যায়, রবীন্দ্রনাথ ঠাকুর, শরৎচন্দ্র চট্টোপাধ্যায়, মানিক বন্দ্যোপাধ্যায়, বিভূতিভূষণ বন্দ্যোপাধ্যায়, তারাশঙ্কর বন্দ্যোপাধ্যায় ও পরবর্তী ধারা।
- নাটক : মধুসূদন দত্ত, দীনবন্ধু মিত্র, গিরিশচন্দ্র ঘোষ, রবীন্দ্রনাথ ঠাকুর, দ্বিজেন্দ্রলাল রায়, বিজন ভট্টাচার্য ও পরবর্তী ধারা।

প্রবন্ধ রচনা –

- রচনা লেখার বিষয়টি এক লাইনে না দিয়ে একটি মানস মানচিত্র এবং তথ্যসম্ভাব দেওয়া হবে। পরীক্ষার্থীরা সেই মানচিত্র এবং প্রয়োজনীয় তথ্য ব্যবহার করে প্রবন্ধ রচনা করবে এবং শিরোনাম দেবে।
- একটি বক্তব্যের স্বপক্ষে বা বক্তব্যের বিপক্ষে একটি অনুচ্ছেদ দেওয়া থাকবে। পরীক্ষার্থীরা প্রদত্ত অনুচ্ছেদের স্বপক্ষে / বিপক্ষে তাদের বক্তব্য লিখবে।

একাদশ শ্রেণি

প্রকল্প

প্রকল্প – ২০ (প্রুফ সংশোধন (০৫) প্রকল্পের অন্তর্ভুক্ত হবে) –

- প্রুফসংশোধন – একটি অনুচ্ছেদ দেওয়া হবে, তার থেকে বিরাম চিহ্ন, সম্পাদনা, অনুচ্ছেদ বিভাজন, বানান, শিরোনাম – এই বিষয়গুলির সম্ভাব্য ভুল-ত্রুটি সংশোধন করতে হবে।

সটীক অনুবাদ –

- মূল ভাষা থেকে অনুবাদ করতে হবে। ১০০০ – ৩০০০ শব্দের মধ্য। সময়সীমা – ৬ মাস।
- অনুবাদ করা যাবে – (ক) প্রবন্ধ, (খ) চিঠি, (গ) ঐতিহাসিক নথি, (ঘ) গল্প এবং (ঙ) নাটক।
(লেখক এবং লেখা সম্পর্কে টীকাসহ অনুবাদ করতে হবে)

সাক্ষাৎকার গ্রহণ –

- লোকায়ত আঙ্গিক / সাহিত্য / রাজনীতি / ক্রীড়া / নাচ / গান / কলা ক্ষেত্রের যে কোনো বিখ্যাত ব্যক্তিত্বের (অন্তত জেলাস্তরে পরিচিতি থাকতে হবে) সাক্ষাৎকার নিতে হবে। অন্তত ২০ টি প্রশ্ন।

প্রতিবেদন রচনা –

- ছ'মাসের দৈনিক সংবাদপত্রের খেলার পাতার কাটিং সংগ্রহ এবং তার ভিত্তিতে নিজের দেশ বা রাজ্যের ক্রীড়াক্ষেত্রে উত্থানপতন বিশ্লেষণ করে প্রতিবেদন লিখতে হবে। ন্যূনতম শব্দ সংখ্যা – ১০০০।

স্বরচিত গল্প লিখন – বিদ্যালয় জীবনের কোনো ঘটনাকে কেন্দ্র করে (মৌলিক) এক বা একাধিক স্বরচিত গল্প লিখতে হবে।

CLASS - XII

SUBJECT : BENGALI – A (BNGA)

SEMESTER – III

FULL MARKS : 40

CONTACT HOURS : 90 Hours

COURSE CODE : THEORY

(MCQ Type Questions)

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	10	08
	প্রবন্ধ	09	05
	কবিতা	12	07
	আন্তর্জাতিক গল্প ও ভারতীয় কবিতা	12	05
	ভাষা	22	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	25	05

CLASS - XII

SUBJECT : BENGALI – A (BNGA)

SEMESTER – IV

FULL MARKS : 40

CONTACT HOURS : 60 Hours

COURSE CODE : THEORY

(SAQ and LAQ Type Questions)

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	10	05
	কবিতা	09	05
	নাটক	06	05
	পূর্ণাঙ্গ সহায়ক গ্রন্থ	12	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	13	05
	প্রবন্ধ রচনা	10	10

PROJECT

CLASS - XII

FULL MARKS : 20

CONTACT HOURS : 30 Hours

দ্বাদশ শ্রেণি

বিষয় : বাংলা – ক

সেমিস্টার – III

পূর্ণমান – ৪০

গল্প –

আদরিণী – প্রভাত কুমার মুখোপাধ্যায়

কবিতা –

১. অন্ধকার লেখাগুচ্ছ – শ্রীজাত

২. দিগ্বিজয়ের রূপকথা – নবনীতা দেবসেন

প্রবন্ধ –

বাঙ্গালা ভাষা – স্বামী বিবেকানন্দ

ভারতীয় গল্প –

পোটরাজ – শঙ্কর রাও খারাট (অনুবাদ : সুনন্দন চক্রবর্তী)

আন্তর্জাতিক কবিতা –

তার সঙ্গে – পাবলো নেরুদা (অনুবাদ : শক্তি চট্টোপাধ্যায়)

ভাষা –

প্রথম অধ্যায় : ভাষাবিজ্ঞান ও তার শাখা-প্রশাখা

দ্বিতীয় অধ্যায় : ধ্বনিতত্ত্ব

তৃতীয় অধ্যায় : শব্দার্থতত্ত্ব

বাংলা ভাষা ও সংস্কৃতি :-

প্রথম অধ্যায় : বাংলাগানের ইতিহাস – সংক্ষিপ্ত রূপরেখা

দ্বিতীয় অধ্যায় : বাঙালির বিজ্ঞানচর্চার সংক্ষিপ্ত পরিচিতি

তৃতীয় অধ্যায় : বাঙালির ক্রীড়া সংস্কৃতি সংক্ষিপ্ত পরিচিতি

দ্বাদশ শ্রেণি
বিষয় : বাংলা – ক
সেমিস্টার – IV
পূর্ণমান – ৪০

গল্প –

১. হলুদ পোঁড়া – মানিক বন্দ্যোপাধ্যায়
২. হারুণ সালেমের মাসি-মহাশ্বেতা দেবী

কবিতা –

১. প্রার্থনা – রবীন্দ্রনাথ ঠাকুর
২. তিমির হননের গান – জীবনানন্দ দাশ
৩. কেন এল না – সুভাষ মুখোপাধ্যায়

নাটক –

নানা রঙের দিন – অজিতেশ বন্দ্যোপাধ্যায়

পূর্ণাঙ্গ সহায়ক গ্রন্থ -

ডাকঘর – রবীন্দ্রনাথ ঠাকুর

বাংলা ভাষা ও সংস্কৃতি :-

চতুর্থ অধ্যায় : বাংলা চিত্রকলার ইতিহাস-সংক্ষিপ্ত রূপরেখা

পঞ্চম অধ্যায় : বাঙালির চলচ্চিত্রের ইতিহাস-সংক্ষিপ্ত রূপরেখা

প্রবন্ধ রচনা –

কোনো একটি বিষয়ে কোনো একজন লেখকের লেখার একটি অংশ দেওয়া থাকবে। প্রদত্ত অনুচ্ছেদটি হল মূল রচনার প্রস্তাবনা বা ভূমিকা। এই প্রস্তাবনা বা ভূমিকাটিকে অবলম্বন করে পরীক্ষার্থী বিষয়বস্তুর গভীরে প্রবেশ করবে এবং পরিণতি দান করবে।

একটি প্রশ্নে কোনো একটি বিষয়ে নানা ধরনের সূত্র ও তথ্য দেওয়া থাকবে। সেগুলিকে ব্যবহার করে পরীক্ষার্থী রচনাটি গড়ে তুলবে।

* বাংলাভাষা ও সংস্কৃতি বইটির বিষয়বস্তুর আংশিক পরিবর্তন, সংযোজন এবং সংক্ষিপ্তকরণ করা হচ্ছে।

দ্বাদশ শ্রেণি

প্রকল্প

প্রকল্প – ২০ (প্রচ্ছদ সংশোধন (০৫) প্রকল্পের অন্তর্ভুক্ত হবে) –

- প্রচ্ছদসংশোধন – একটি অনুচ্ছেদ দেওয়া হবে, তার থেকে বিরাম চিহ্ন, সম্পাদনা, অনুচ্ছেদ বিভাজন, বানান, শিরোনাম – এই বিষয়গুলির সম্ভাব্য ভুল ত্রুটি সংশোধন করতে হবে।

প্রকল্প (প্রদত্ত প্রকল্পের মধ্যে যে কোনো একটি)

সমীক্ষা পত্র –

ন্যূনতম ৫০ টি নমুনার ভিত্তিতে তৈরি করতে হবে।

নিজের গ্রাম / পাড়া / অঞ্চলকে কেন্দ্র করে সাধারণ সমাজ ভিত্তিক সমীক্ষা নারী / পুরুষের হার, সাক্ষরতা, ধর্মসম্প্রদায়, খেলাধুলার প্রবণতা (ইন্ডোর, আউটডোর, ফুটবল, ক্রিকেট, হকি) প্রভৃতি। অন্যদিকে, আদিবাসী, জনজাতি, জনগোষ্ঠী ও সম্প্রদায়ভিত্তিক বিবিধ সমীক্ষা।

নিজের বিদ্যালয় নিয়েও করা যাবে, আবার অন্যান্য প্রতিবেশী বিদ্যালয়কে অন্তর্ভুক্ত করা যেতে পারে। সমীক্ষার বিষয় হবে বিদ্যালয় সংশ্লিষ্ট কোনো তথ্য উদ্ঘাটন। হতে পারে বিদ্যালয় শিক্ষার্থীদের জাতিগত / সম্প্রদায় নকশা, হতে পারে শিক্ষক-শিক্ষিকাদের বয়স, সারস্বত ক্রিয়াকর্ম, বিদ্যালয়ের ইতিহাস (মৌখিক সাক্ষাৎকার-সমীক্ষা), বিদ্যালয়ের শ্রেণিবিভাজন – শিক্ষার্থীসামর্থ্য প্রভৃতি।

গল্পের নাট্যরূপ –

ন্যূনতম শব্দসংখ্যা ২০০০-৪০০০ (কোনো প্রথিতযশা লেখক / লেখিকার লেখা থেকে তৈরী করতে হবে)।

গল্পের চিত্রনাট্য –

শব্দসংখ্যা ২০০০-৪০০০ (কোনো প্রথিতযশা লেখক / লেখিকার লেখা থেকে তৈরী করতে হবে)।

গ্রন্থ সমালোচনা –

প্রথিতযশা কবি, গল্পকার, নাট্যকার ও প্রাবন্ধিকের কোনো ১টি পূর্ণাঙ্গ বইয়ের সমালোচনা। ন্যূনতম শব্দসংখ্যা-১০০০।

নির্বাচিত রচনাকারের সাহিত্যশৈলী বিচার –

বাংলা ‘ক’ ভাষার ছাত্র-ছাত্রীরা ইংরাজি পাঠক্রমের অন্তর্ভুক্ত কোনো একজন লেখকের গল্প (চারটি) বা কবিতা (ছয়টি) আলোচনা করে রচনার শৈলী, বৈশিষ্ট্য ও অভিনবত্ব সম্পর্কে একটি মৌলিক প্রবন্ধ লিখবে।

বাংলা ‘খ’ ভাষার ছাত্র-ছাত্রীরা বাংলা পাঠক্রমের অন্তর্ভুক্ত কোনো একজন লেখকের গল্প (চারটি) বা কবিতা (ছয়টি) আলোচনা করে তাঁর শৈলী বৈশিষ্ট্য ও অভিনবত্ব সম্পর্কে একটি মৌলিক প্রবন্ধ লিখবে। (৫০০-১০০০ শব্দ)

নির্দিষ্ট ঐতিহাসিক কালপর্বে সাহিত্য সংস্কৃতির বিকাশ ও বিবর্তনের তথ্যানুসন্ধান –

সাহিত্য সংস্কৃতির ইতিহাস থেকে ঊনবিংশ বা বিংশ শতকের বাংলার কোনো একটি কাল ও পর্বকে কেন্দ্র করে (১) ভাষা (২) খাদ্যাভ্যাস (৩) পোশাক (৪) গৃহনির্মাণ (৫) সংগীত (৬) নাটক (৭) চলচ্চিত্র (৮) চিত্রকলা ও ভাস্কর্য (৯) খেলাধুলা এইসব বিষয়ে যে কোনো একটির বিকাশ, বিবর্তন ও বৈশিষ্ট্যের তথ্যসহ অনুসন্ধান ও আলোচনা।

নির্বাচিত সাহিত্য-সৃষ্ট চরিত্রের জীবনী নির্মাণ –

ফেলুদা / ঘনাদা / টেনিদা / শঙ্কু / সদাশিব / ঋজুদা / পাগলা দাশু / গোগোল / কাকাবাবু / কিকিরা / হাঁদা-ভোঁদা বা নন্টে-ফন্টে / হর্ষবর্ধন-গোবর্ধন।

নির্বাচিত সাহিত্যিকের সাহিত্য-অবদান সম্পর্কিত প্রকল্প নির্মাণ –

উপেন্দ্রকিশোর রায়চৌধুরী, দক্ষিণারঞ্জন মিত্র মজুমদার, সুকুমার রায়, অবনীন্দ্রনাথ ঠাকুর, প্রেমেন্দ্র মিত্র, শরদিন্দু বন্দ্যোপাধ্যায়, লীলা মজুমদার, নারায়ণ গঙ্গোপাধ্যায়, শিবরাম চক্রবর্তী, সত্যজিত রায়, শীর্ষেন্দু মুখোপাধ্যায়, হিমালীশ গোস্বামী, নারায়ন দেবনাথ, শৈলেন ঘোষ।

উল্লিখিত লেখকদের মধ্যে থেকে যে কোনো একজনের বাংলা সাহিত্যে অবদান ও সেই অবদান অপরিহার্য প্রতিপন্ন করে প্রকল্প নির্মাণ (ন্যূনতম ২০০০ শব্দ)

BNGB**SUBJECT : BENGALI – B****CLASS - XI****SUBJECT : BENGALI – B (BNGB)****SEMESTER – I****FULL MARKS : 40****CONTACT HOURS : 90 Hours****COURSE CODE : THEORY****(MCQ Type Questions)**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	10	08
	প্রবন্ধ	10	05
	কবিতা	15	07
	আন্তর্জাতিক গল্প ও কবিতা	20	05
	ভাষা	15	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	20	05

CLASS - XI**SUBJECT : BENGALI – B (BNGB)****SEMESTER – II****FULL MARKS : 40****CONTACT HOURS : 60 Hours****COURSE CODE : THEORY****(SAQ and LAQ Type Questions)**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	08	05
	প্রবন্ধ	08	05
	কবিতা	08	05
	পূর্ণঙ্গ সহায়ক গ্রন্থ	15	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	12	05
	প্রবন্ধ রচনা	09	10

PROJECT**CLASS - XI****FULL MARKS : 20****CONTACT HOURS : 30 Hours**

একাদশ শ্রেণি
বিষয় : বাংলা – খ
সেমিস্টার – I
পূর্ণমান – ৪০

গল্প –

পৌরাণিক-আধুনিক – বনফুল

প্রবন্ধ –

আড্ডা – বুদ্ধদেব বসু

কবিতা –

১. কপোতাক্ষনদ – মাইকেল মধুসূদন দত্ত

২. ছেলেটা – রবীন্দ্রনাথ ঠাকুর

আন্তর্জাতিক গল্প –

বুমবুম-জুল ক্লারেতি (অনুবাদ – উৎপল দত্ত)

আন্তর্জাতিক কবিতা –

আমার মাকে – ডেভিড দিয়োপ (অনুবাদ : শঙ্খ ঘোষ)

ভাষা –

- বাংলা ভাষার উদ্ভব ও ক্রমবিকাশ
- বাংলা ভাষার বৈচিত্র্য

বাংলা শিল্প – সাহিত্য ও সংস্কৃতির ইতিহাস –

পর্ব ১ – প্রাচীন বাংলা : সমাজ ও সাহিত্য

পর্ব ২ – মধ্যযুগের বাংলা সমাজ ও সাহিত্যের প্রধান ধারা : শ্রীকৃষ্ণকীর্তন, বৈষ্ণব পদাবলি, চৈতন্য ও চৈতন্য
জীবনী, মঙ্গলকাব্য, অনুবাদ, ইসলামীয় ধারা, শাক্ত পদাবলি।

একাদশ শ্রেণি
বিষয় : বাংলা – খ
সেমিস্টার – II
পূর্ণমান – ৪০

গল্প –

ঈর্ষা – আশাপূর্ণা দেবী

প্রবন্ধ –

মুনশীজী – শ্রীপাছ

কবিতা –

১. মানুষের নামে – বীরেন্দ্র চট্টোপাধ্যায়

২. জনমদুখিনী – সুনীল গঙ্গোপাধ্যায়

পূর্ণাঙ্গ সহায়ক গ্রন্থ – রবীন্দ্রনাথের গল্প

১. দেনাপাওনা

২. কাবুলিওয়ালা

৩. চিত্রকর

বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস –

পর্ব ৩ –

- আধুনিক বাংলা সাহিত্য : যুগের আধুনিকতা ও ঔপনিবেশিক বাংলার সামাজিক ও ধর্মীয় আন্দোলন, শিক্ষা- সংস্কার, গদ্যের উদ্ভব, কবিতা, উপন্যাস ও ছোটগল্প, নাটক, যাত্রা ও নাট্যমঞ্চ, প্রবন্ধ (স্বাধীনতার সময় পর্যন্ত)।
- কবিতায় মহাকাব্য, আখ্যানকাব্য, গীতিকবিতার ধারা
- কবিতা : মধুসূদন দত্ত, রবীন্দ্রনাথ ঠাকুর, কাজী নজরুল ইসলাম, জীবনানন্দ দাশ।
- কথাসাহিত্য : বঙ্কিমচন্দ্র চট্টোপাধ্যায়, রবীন্দ্রনাথ ঠাকুর, শরৎচন্দ্র চট্টোপাধ্যায়, মানিক বন্দ্যোপাধ্যায়, বিভূতিভূষণ বন্দ্যোপাধ্যায়, তারাশঙ্কর বন্দ্যোপাধ্যায়।
- নাটক : মধুসূদন দত্ত, দীনবন্ধু মিত্র, গিরিশচন্দ্র ঘোষ, রবীন্দ্রনাথ ঠাকুর, দ্বিজেন্দ্রলাল রায়, বিজন ভট্টাচার্য।

প্রবন্ধ রচনা –

- রচনা লেখার বিষয়টি এক লাইনে না দিয়ে একটি মানস মানচিত্র এবং তথ্যসম্ভাব দেওয়া হবে। পরীক্ষার্থীরা সেই মানচিত্র এবং প্রয়োজনীয় তথ্য ব্যবহার করে প্রবন্ধ রচনা করবে এবং শিরোনাম দেবে।
- একটি বক্তব্যের স্বপক্ষে বা বক্তব্যের বিপক্ষে একটি অনুচ্ছেদ দেওয়া থাকবে। পরীক্ষার্থীরা প্রদত্ত অনুচ্ছেদের স্বপক্ষে / বিপক্ষে তাদের বক্তব্য লিখবে।

একাদশ শ্রেণি

প্রকল্প

প্রকল্প – ২০ (প্রুফ সংশোধন (০৫) প্রকল্পের অন্তর্ভুক্ত হবে) –

- প্রুফ সংশোধন – একটি অনুচ্ছেদ দেওয়া হবে, তার থেকে বিরাম চিহ্ন, সম্পাদনা, অনুচ্ছেদ বিভাজন, বানান, শিরোনাম – এই বিষয়গুলির সম্ভাব্য ভুল-ত্রুটি সংশোধন করতে হবে।

সটীক অনুবাদ –

- মূল ভাষা থেকে অনুবাদ করতে হবে। ১০০০ – ৩০০০ শব্দের মধ্যে। সময়সীমা – ৬ মাস।
- অনুবাদ করা যাবে – (ক) প্রবন্ধ, (খ) চিঠি, (গ) ঐতিহাসিক নথি, (ঘ) গল্প এবং (ঙ) নাটক।
(লেখক এবং লেখা সম্পর্কে টীকাসহ অনুবাদ করতে হবে)

সাক্ষাৎকার গ্রহণ –

- লোকায়ত আঙ্গিক / সাহিত্য / রাজনীতি / ক্রীড়া / নাচ / গান / কলা ক্ষেত্রের যে কোনো বিখ্যাত ব্যক্তিত্বের (অন্তত জেলাস্তরে পরিচিতি থাকতে হবে) সাক্ষাৎকার নিতে হবে। অন্তত ২০ টি প্রশ্ন।

প্রতিবেদন রচনা –

- ছ'মাসের দৈনিক সংবাদপত্রের খেলার পাতার কাটিং সংগ্রহ এবং তার ভিত্তিতে নিজের দেশ বা রাজ্যের ক্রীড়াক্ষেত্রে উত্থানপতন বিশ্লেষণ করে প্রতিবেদন লিখতে হবে। ন্যূনতম শব্দ সংখ্যা – ১০০০।

স্বরচিত গল্প লিখন – বিদ্যালয় জীবনের কোনো ঘটনাকে কেন্দ্র করে (মৌলিক) এক বা একাধিক স্বরচিত গল্প লিখতে হবে।

CLASS - XII

SUBJECT : BENGALI – B (BNGB)

SEMESTER – III

FULL MARKS : 40

CONTACT HOURS : 90 Hours

COURSE CODE : THEORY

(MCQ Type Questions)

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	12	08
	কবিতা	10	07
	নাটক	15	05
	ভারতীয় গল্প ও কবিতা	20	05
	ভাষা	18	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	15	05

CLASS - XII

SUBJECT : BENGALI – B (BNGB)

SEMESTER – IV

FULL MARKS : 40

CONTACT HOURS : 60 Hours

COURSE CODE : THEORY

(SAQ and LAQ Type Questions)

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	গল্প	10	05
	কবিতা	08	05
	নাটক	07	05
	পূর্ণাঙ্গ সহায়ক গ্রন্থ	16	10
	বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস	10	05
	প্রবন্ধ রচনা	09	10

PROJECT

CLASS - XII

FULL MARKS : 20

CONTACT HOURS : 30 Hours

দ্বাদশ শ্রেণি

বিষয় : বাংলা – খ

সেমিস্টার – III

পূর্ণমান – ৪০

গল্প –

মহেশ – শরৎচন্দ্র চট্টোপাধ্যায়

কবিতা –

কান্ডারী হুঁশিয়ার – কাজী নজরুল ইসলাম

ভারতীয় গল্প –

নেমকের দারোগা – প্রেমচন্দ্র

ভারতীয় কবিতা –

পৃথিবী আমার কবিতা – হীরেন ভট্টাচার্য

নাটক –

লক্ষ্মণের শক্তিশেল – সুকুমার রায়

ভাষা –

১. বাংলা ধ্বনিতত্ত্ব : বাংলা স্বরধ্বনি ও উচ্চারণ অনুযায়ী তার শ্রেণিবিভাগ

২. বাংলা শব্দভাণ্ডার

৩. শব্দার্থতত্ত্ব

বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস –

পর্ব ৪ – বাংলা গানের ইতিহাস : সংক্ষিপ্ত রূপরেখা

পর্ব ৬ – বাঙালির চলচ্চিত্রের ইতিহাস – সংক্ষিপ্ত রূপরেখা

পর্ব ৮ – বাঙালির ক্রীড়া সংস্কৃতি

দ্বাদশ শ্রেণি
বিষয় : বাংলা – খ
সেমিস্টার – IV
পূর্ণমান – ৪০

গল্প –

১. নাম নেই – সুনীল গঙ্গোপাধ্যায়
২. সম্প্রদায়ের ভাষা – শঙ্খ ঘোষ

কবিতা –

১. খুকু ও খোকা – অন্নদাশঙ্কর রায়
২. কন্যাশ্লোক – মল্লিকা সেনগুপ্ত

নাটক –

সূক্ষ্ম বিচার – রবীন্দ্রনাথ ঠাকুর

পূর্ণাঙ্গ সহায়ক গ্রন্থ - রবীন্দ্রনাথের গল্প

১. খোকাবাবুর প্রত্যাবর্তন
২. সুভা
৩. গুপ্তধন
৪. দর্পহরণ

বাংলা শিল্প-সাহিত্য ও সংস্কৃতির ইতিহাস –

পর্ব ৫ – বাংলা চিত্রকলার ইতিহাস – সংক্ষিপ্ত রূপরেখা

পর্ব ৭ – বাঙালির বিজ্ঞানচর্চার সংক্ষিপ্ত পরিচিতি

প্রবন্ধ রচনা –

- কোনো একটি বিষয়ে কোনো একজন লেখকের লেখার একটি অংশ দেওয়া থাকবে। প্রদত্ত অনুচ্ছেদটি হল মূল রচনার প্রস্তাবনা বা ভূমিকা। এই প্রস্তাবনা বা ভূমিকাটিকে অবলম্বন করে পরীক্ষার্থী বিষয়বস্তুর গভীরে প্রবেশ করবে এবং পরিণতি দান করবে।
- একটি প্রশ্নে কোনো একটি বিষয়ে নানা ধরনের সূত্র ও তথ্য দেওয়া থাকবে। সেগুলিকে ব্যবহার করে পরীক্ষার্থী রচনাটি গড়ে তুলবে।

দ্বাদশ শ্রেণি

প্রকল্প

প্রকল্প – ২০ (প্রফ সংশোধন (০৫) প্রকল্পের অন্তর্ভুক্ত হবে) –

- প্রফসংশোধন – একটি অনুচ্ছেদ দেওয়া হবে, তার থেকে বিরাম চিহ্ন, সম্পাদনা, অনুচ্ছেদ বিভাজন, বানান, শিরোনাম – এই বিষয়গুলির সম্ভাব্য ভুল-ত্রুটি সংশোধন করতে হবে।

প্রকল্প (প্রদত্ত প্রকল্পের মধ্যে যে কোনো একটি)

সমীক্ষা পত্র –

ন্যূনতম ৫০ টি নমুনার ভিত্তিতে তৈরি করতে হবে।

নিজের গ্রাম / পাড়া / অঞ্চলকে কেন্দ্র করে সাধারণ সমাজভিত্তিক সমীক্ষা নারী / পুরুষের হার, সাক্ষরতা, ধর্মসম্প্রদায়, খেলাধুলার প্রবণতা (ইন্ডোর, আউটডোর, ফুটবল, ক্রিকেট, হকি) প্রভৃতি। অন্যদিকে, আদিবাসী, জনজাতি, জনগোষ্ঠী ও সম্প্রদায়ভিত্তিক বিবিধ সমীক্ষা।

নিজের বিদ্যালয় নিয়েও করা যাবে, আবার অন্যান্য প্রতিবেশী বিদ্যালয়কে অন্তর্ভুক্ত করা যেতে পারে। সমীক্ষার বিষয় হবে বিদ্যালয় সংশ্লিষ্ট কোনো তথ্য উদঘাটন। হতে পারে বিদ্যালয় শিক্ষার্থীদের জাতিগত / সম্প্রদায় নকশা, হতে পারে শিক্ষক-শিক্ষিকাদের বয়স, সারস্বত ক্রিয়াকর্ম, বিদ্যালয়ের ইতিহাস (মৌখিক সাক্ষাৎকার-সমীক্ষা), বিদ্যালয়ের শ্রেণিবিভাজন – শিক্ষার্থীসামর্থ্য প্রভৃতি।

গল্পের নাট্যরূপ –

ন্যূনতম শব্দসংখ্যা ২০০০-৪০০০ (কোনো প্রথিতযশা লেখক / লেখিকার লেখা থেকে তৈরী করতে হবে)।

গল্পের চিত্রনাট্য –

শব্দসংখ্যা ২০০০-৪০০০ (কোনো প্রথিতযশা লেখক / লেখিকার লেখা থেকে তৈরী করতে হবে)।

গ্রন্থ সমালোচনা –

প্রথিতযশা কবি, গল্পকার, নাট্যকার ও প্রাবন্ধিকের কোনো ১টি পূর্ণাঙ্গ বইয়ের সমালোচনা। ন্যূনতম শব্দসংখ্যা-১০০০।

নির্বাচিত রচনাকারের সাহিত্যশৈলী বিচার –

বাংলা ‘ক’ ভাষার ছাত্র-ছাত্রীরা ইংরাজি পাঠক্রমের অন্তর্ভুক্ত কোনো একজন লেখকের গল্প (চারটি) বা কবিতা (ছয়টি) আলোচনা করে রচনার শৈলী, বৈশিষ্ট্য ও অভিনবত্ব সম্পর্কে একটি মৌলিক প্রবন্ধ লিখবে।

বাংলা ‘খ’ ভাষার ছাত্র-ছাত্রীরা বাংলা পাঠক্রমের অন্তর্ভুক্ত কোনো একজন লেখকের গল্প (চারটি) বা কবিতা (ছয়টি) আলোচনা করে তাঁর শৈলী, বৈশিষ্ট্য ও অভিনবত্ব সম্পর্কে একটি মৌলিক প্রবন্ধ লিখবে। (৫০০-১০০০ শব্দ)

নির্দিষ্ট ঐতিহাসিক কালপর্বে সাহিত্য সংস্কৃতির বিকাশ ও বিবর্তনের তথ্যানুসন্ধান –

সাহিত্য সংস্কৃতির ইতিহাস থেকে ঊনবিংশ বা বিংশ শতকের বাংলার কোনো একটি কাল ও পর্বকে কেন্দ্র করে (১) ভাষা (২) খাদ্যাভ্যাস (৩) পোষাক (৪) গৃহনির্মাণ (৫) সংগীত (৬) নাটক (৭) চলচ্চিত্র (৮) চিত্রকলা ও ভাস্কর্য (৯) খেলাধুলা এইসব বিষয়ে যে কোনো একটির বিকাশ, বিবর্তন ও বৈশিষ্ট্যের তথ্যসহ অনুসন্ধান ও আলোচনা।

নির্বাচিত সাহিত্য-সৃষ্ট চরিত্রের জীবনী নির্মাণ –

ফেলুদা / ঘনাদা / টেনিদা / শঙ্কু / সদাশিব / ঋজুদা / পাগলা দাশু / গোগোল / কাকাবাবু / কিকিরা / হাঁদা-ভোঁদা বা নন্টে-ফন্টে / হর্ষবর্ধন-গোবর্ধন।

নির্বাচিত সাহিত্যিকের সাহিত্য-অবদান সম্পর্কিত প্রকল্প নির্মাণ –

উপেন্দ্রকিশোর রায়চৌধুরী, দক্ষিণারঞ্জন মিত্র মজুমদার, সুকুমার রায়, অবনীন্দ্রনাথ ঠাকুর, প্রমেন্দ্র মিত্র, শরদিন্দু বন্দ্যোপাধ্যায়, লীলা মজুমদার, নারায়ণ গঙ্গোপাধ্যায়, শিবরাম চক্রবর্তী, সত্যজিত রায়, শীর্ষেন্দু মুখোপাধ্যায়, হিমালীশ গোস্বামী, নারায়ণ দেবনাথ, শৈলেন ঘোষ।

উল্লিখিত লেখকদের মধ্যে থেকে যে কোনো একজনের বাংলা সাহিত্যে অবদান ও সেই অবদান অপরিহার্য প্রতিপন্ন করে প্রকল্প নির্মাণ (ন্যূনতম ২০০০ শব্দ)

CLASS - XI**SUBJECT : CHEMISTRY (CHEM)****SEMESTER – I****FULL MARKS : 35****CONTACT HOURS : 70 Hours****COURSE CODE : THEORY****Sub-topics**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 1	Some Basic Concepts of Chemistry: Laws of chemical combination. Concept of elements, atoms and molecules. Atomic and molecular masses. Mole concept and molar mass, percentage composition, empirical and molecular formula, chemical reactions, stoichiometry and calculations based on stoichiometry. Different concentration terms of solutions and related calculations.	07	03
Unit - 2	Structure of Atom: Bohr's model and its limitations, concept of shell and sub-shells, the dual nature of matter and light, de Broglie's relationship. Heisenberg uncertainty principle, Schrödinger wave equation (elementary idea only). Concept of orbitals, quantum numbers, shapes of <i>s</i> , <i>p</i> and <i>d</i> orbitals, rules for filling electrons in orbitals: Aufbau principle, Pauli exclusion principle and Hund's rule, exchange energy, electronic configuration of atom, stability of half-filled, completely filled orbitals.	12	06
Unit - 3	Classification of Elements and Periodicity in Properties: Modern periodic law and the present form of the periodic table, periodic trends in properties of elements – atomic radii, ionic radii, van der Waals' radii, ionization enthalpy, electron gain enthalpy, electronegativity, valency. Nomenclature of elements with atomic number greater than 100.	07	04

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 4	Chemical Bonding and Molecular Structure: Valence electrons, ionic bond, bond parameters, covalent bond, Lewis structure, polar character of covalent bond, covalent character of ionic bond, valence bond theory, resonance, geometry of covalent molecules, VSEPR theory, concept of hybridisation, involving <i>s</i> , <i>p</i> and <i>d</i> orbitals and shapes of some simple molecules, intermolecular interactions, Hydrogen bonding, Molecular orbital theory of homonuclear diatomic molecules (H_2 , He_2 , O_2 , N_2 , F_2 – qualitative idea only)	13	06
Unit - 5	States of Matter — Solids and Gases: Classification of solids (elementary idea): molecular, ionic, covalent and metallic solids, amorphous and crystalline solids (elementary idea), unit cell in two-dimensional and three-dimensional lattices, packing efficiency, calculation of density of unit cell, packing in solids, voids, number of atoms per unit cell in a cubic unit cell, point defects. Kinetic theory of gas, molecular speeds, Dalton's law of partial pressure, Graham's law, deviation of ideal behaviour and van der Waals' equation, Liquefaction of gases, critical temperature.	09	04
Unit - 6	s-Block Elements (Group 1 and Group 2 elements): Electronic configuration, occurrence, trends in the variation of properties (such as ionization enthalpy, atomic and ionic radii), trends in chemical reactivity with oxygen, water, hydrogen and halogens, hydrides (ionic, covalent and interstitial), hydrogen peroxide (preparation, properties, structure & use.), hydrogen as a fuel. Biological importance of Na, K, Mg, Ca.	10	05
Unit - 7	p-Block Elements (Group 13 and Group 14 elements): General introduction to <i>p</i> -block elements, electronic configuration, occurrence, variation in properties, oxidation states, and trends in chemical reactivity of group 13 and 14 elements. Group 13: Boron: physical and chemical properties of compounds of Boron: Boron oxides, boric acid, borates and B_2H_6 Aluminium: Reactions of Al with acid and alkali, uses of Al, Preparation and uses of $LiAlH_4$ and Al_2O_3 . Group 14: Carbon: catenation, allotropic forms, nano carbon, graphene, physical and chemical properties of two oxides of carbon- CO and CO_2 , Silicon: some compounds of silicon and their important uses – Silicon tetrachloride (Structure, preparation, hydrolysis and reduction reaction only), silicates [structure of open chain silicates constructing of $(SiO_3)_n^{2n-}$ ions], use of zeolites,	12	07

CLASS - XI

SUBJECT : CHEMISTRY (CHEM)

SEMESTER – II

FULL MARKS : 35

CONTACT HOURS : 60 HOURS

COURSE CODE : THEORY

Sub-topics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 1	Thermodynamics : Concepts of system (including types of system), surroundings. Work, heat, energy, extensive and intensive properties, state function, Zeroth law of thermodynamics and definition of temperature. The first law of thermodynamics – internal energy change (ΔU) and enthalpy change (ΔH), Enthalpy of bond dissociation, combustion, formation, atomization, ionization, solution and sublimation. Transformation of state. Hess's law of constant heat summation, Born Haber Cycle and its application. 2 nd law of thermodynamics, the introduction of entropy as a state function, Gibbs energy change for spontaneous and non-spontaneous processes, criteria for equilibrium.	12	07
Unit - 2	Equilibrium: Equilibrium in physical and chemical processes, dynamic nature of equilibrium, law of mass reaction, equilibrium constant, factors affecting equilibrium – Le Chatelier's principle; ionic equilibrium, ionization of acids and bases, strong and weak electrolytes, degree of ionization of polybasic acids, acid strength, concept of pH Henderson Equation. Hydrolysis of salts (elementary idea). Buffer solutions, solubility product, common ion effect (with illustrative examples).	10	06
Unit - 3	Redox Reactions: Concept of oxidation and reduction, redox reactions, oxidation number, balancing redox reactions in terms of loss and gain of electrons and change in oxidation number, applications of redox reactions in permanganometry and dichromatometry	05	03

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 4	Organic Chemistry: Some basic principles : General introduction, classification and IUPAC nomenclature of organic compounds. Electronic displacements in a covalent bond: inductive effect, resonance and hyperconjugation. Homolytic and Heterolytic fission of a covalent bond: free radicals, carbocations, carbanions electrophiles and nucleophiles, types of organic reactions. Elementary idea of addition, elimination and substitution reactions.	12	07
Unit - 5	Hydrocarbons: Classification of hydrocarbons Alkanes – Nomenclature, isomerism, conformations (ethane only), physical properties (up to 6 carbons) and chemical reactions including halogenations, free radical mechanism, combustion and pyrolysis. Alkenes – Nomenclature, structure of double bond (ethene), geometrical isomerism, physical properties (up to 3 carbons) methods of preparation; chemical reactions; addition of hydrogen, halogen, water hydrogen halides (Markovnikov's addition and peroxide effect), ozonolysis, oxidation, mechanism of electrophilic addition. Alkynes – Nomenclature, structure of triple bond (ethyne), physical properties (up to 3 carbons) preparation, chemical reactions; acidic character of Alkynes, addition reaction of – hydrogen, halogens, hydrogen halides and water. Aromatic hydrocarbons; Introduction, IUPAC nomenclature; Benzene; resonance aromaticity; chemical properties; mechanism of electrophilic substitution – nitration, sulphonation, halogenations, Friedel-Crafts alkylation and acylation, carcinogenicity and toxicity.	14	08
Unit - 6	Environmental Chemistry : Environmental pollution – air, water and soil pollution (cause and effects), Primary and secondary pollutants (solid and liquid), chemical reactions in the atmosphere, smog, pollution due to industrial wastes; solid waste management (elementary idea only), SPM, RSPM, green chemistry as an alternative tool for reducing pollution. Water preservation and protection, Strategy for control of environmental pollution.	07	04

CLASS - XII

SUBJECT : CHEMISTRY (CHEM)

SEMESTER – III

FULL MARKS : 35

CONTACT HOURS : 70 HOURS

COURSE CODE : THEORY

Sub-topics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 1	Liquid State Introduction, Solubility of gases in liquids, solid solutions, Vapour pressure and Raoult's law. Colligative properties; relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, osmotic pressure. Determination of molecular mass using colligative properties. Abnormal molecular mass, van't Hoff factor and calculations involving it. Colloidal solution, the difference between true solutions, colloids and suspensions; lyophilic, lyophobic, multi-molecular colloids; properties of colloids; Tyndal effect, Brownian movement, electrophoresis, coagulation, emulsions and types of emulsions.	16	08
Unit - 2	p-Block Elements (Groups 15, 16, 17 and 18) Group 15 elements: general introduction, electronic configuration, occurrence, oxidation states, Structure and reaction of NH_3 , HNO_3 , NCl_3 , oxides of nitrogen (structure only); Phosphorus – allotropic forms(White and Red), preparation and properties of phosphine, phosphorus halides (PCl_3 , PCl_5) and oxoacids (elementary idea only) Group 16 elements: General introduction, electronic configuration, occurrence, oxidation states; Oxygen: classification of oxides. Preparation and properties of Ozone. Sulphur: allotropic forms (rhombic and monoclinic). Properties and uses of oxides, oxoacids and peracids of sulphur. Group 17 elements: General introduction, electronic configuration, oxidation states, occurrence, trends in physical and chemical properties; Compounds of halogen; preparation, structure and uses of oxides, oxoacids of halogens, interhalogen compounds. Elementary idea of pseudohalogens and polyhalides. Group 18 elements : General introduction, electronic configuration, occurrence, uses of noble gases. Preparation, structure and chemical reactions of XeO_2 , XeO_3 , XeF_2 , XeF_4 , XeF_6 , XeOF_2 .	18	08

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 3	Haloalkanes and Haloarenes Haloalkanes : Nomenclature, nature of C-X bond, physical and chemical properties, mechanism of substitution reactions. Stability of carbocations. <i>R/S</i> and <i>D/L</i> configurations Uses and environmental effects of – dichloromethane, trichloromethane, tetrachloromethane, iodoform, freons, Haloarenes: Nature of C-X bond, substitution reaction (directive influence of halogen for monosubstituted compounds only), stability of carbocations, <i>R/S</i> and <i>D/L</i> configurations. Uses and environmental effects of DDT.	10	05
Unit - 4	Alcohols, Phenols and Ethers Alcohols : Nomenclature, methods of preparation, physical and chemical properties (primary alcohols only); identification of primary, secondary and tertiary alcohols; mechanism of dehydration, uses of methanol and ethanol. Phenols: Nomenclature, methods of preparation, physical and chemical properties, acidic nature of phenol, electrophilic substitution reaction, uses of phenol. Ethers: Nomenclature, methods of preparation, physical and chemical properties, uses.	10	05
Unit - 5	Biomolecules : Carbohydrates Classification (aldoses and ketoses), monosaccharides (glucose and fructose), <i>D/L</i> configuration, oligosaccharides (sucrose), polysaccharides (starch, cellulose) Proteins Elementary idea of α -amino acids, peptide bonds, polypeptides, structure of proteins (primary structure only), denaturation of proteins; enzymes. Nucleic Acids: DNA & RNA (introduction and basic concept)	08	05
Unit - 6	Polymers : Classification- (natural and synthetic), methods of polymerization (addition and condensation), copolymerization. Some important polymers; like polythene, nylon, polyesters, bakelite, and rubber. Biodegradable and non-biodegradable polymers	08	04

CLASS - XII**SUBJECT : CHEMISTRY (CHEM)****SEMESTER – IV****FULL MARKS : 35****CONTACT HOURS : 60 HOURS****COURSE CODE : THEORY****Sub-topics**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 1	Electrochemistry Redox reactions, conductance in electrolytic solutions, specific and molar conductivity, variation of conductivity with concentration, Kohlrausch's law, electrolysis and laws of electrolysis (elementary idea), dry cell – electrolytic cells and Galvanic cells, emf of a cell, standard electrode potential, Nernst equation and its application to chemical cells, relation between Gibbs energy change and emf of a cell, fuel cells, Li-ion battery.	08	05
Unit - 2	Chemical Kinetics Rate of a reaction (average and instantaneous), factors affecting rate of reactions- concentration, temperature and catalyst. Order and molecularity of a reaction; rate law and specific rate constant, integrated rate equations and half-life (only for zero and first order reactions); the concept of collision theory (elementary idea, no mathematical treatment) activation energy, Arrhenius equation Catalysis, homogeneous and heterogeneous catalysis, enzyme catalysis.	10	07
Unit - 3	d and f Block elements General introduction, electronic configuration, occurrence and characteristics of transition metals, general trends in properties of the first-row transition metals – ionic radii, ionization enthalpy, oxidation states, colour, catalytic property, magnetic property. Preparation and properties of $K_2Cr_2O_7$ and $KMnO_4$. Lanthanoids Electronic configuration, oxidation states, chemical reactivity, lanthanoid contraction and its consequences, uses. Actinoids Electronic configuration, oxidation states, comparison with lanthanoids, uses.	10	06

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit - 4	Coordination compounds Introduction, ligands, classification of ligands based on denticity and field intensity, coordination number, colour, magnetic properties and shape, IUPAC nomenclature of mononuclear coordination compounds, EAN rule, Bonding (Werner's theory, VBT and CFT), CFSE, structural-isomerism and stereo-isomerism, importance of coordination compounds (in qualitative analysis, extraction of metals and biological systems)	08	05
Unit - 5	Aldehydes, Ketones and Carboxylic Acids Aldehydes and Ketones : Nomenclature, nature of carbonyl group, methods of preparation, physical and chemical properties, mechanism of nucleophilic addition, reactivity of alpha hydrogen in aldehydes; uses. Carboxylic Acids: Nomenclature, acidic nature, methods of preparation, physical and chemical properties, uses	10	05
Unit - 6	Organic compounds containing Nitrogen Nitro compounds: General methods of preparation and reduction reactions. Amines: Nomenclature, classification, structure, methods of preparation, physical and chemical properties, uses, identification of primary, secondary and tertiary amines. Cyanides and Isocyanides – Nomenclature, structure, methods of preparation, chemical reactions (hydrolysis and reduction reactions only). Diazonium salts: Preparations, chemical reactions and importance in synthetic organic chemistry	14	07

PRACTICAL FOR CLASSES XI AND XII

SUBJECT : CHEMISTRY (CHEM)

CLASS – XI

COURSE CODE : PRACTICAL

FULL MARKS : 30

Evaluation Scheme for Examination	Marks
Volumetric analysis	10
Environment-related experiments	08
Characterization and purification of chemical substances	06
Class Record, Project and Viva	06
Total	30

Practical Syllabus

A. Basic Laboratory Techniques

- Cutting glass tube and glass rod
- Bending a glass tube
- Drawing out a glass jet
- Boring a cork

B. Characterization and purification of chemical substances

- Determination of the melting point of an organic compound
- Determination of the boiling point of an organic compound
- Crystallization of impure sample of any of the following: Alum, Copper, Sulphate, Benzoic acid.

C. Environment-related experiments

- Calculation of pH of soil sample.
- Determination of turbidity for a given sample of water
- Determination of dissolved oxygen in a given sample of water
- Determination of TDS of water sample

D. Quantitative estimation (Use of digital balance (precession up to 3 decimal points)) (Volumetric analysis)

- Determination of strength of a given sodium hydroxide solution by titrating it against a standard oxalic acid solution.
- Determination of strength of a given hydrochloric acid solution by titrating it against standard sodium carbonate solution.
- Standardisation of KMnO_4 solution by using standard Oxalic acid solution.
- Estimation of Fe in Mohr's salt solution using standard KMnO_4 solution or standard $\text{K}_2\text{Cr}_2\text{O}_7$ solution.

Project Work

a) Preparation of standard solutions :

- i) Preparation of (N/10) Oxalic acid solution.
- ii) Preparation of (N/10) Mohr's salt solution.
- iii) Preparation of (N/10) Sodium carbonate solution.
- iv) Preparation of (N/10) Hydrochloric acid solution.
- v) Preparation of (N/10) Sodium hydroxide solution.

b) Preparation of inorganic compounds :

- i) Preparation of potash alum.
- ii) Preparation of potassium ferric oxalate.

c) Study of acidity of-

- i) Different samples of tea leaves.
- ii) Fruit and vegetable juices.

CLASS – XII

COURSE CODE : PRACTICAL

FULL MARKS : 30

Evaluation Scheme for Examination	MARKS
Potentiometric Analysis	06
Salt Analysis	08
Detection of functional groups in Organic compounds	04
Content-Based Experiment (Chemical Kinetics/Thermochemistry/ Preparation of Organic Compounds)	06
Class record, Viva and Project work	06
Total	30

Practical Syllabus

A. Chemical kinetics

- (i) Study of the rate of reaction of iodide ions with hydrogen peroxide at room temperature using different concentrations of iodide ions. (with Excel plot)
- (ii) Study of the reaction rate of hydrolysis of ester in an acidic medium (with Excel plot)

B. Thermochemistry :

Any one of the following experiments :

- (i) Enthalpy of dissolution of copper sulphate or potassium nitrate.
- (ii) Enthalpy of neutralization of strong acid (HCl) and strong base (NaOH)
- (iii) Determination of enthalpy change during interaction (hydrogen bond formation) between acetone and chloroform.

C. Electrochemistry

- (i) Potentiometric titration of $\text{Fe}^{3+}/\text{Fe}^{2+}$ system with Potassium dichromate and Potassium permanganate solutions.
- (ii) Potentiometric determination of concentration of AgNO_3 solution (N/100 or N/200) using standard KCl solution (N/10).

D. Tests for the functional groups present in organic compounds:

Unsaturation, alcoholic $-\text{OH}$ (1°), phenolic $-\text{OH}$, aldehyde, ketone, carboxylic acid and primary aromatic amine groups.

E. Preparation of Organic compounds:

Preparation of any two of the following compounds :

- (i) Benzilic acid (From Benzil)
- (ii) Aniline yellow or 2-Naphthol aniline dye.
- (iii) Iodoform.

F. Characteristic test of carbohydrates, fats and proteins in pure samples and their detection in given foodstuffs.**G. Qualitative analysis**

Determination of one cation and anion in a given salt.

Cations - Pb^{2+} , Cu^{2+} , Al^{3+} , Fe^{3+} , Cr^{3+} , Mn^{2+} , Ni^{2+} , Zn^{2+} , Co^{2+} , Ca^{2+} , Sr^{2+} , Ba^{2+} , Mg^{2+} , NH_4^+

Anions – CO_3^{2-} , S^{2-} , SO_4^{2-} , $\text{S}_2\text{O}_3^{2-}$, NO_2^- , NO_3^- , Cl^- , Br^- , I^- , PO_4^{3-}

(Note: Insoluble salts excluded)

Project work – where feasible may include

- (i) Model preparation
- (ii) Investigatory project
- (iii) Science exhibits
- (iv) Participation in science fairs
- (v) Testing purity of food articles like butter, pulse, milk etc.

Course Overview :

This course covers the fundamental concepts of computer system organization, programming fundamental along with its theoretical analysis, data structure, computer networks, the value of technology in societies, e-commerce, python programming, database management system, and artificial intelligence for the students from science background only.

Course Objective:

This course enables students to-

- develop an understanding of how computer system works; the components of computer systems and how they interrelate, including software, data, hardware, communications and users.
- analyse a computing problem and to apply principles of computing to identify solutions.
- use of efficient data storing and retrieval technique along with basic programming skill.
- gather the fundamental knowledge on computer networks and web page designing.
- appreciate the ethical implications relating to the use of computing technology and information and identify the impact of technology on personal life and society.
- develop the knowledge, skills, and competencies needed to leverage the opportunities presented by the digital economy and to navigate the challenges and risks associated with online business operations.
- understand the basics of artificial intelligence and its subfields.
- develop an understanding of database management systems, with an emphasis on how to organize, maintain and retrieve - efficiently, and effectively.

Class XI**SUBJECT : COMPUTER SCIENCE (COMS)****SEMESTER – I**

Total Contact Hours: 200 (Theory & Practical: 180 ; Remedial & Home Assignment:20)

Course Code: COMS (Theory)

Full Marks: 35

Contact Hours: 100

Unit – 1	Computer System and Organisation	15 Marks	Total 30 Hours
	• Basic Computer Organisation ➤ CPU, Primary Memory (RAM, ROM, Cache), Secondary storage device, I/O devices, units of memory (bit, byte, KB, MB, GB, TB, PB). • Classification of Computers ➤ Super, Mainframe, Mini, PC.		4 hours
	• Concepts of Software ➤ Definition of software, types of software – System Software (Translator: assembler, interpreter, compiler, Loader, Linker, Operating System: Definition and functions, types of OS- Single use, Multiuse, Multiprogramming, Multiprocessing, Time sharing), Application Software (Definition and example), Utility Software, concept of GUI and CUI with examples using LINUX (Basic Commands).		9 hours

	- Number System ➤ Binary, Octal, Decimal, Hexadecimal number system, conversion between number system, Weighted Code (BCD, Binary, 84-2-1 code), non-weighted code (GREY, Excess-3), encoding schemes (ASCII, ISCII, unicode), 1's complement, 2's complement.	7 Hours
	- Boolean Algebra ➤ Postulates, logic gates: NOT, AND, OR, NAND, XOR, XNOR, truth tables, De Morgan theorem, SOP, POS, Simplifications using K-Map and Boolean algebra, logic circuits.	10 Hours
Unit – 2	Programming Fundamentals	10 Marks
	- Concept of Programming ➤ Instruction (Definition, Example), Program (definition, example), Programming Language (concept of high level, low level and assembly language), Procedural and Non-procedural programming, Concept of Structured Programming, Object Oriented Programming	2 Hours
	- Algorithm fundamentals ➤ Definition, characteristic of algorithm, recursive and non-recursive algorithms, representation of algorithm using flowchart, pseudo code, efficiency of algorithm, space complexity, time complexity, asymptotic notation- big O, big Omega, big Theta.	18 Hours
	- Introduction to Problem Solving ➤ Steps for Problem Solving (analysing the problem, developing an algorithm, coding, testing, debugging).	5 Hours
Unit – 3	Introduction to C	10 Marks
	- Basic Structure ➤ Character set, keywords, identifiers, constants, variables and type declaration, Sample programs, pre-processor.	2 Hours
	- Operators ➤ Arithmetic, Relational, Logical, Assignment, Increment and Decrement, Conditional, comma; operator precedence and associativity; arithmetic expression-evaluation and type conversion. Character I/O, Escape sequence and formatted I/O.	3 Hours
	- Branching and Looping ➤ if, if-else, while, do-while, for.	3 Hours
	- Arrays and Structure ➤ One-dimensional and Two-dimensional, Different types of uses. String handling with arrays – read and write, concatenation, comparison, string functions. ➤ Structures: Initialization; arrays of a structure, arrays within structures, structure within structure.	12 Hours
	- User defined functions ➤ Need, Call by Reference, call by value, return value and types, nesting of functions, recursion.	10 Hours
	- Pointers ➤ Declaration and initialization, operators, pointer arithmetic's, accessing variables, pointer & arrays, strings, functions.	15 Hours

Class XI
SUBJECT : COMPUTER SCIENCE (COMS)
SEMESTER – II

Course Code: COMS (Theory)

Full Marks : 35

Contact Hours : 80

Unit – 1	Data Structure	15 Marks	Total 45 Hours
	Definition, types of data structure-linear and non-linear.		1 Hour
	Abstract Data types.		1 Hour
	Arrays: 1D, 2D and their applications.		7 Hours
	Linked List: Single, circular and double link list.		10 Hours
	- Stack - Stack operations (push and pop), implementation using array and list, application of Stack.		6 Hours
	- Queue - Queue operation implementation using array and list, circular queue, de-queue, priority queue.		6 Hours
	- Recursion - Definition. - Advantages and limitations of recursion. - Understanding what goes behind recursion (internal stack implementation), tail recursion.		4 Hours
	- Searching and Sorting - Linear Search, Binary Search and their comparison. - Bubble Sort and its implementation.		10 Hours
Unit – 2	Computer Networks	10 Marks	Total 20 Hours
	- Introduction to Networking - Analogue and digital Communication. - Mode of Communication- Simplex, half duplex and full duplex. - Network Architecture- Client server, Peer to Peer. - Serial and Parallel Communication. - Measuring Capacity of Communication Media (bandwidth, channel capacity, baud). - Synchronous and asynchronous Transmission Mode. - Baseband and Broadband network.		6 Hours
	- Transmission Media - Wired Communication Media (Twisted Pair, Co-axial cable, Fiber Optic). - Wireless Communication Media (Radio wave, Microwave, Infrared, Satellite).		3 Hours

	• Network Connecting Devices ➤ Modem, Ethernet Card, RJ45, Repeater, Hub, Switch, Router, Gateway, Wifi card.	2 Hours	
	• Network Type and Topologies ➤ Types of Network-LAN, MAN, WAN. ➤ Network Topologies- Bus, Star, Ring, Tree.	3 Hours	
	• Network Protocols -HTTP, FTP, PPP, SMTP, TCP/IP, POP3, TELNET, HTTPS, VoIP.	2 Hours	
	• Referential Model- OSI Model (Basic Concept, use of devices and protocols at different layers).	1 Hour	
	• Introduction to Web Services: WWW, HTML, XML, IP Addresses, Domainnames, URL, ISP, Website, Web browser, Web Server, Web Hosting.	3 Hours	
Unit – 3	Ethics	10 Marks	Total 15 Hours
	• Digital Footprints.		1 Hour
	• Data Protection: Intellectual property rights (copyright, patent, trademark), violation of IPR (plagiarism, copyright infringement, trademark infringement),open-source software and licensing (Creative Commons, GPL and Apache).		5 Hours
	• Cyber Crime: Definition, hacking, eavesdropping, phishing and fraud emails, ransomware, cyber trolls, cyber bullying.		3 Hours
	• Cyber safety: Safely browsing the web, identity protection, confidentiality.		2 Hours
	• Malware: Viruses, trojans, adware.		1 Hour
	• E-waste management: Proper disposal of used electronic gadgets.		2 Hours
	• Information Technology Act: (IT Act).		1 Hour

Class XII
SUBJECT : COMPUTER SCIENCE (COMS)
SEMESTER – III

Total Contact Hours : 200 (Theory & Practical: 180; Remedial & Home Assignment : 20)

Course Code: COMS (Theory)

Full Marks : 35

Contact Hours : 100

Unit – 1	Python Programming	25 Marks	Total 80 Hours
	Familiarization with the basics of Python programming ➤ Introduction to Python, Features of Python, executing a simple “hello world” program, execution modes: interactive mode and script mode, Python character set, Python tokens (keyword, identifier, literal, operator, punctuator), variables, concept of l-value and r-value, use of comments.		2 Hours
	Knowledge of data types ➤ Number(integer, floating point, complex), boolean, sequence(string, list, tuple), None, Mapping(dictionary), mutable and immutable data types.		1 Hour
	Operators ➤ Arithmetic operators, relational operators, logical operators, assignment operators, augmented assignment operators, identity operators (is, is not), membership operators (in not in).		2 Hours
	Expressions, statement, type conversion, and input/output ➤ Precedence of operators, expression, evaluation of an expression, type-conversion (explicit and implicit conversion), accepting data as input from the console and displaying output.		3 Hours
	Errors: Syntax errors, logical errors, and run-time errors.		2 Hours
	Flow of Control ➤ Introduction, use of indentation, sequential flow, conditional and iterative flow.		4 Hours
	Conditional statements ➤ if, if-else, if-elif-else, flowcharts, simple programs: e.g.: absolute value, sort 3 numbers and divisibility of a number.		5 Hours
	Iterative Statement ➤ For loop, range(), while loop, flowcharts, break and continue statements, nested loops, suggested programs: generating pattern, summation of series, finding the factorial of a positive number, etc.		7 Hours

	• Strings ➤ Introduction, string operations (concatenation, repetition, membership and slicing), traversing a string using loops, built-in functions/methods—len(), capitalize(), title(), lower(), upper(), count(), find(), index(), endswith(), startswith(), isalnum(), isalpha(), isdigit(), islower(), isupper(), isspace(), lstrip(), rstrip(), strip(), replace(), join(), partition(), split().	10 Hours
	• Lists ➤ Introduction, indexing, list operations (concatenation, repetition, membership and slicing), traversing a list using loops, built-in functions/methods—len(), list(), append(), extend(), insert(), count(), index(), remove(), pop(), reverse(), sort(), sorted(), min(), max(), sum(); nested lists, suggested programs: finding the maximum, minimum, mean of numeric values stored in a list; linear search on list of numbers and counting the frequency of elements in a list.	10 Hours
	• Tuples ➤ Introduction, indexing, tuple operations (concatenation, repetition, membership and slicing); built-in functions/methods – len(), tuple(), count(), index(), sorted(), min(), max(), sum(); tuple assignment, nested tuple.	5 Hours
	• Dictionary ➤ Introduction, accessing items in a dictionary using keys, mutability of a dictionary (adding a new term, modifying an existing item), traversing a dictionary, built-in functions/methods – len(), dict(), keys(), values(), items(), get(), update(), del(), del, clear(), fromkeys(), copy(), pop(), popitem(), setdefault(), max(), min(), sorted().	5 Hours
	• Introduction to Python modules ➤ Importing module using 'import <module>' and using from statement, importing math module (pi, e, sqrt(), ceil(), floor(), pow(), fabs(), sin(), cos(), tan()); random module (random(), randint(), randrange()), statistics module (mean(), median(), mode()).	10 Hours
	• Functions ➤ Types of function (built-in functions, functions defined in module, user defined functions), creating user defined function, arguments and parameters, default parameters, positional parameters, function returning value(s), flow of execution, scope of a variable (global scope, local scope).	7 Hours

	• Exception Handling ➤ Introduction, handling exceptions using try-except-finally blocks.	7 Hours
Unit – 2	E-Commerce	10 Marks
	• An introduction to Electronic Commerce ➤ What is E-Commerce (Introduction And Definition), Main activities E-Commerce, Goals of E-Commerce, Technical Components of E-Commerce, Functions of E-Commerce, Advantages and disadvantages of E-Commerce, Scope of E-Commerce, Electronic Commerce Applications, Electronic Commerce and Electronic Business (C2C, C2G, G2G, B2G, B2P, B2A, P2P, B2A, C2A, B2B, B2C). ➤ Internet, Intranet & Extranet, Role of Internet in B2B Application, Web promotion, Banner, Exchange, Shopping Bots.	8 Hours
	• Electronic Data Exchange ➤ Introduction, Concepts of EDI and Limitation, Applications of EDI, Disadvantages of EDI, EDI model.	4 Hours
	• Electronic Payment System ➤ Introduction, Types of Electronic Payment System, Payment Types, Value Exchange System, Credit Card System, Electronic Fund Transfer, Paperless bill, Modern Payment Cash, Electronic Cash.	4 Hours
	• Internet Marketing ➤ The PROS and CONS of online shopping, The cons of online shopping, Justify an Internet business, Internet marketing techniques, The E-cycle of Internet marketing, Personalization e-commerce.	4 Hours

Class XII
SUBJECT : COMPUTER SCIENCE (COMS)
SEMESTER – IV
Course Code : COMS (Theory)

Full Marks : 35

Contact Hours : 80

Unit – 1	Database Management System	20 Marks	Total 50 Hours
	• Introduction ➤ Drawbacks of Legacy System, Advantages of DBMS, Layered Architecture of Database, Data Independence, Data Models, Schemas and Instances, Database Languages, Database Users, DBA, Data Dictionary.		3 Hours
	• Entity Relationship (ER) Modelling ➤ Entity, Attributes and Relationship, Structural Constraints, Keys (Super Key, Key, Candidate Key, Alternate Key, Primary Key), ER Diagram of Some Example Database, Weak and strong Entity Set, Specialization and Generalization, Constraints of Specialization and Generalization, Aggregation.		10 Hours
	• Relational Model ➤ Basic Concepts of Relational Model, Relational Algebra.		10 Hours
	• Integrity Constraints ➤ Domain Constraints, Referential Integrity, View.		2 Hour
	• SQL ➤ Introduction, Data Definition Language and Data Manipulation Language, Data type (char(n), varchar(n), int, float, date), constraints (not null, unique, primary key), create database, use database, show databases, drop database, show tables, create table, describe table, alter table (add and remove an attribute, add and remove primary key), drop table, insert, delete, select, operators (mathematical, relational and logical), aliasing, distinct clause, where clause, in, between, order by, meaning of null, is null, is not null, like, update command, delete command, aggregate functions (max, min, avg, sum, count), group by, having clause, joins: cartesian product on two tables, equi-join and natural join .		25 Hours

Unit – 2	Foundation of Artificial Intelligence (AI)	15 Marks	Total 30 Hours
	• Introduction to Artificial Intelligence ➤ Definition and scope of AI. ➤ Historical overview and key milestones. ➤ Differentiating AI from human intelligence.		4 Hours
	• AI Subfields and Technologies ➤ Machine learning: Supervised, unsupervised, and reinforcement learning. ➤ Deep learning and neural networks. ➤ Natural language processing (NLP) and computer vision.		10 Hours
	• Search as Optimization (only Basic Concepts) ➤ Strategies for State Space Search. ➤ Data Driven and Gold Driven Search. ➤ Heuristic Search, Breadth First Search and Depth First Search. ➤ A* Search.		10 Hours
	• Applications of AI ➤ AI in finance: Fraud detection, algorithmic trading, and risk assessment. ➤ AI in customer service and chatbots. ➤ AI in education: Personalized learning and intelligent tutoring systems.		3 Hours
	• Ethical and Social Implications of AI ➤ Bias and fairness in AI systems. ➤ Impact of AI on employment and the workforce. ➤ AI and social inequality.		3 Hours

CLASS : XI

SUBJECT : COMPUTER SCIENCE (COMS)

COURSE CODE : PRACTICAL

Full Marks: 30

1.	Experiments with basic Linux Commands • Use of Simple commands-ls, cd, cp, cat, mv, mkdir, rmdir, who, use of wildcard etc.	5Marks
2.	C programming • Flowchart 3 marks • Program coding 10 marks • Output 4 marks	17Marks
3.	Lab Book	3Marks
4.	Viva	5Marks

List of Suggestive C Programs

1. WRITE A PROGRAM to print the sum and product of digits of an integer.
2. WRITE A PROGRAM to reverse a number.
3. WRITE A PROGRAM to compute the sum of the first n terms of the following series,
 $S=1+1/2+1/3+1/4+.....$
4. WRITE A PROGRAM to compute the sum of the first n terms of the following series,
 $S=1-2+3-4+5+.....$
5. Write a function that checks whether a given string is Palindrome or not. Use this function to find whether the string entered by user is Palindrome or not.
6. Write a function to find whether a given no. is prime or not. Use the same to generate the prime numbers less than 100.
7. WRITE A PROGRAM to compute the factors of a given number.
8. Write a macro that swaps two numbers. WRITE A PROGRAM to use it.
9. WRITE A PROGRAM to print a triangle of stars as follows (take number of lines from user):
*

10. WRITE A PROGRAM to perform following actions on an array entered by the user :
 - i) Print the even-valued elements
 - ii) Print the odd-valued elements
 - iii) Calculate and print the sum and average of the elements of array
 - iv) Print the maximum and minimum element of array

v) Remove the duplicates from the array

vi) Print the array in reverse order

The program should present a menu to the user and ask for one of the options. The menu should also include options to re-enter array and to quit the program.

11. WRITE A PROGRAM that prints a table indicating the number of occurrences of each alphabet in the text entered as command line arguments.
12. Write a program that swaps two numbers using pointers.
13. Write a program in which a function is passed address of two variables and then alter its contents.
14. Write a program which takes the radius of a circle as input from the user, passes it to another function that computes the area and the circumference of the circle and displays the value of area and circumference from the main() function.
15. Write a program to find sum of n elements entered by the user. To write this program, allocate memory dynamically using malloc() / calloc() functions or new operator.
16. Write a menu driven program to perform following operations on strings:
 - a) Show address of each character in string
 - b) Concatenate two strings without using strcat function.
 - c) Concatenate two strings using strcat function.
 - d) Compare two strings
 - e) Calculate length of the string (use pointers)
 - f) Convert all lowercase characters to uppercase
 - g) Convert all uppercase characters to lowercase
 - h) Calculate number of vowels
 - i) Reverse the string
17. Given two ordered arrays of integers, write a program to merge the two-arrays to get an ordered array.
18. WRITE A PROGRAM to display Fibonacci series (i) using recursion, (ii) using iteration.
19. WRITE A PROGRAM to calculate Factorial of a number (i) using recursion, (ii) using iteration.
20. WRITE A PROGRAM to calculate GCD of two numbers (i) with recursion (ii) without recursion.
21. Write a menu-driven program to perform following Matrix operations (2-D array implementation): a) Sum b) Difference c) Product d) Transpose
22. Copy the contents of one text file to another file, after removing all whitespaces.
23. Write a function that reverses the elements of an array in place. The function must accept only one pointer value and return void.
24. Write a program that will read 10 integers from user and store them in an array. Implement array using pointers. The program will print the array elements in ascending and descending order.
25. Write a Program to implement stack operations using array.
26. Write a Program to implement queue operations using array.
27. Write a Program to search an element from a collection using Linear Search.
28. Write a Program to sort a collection of elements using Bubble Sort.

CLASS : XII

SUBJECT : COMPUTER SCIENCE (COMS)

COURSE CODE : PRACTICAL

Full Marks : 30

1.	MYSQL 10 Marks - Table creation 3 Marks - Data insertion in table 2 Marks 2 Queries 5 Marks
2.	Python Program 12 Marks - Coding 9 Marks - Output 3 Marks
3.	Lab Book 3 Marks
4.	Viva 5 Marks

Suggested Practical List

List of Suggestive Python Programs

- Input a welcome message and display it.
- Input two numbers and display the larger / smaller number.
- Input three numbers and display the largest / smallest number.
- Generate the following patterns using nested loops:

Pattern-1	Pattern-2	Pattern-3
*	12345	A
**	1234	AB
***	123	ABC
****	12	ABCD
*****	1	ABCDE

- Write a program to input the value of x and n and print the sum of the following series:

➤ $1 + x + x^2 + x^3 + x^4 + \dots x^n$

➤ $1 - x + x^2 - x^3 + x^4 - \dots x^n$

➤ $x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \frac{x^n}{n}$

➤ $x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \frac{x^n}{n!}$

- Determine whether a number is a perfect number, an Armstrong number or a palindrome.

- Input a number and check if the number is a prime or composite number.
- Display the terms of a Fibonacci series.
- Compute the greatest common divisor and least common multiple of two integers.
- Count and display the number of vowels, consonants, uppercase, lowercase characters in string.
- Input a string and determine whether it is a palindrome or not; convert the case of characters in a string.
- Find the largest/smallest number in a list/tuple
- Input a list of numbers and swap elements at the even location with the elements at the odd location.
- Input a list/tuple of elements, search for a given element in the list/tuple.
- Create a dictionary with the roll number, name and marks of n students in a class and display the names of students who have marks above 75.
- Read a text file line by line and display each word separated by a #.
- Read a text file and display the number of vowels/consonants/uppercase/lowercase characters in the file.
- Remove all the lines that contain the character 'a' in a file and write it to another file.
- Create a binary file with name and roll number. Search for a given roll number and display the name, if not found display appropriate message.
- Create a binary file with roll number, name and marks. Input a roll number and update the marks.
- Write a random number generator that generates random numbers between 1 and 6 (simulates a dice).

Database Management System

- Create a student table and insert data. Implement the following SQL commands on the student table:
 - ALTER table to add new attributes / modify data type / drop attribute
 - UPDATE table to modify data
 - ORDER By to display data in ascending / descending order
 - DELETE to remove tuple(s)
 - GROUP BY and find the min, max, sum, count and average.
 - LIKE, DISTINCT, NOT IN, IN, JOIN, CARTISIAN PRODUCT SET DIFFERENCE, UNION, INTERSECTION etc.

Course Objective :

The objective of any course is primarily twofold: first, to improve the understanding of the world around us, and to impart knowledge & develop skills in the students, which improves their employability. The HS Course in Economics has also been framed keeping these two primary objectives in mind.

The course is designed to ensure that the students develop a basic knowledge of the economic concepts, terms, theories and processes so that they can understand literatures and articles on economic issues. The course also aims at helping the students to understand the problems and challenges faced by the Indian economy as well as world economy in general and to apply the knowledge of the economics to make informed choices. The course is also intended to inculcate analytical, logical and statistical skills in a student so that they are able to analyse economic problems applying the concepts and processes they have learnt. The course should also equip them to draw conclusions and present the conclusions in an innovative way.

Course Overview:

The course consists of four modules. The first two modules offered in class XI are Microeconomics and Statistics. The next two modules offered in class XII are Macroeconomics and Indian Economic Development.

The course on Microeconomics familiarizes the students with different basic concepts of choice, demand, supply, revenue, cost and profit. The course is intended to develop an understanding about different forms of market structures their working process as well as basic concepts and processes about international trade.

The course on Statistics introduces the students to the basic concepts and techniques used in Statistical analysis of data. The course is developed with the aim of equipping the students to apply the statistical techniques in analyzing economic problems. This course will be particularly useful in conducting the project work, which is also part of the course. The students can apply statistical techniques on economic data along with the theories and concepts they learn in theoretical courses to analyse problems relating to economic issues.

The course on Macroeconomics likewise introduces the basic concepts of national income accounting and simple Keynesian Economic system to the students. The course also includes topics on Balance of payments, money & banking, government budget and taxation. All these topics are introduced at a basic level with the aim of enabling the students to understand and analyse Macroeconomic issues in an integrated framework.

Finally, the course on Indian Economic Development attempts to provide the students with an overall view of Indian Economy, the nature and path of development that the Economy has experienced and the factors that have contributed to the process of development. The topics include sectoral distribution of income and occupation, evolution of the structure of Agricultural sector, Industrial sector and Services sector and the topical issue of inclusive growth. The necessary concepts like poverty, inequality, unemployment, human development have been introduced to provide the necessary perspective for understanding and analysing Indian Economic problems.

All the courses are structured in such a way that the student can understand, remember and apply the different concepts, processes and theories. The ultimate aim is to develop the students' analytical skills and ability to develop innovative ways to address the economic problems.

CLASS - XI
SUBJECT : ECONOMICS (ECON)
SEMESTER – I

COURSE CODE : THEORY

FULL MARKS : 40

CONTACT HOURS : 100 HOURS

Sub Topics

Microeconomics & Statistics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1: Microeconomics	1. Microeconomics Unit 1: Introduction a. Problems addressed in Economics (both Micro & Macro): Basic Problems: for whom to produce, How to Produce, what to produce; Problems of Economic Growth, Economic Development and Problems of Sustainable Development. b. How the Problems are addressed: Free Market Economy; Government Intervention; Socialist Economic system; Mixed Economic system; Broad Social Goals: Efficiency, Equity, Freedom of Choice, Economic Growth, Sustainable Development. c. Indicators - Indicators of Economic Growth (Only definition) a. Income, per capita income b. Savings c. Investment i. Physical Capital ii. Human Capital d. Development Indicators (Concepts & Definitions) a. Inequality b. Poverty c. Unemployment d. Gender Inequality e. Environment	10	4

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 2 Microeconomics	Consumer Behaviour - Demand & Supply-Concept of Equilibrium Price-Invisible Hand –laissez faire - Concept of Elasticity of Demand – Price Elasticity of Demand; Income Elasticity of Demand; Cross Price Elasticity of Demand; Elasticity of Supply (Numerical Problems can be Introduced); Comparative Statistics-Price Ceiling& Price Floor. - Basics of Theory of Consumer Behaviour, Cardinal Utility, Ordinal Utility; Budget Constraint, Indifference Curve & Its Properties, Utility Maximization Using Indifference Curve; Derivation of Demand Function.	20	8
Unit – 3 Microeconomics	Producer Behaviour - Short Run and Long Run Production Function, Law of Variable Proportions ,Returns to Scale, Economies of Scale - Basic Concepts of Cost: STC, SAC, SMC, LTC, LAC, LMC, Implicit Cost, Opportunity Cost - Concepts of Revenue and Profit: TR, AR, MR, Profit	20	8
Unit – 4 Statistics	Introduction to Data - Scope of Statistics - Introduction to Different Types of Data: Primary vs Secondary, Time Series vs Cross-Section, Qualitative vs Quantitative; Features of a Good Questionnaire - Data Sources for the Indian Economy	10	4
Unit – 5 Statistics	Organization & Representation of Data - Classification of Data: Continuous & Discrete Variables - Textual Presentation of Data - Tabular form Presentation of Data - Graphical Representation of Data:- Bar Chart, Histogram, Frequency Curve, Pie Chart, Ogive, Arithmetic Line Graph	20	8

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit- 6 Statistics	Measures of Central Tendency a. Mean: Arithmetic Mean, Geometric Mean b. Quartile & Median, Percentile (definition only) c. Mode	20	8

CLASS - XI

SUBJECT : ECONOMICS (ECON)

SEMESTER – II

COURSE CODE : THEORY

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

Sub Topics

Microeconomics & Statistics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Microeconomics	Introduction to Concepts of Market Structure a. Basic Concepts of Forms of Markets: Perfect Competition, Monopoly, Natural Monopoly, Monopolistic Competition, Oligopoly. (Definition & Features) b. TR, AR & MR under Fixed Price & Variable Price, Equilibrium Condition under Perfect Competition & Monopoly c. Financing through Debt & Equity: Basics of Financial Instruments (Stocks & Bonds) (Definitions & Differences) d. Problems of Externality: Social Cost & Private Cost, Problems of Market Failure, with special reference to Environmental Problems e. Public Goods & Private Goods	20	10
Unit – 2 Microeconomics	Basics of International Trade a. Difference Between Internal Trade & International Trade b. Concept of Absolute Advantage & Concept of Comparative Advantage c. Gains from Trade	20	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	d. Concepts of Tariff & Quota (Only Definition) e. Basic Ideas about Trade Liberalization, Bilateral Trade Agreements, Regional Trade Agreements.		
Unit – 3 Statistics	Measures of Dispersion a. Absolute Measures of Dispersion Range, Standard Deviation and Variance Quartile Deviation b. Relative Measure of Dispersion: Coefficient of Variation	17	8
Unit – 4 Statistics	Correlation a. Concept b. Scatter Diagram c. Spearman's Rank Correlation d. Pearson's Correlation Coefficient	17	8
Unit – 5 Statistics	Index Numbers: a. Concept & Use b. CPI & WPI	6	4

CLASS - XII

SUBJECT : ECONOMICS ECON)

SEMESTER – III

COURSE CODE : THEORY

FULL MARKS : 40

CONTACT HOURS : 100 HOURS

Sub Topics

Macroeconomics & Indian Economic Development

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Macroeconomics	National Income Accounting: a. i) Concept of Stocks & Flows, Concept of Transfer Payments, Concept of Closed Economy & Open Economy, Concepts of GDP, GNP, NDP, NNP, NI, Concept of factor cost and Market Price, Concept on intermediate and final goods, Real Income vs Nominal Income ii) Implementation of factor cost and Market Price concept for GDP, GNP, NDP, NNP, NI and simple numerical problem on conversion. b. Methods of Measuring NI: i) Product Census Method (mentioning Value Added Method) ii) Income Method iii) Expenditure Method iv) Simple numerical problems on all methods. c. Difficulties of measuring NI.	20	8
Unit – 2 Macroeconomics	Simple Keynesian system (Closed Economy Model with Government) a. Keynesian Consumption Function & Savings Function (mentioning MPC, APC, MPS and APS and their relationship) b. Cross-Diagonal Model, i) Determination of equilibrium income, impact on equilibrium due to the change in each of the following factors-investment, Government, expenditure, taxation. ii) Concept of paradox of thrift. c. Basic concept of Government Budget and taxation. d. Alternative Forms of Multiplier : Investment Multiplier, Government Expenditure Multiplier, Tax-Multiplier, Balanced Budget Multiplier.	20	8

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit-3 Macroeconomics	Balance of Payments & Exchange Rate a. Concepts of BOP : i) Balance of Current Account, Balance of Capital Account, Autonomous & Accommodating Transactions. ii) Difference between balance of payment and balance of trade, Capital Account and Current Account, Autonomous & Accommodating Transactions. iii) Causes and solution of balance of payment disequilibrium. b. i) Basic Concepts and Types of Exchange Rates. ii) Types of exchange rate (Fixed, Flexible and managed float) iii) Advantages and disadvantages of Fixed and Flexible exchanges.	10	4
Unit – 4 Indian Economic Development	Brief Reference to Colonial Past : i. Impact of British rule on agricultural sector, factors causing backwardness and stagnation of Indian agriculture during the British rule. a) 1. Low level of productivity 2. High Degree of vulnerability 3. Absence of organize market for agricultural product 4. Labour intensive technique of production 5. Feudalism in agriculture 6. Sub-division of Landholding. b) Introduction of land revenue system under the British rule and forced commercialization of agriculture sector. ii. Impact of British rule on industrial sector. 1. Decay of handicraft 2. New pattern of demand 3. Introduction of Railways in India iii. Foreign trade under British rule. 1. Net exporter of primary product and importer of finished goods 2. Monopoly control of Indian's Foreign Trade (Mentioning Suez canal and axis to Indian market) 3. Surplus trade but only to benefit Britishers iv) Drainage of Indian economic recourses (Dadavai Nauroji's theory)	10	4

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 5 Indian Economic Development	Evolving Features of Indian Economy : - Sectoral Composition of NI and changes over time during Pre-Liberalization and Post Liberalization Period - Occupation Distribution of NI during Pre-Liberalization and Post-Liberalization Period, Post-Independence Period & Changes thereof - Types of Unemployment – Changes in Occupational Patterns during Pre-Liberalization & Post-Liberalization Period – Problem of Unemployment in India – Different Employment Generating Programs in the Post-Liberalization Period – Jobless Growth. - India & Its Neighbours – a Comparison of Economic Growth, Population & Sectoral Development & other Human Development Indicators (Comparison with China)	20	8
Unit-6 Indian Economic Development	Agriculture : - Problems in Agricultural in the Post-Independence Era-1st Plan and Green Revolution - Institutional Reforms: Land Reforms & Operation Barga – Reforms in Rural Credit. - Agricultural Reforms in the Post-Liberalization Era – Impact of Globalization in Indian Agriculture - Food Security & Public Distribution System	20	8

CLASS - XII

SUBJECT : ECONOMICS (ECON)

SEMESTER – IV

COURSE CODE : THEORY

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

Sub Topics

Macroeconomics & Indian Economic Development

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Macroeconomics	Money, Banking & Inflation : a. Concept & Functions of Money b. Basics of Supply of Money, Central Bank Money, Process of Credit Creation & Role of Commercial Banks, Concept of Money Multiplier c. Concept of Inflation, Demand Pull Inflation, Cost Push Inflation, Stagflation. d. Functions of Central Bank & Methods of Credit Control: Direct Control vs Indirect Control; Control Rates: CRR, SLR, Repo Rate, Bank Rate, Prime Lending Rate	20	10
Unit – 2 Macroeconomics	Financing of Government Budget: a. Public Finance vs Private Finance b. Sources of Government Revenue: Tax & Non-Tax Revenue, Alternative Forms of Taxes – Progressive & Proportional Taxes, Direct and Indirect Taxes, Concept of GST, Sources of Non-Tax revenue c. Concept of Government Budget & Budget Deficit: Revenue Deficit, Fiscal Deficit, Primary Deficit, Monetization of Fiscal Deficit & Inflation, Balanced Budget Multiplier d. Public Debt: Internal vs External Debt; Short Term vs Long Term Debt	20	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 3 Indian Economic Problems	Industry : 2nd 5-year Plan, Mahalanobish Model & Industrial Growth - Industrial Liberalization in 1991 – Its Impact on Industrial Growth - Trade Policy & Impact of Globalization on Indian Industry – Shift in Trade Policy Regime from Import Substitution to Export Promotion – India as a member of WTO - Importance of Small Scale and Village Industries	10	4
Unit – 4 Indian Economic Problems	Service Sector : - Growth in Service Sector in India - Growth in Trade of Services (Financial Sector, Information Technology Sector, Outsourcing) - Impact of Economic Liberalization & Globalization on Service Sector Growth	15	8
Unit – 5 Indian Economic Problems	Inclusive Growth : - Inclusive Growth –Definition & Measurement of Inclusive Growth - Sustainable Development: 17 SDGs of UNDP - Inequality & Poverty: Basic Measurements – Head Count Ratio, Poverty Gap, Lorenz Curve; Changes in Indian Scenario during 1950-1990, 1991-2019; Special Programs & Policies Adopted in India to Eradicate Poverty with special reference to West Bengal - Environmental Problems of Economic Growth with special reference to India - Human Development : Concept of HDI, Indian Scenario - NITI Aayog : - Concept of NITI Aayog : The National Institute for Transforming India (NITI Aayog) is a policy think tank that provides strategic advice to the Indian Government. It was established on 1st January 2015 in New Delhi, replacing the Planning Commission. It is supported by Development Monitoring and Evaluation Organisation (DMEO), the Atal Innovation Mission (AIM) and National Institute of Labour Economics Research and Development (NILERD). - Objectives : Economic development Co-operative Federalism, Policy support, Research and innovation, Resources centre, Technology advancement. - Functions : It includes formulating strategic and long term policy framework and monitoring the implementation of Government program. It also provides technical advice to both Central and State Government. It also aims to fulfil the goals of SDG.	15	8

CLASS : XII

SUBJECT : ECONOMICS (ECON)

COURSE CODE : PROJECT WORK

FULL MARKS : 20 For Classes XI and XII Separately

MARKS DIVISION : 12 [PROJECT REPORT] + 8 [VIVA]

A project work in economics is defined as a sequence of tasks that must be conducted and completed to attain a certain outcome either based on primary data or on secondary data. Learners will undertake only one project in each academic session and that project should be within 3,500-4,000 words (excluding diagrams, tables and graphs). Own observations and conclusions, depending on self-understanding of the selected topic should be incorporated while conducting and writing the project.

The Objectives of the Project Work are to enable students to :

- explore and elucidate theoretical concepts learnt in class XI and XII.
- analyse and evaluate empirically and theoretically real-world economic scenarios using theoretical constructs and arguments.
- nurture critical thinking skills.
- develop better communication skills while collecting primary data to focus on the problems of the local economy either individually or collectively.
- help to improve cognitive abilities with a view to making them sharper and more profound over time.
- follow up different aspects of economics in which learners have interest.

Topics for Secondary Data based Project Work for Class XI

1. Study of growth of GDP/ Per capita income in India for a specific period
2. Study of changing pattern of employment in India during plan period / post-liberalization period
3. Changing pattern of export/ Import during pre-liberalization & post-liberalization period
4. Study of government expenditure on education during post-liberalization period
5. Any other relevant topic from the topics covered

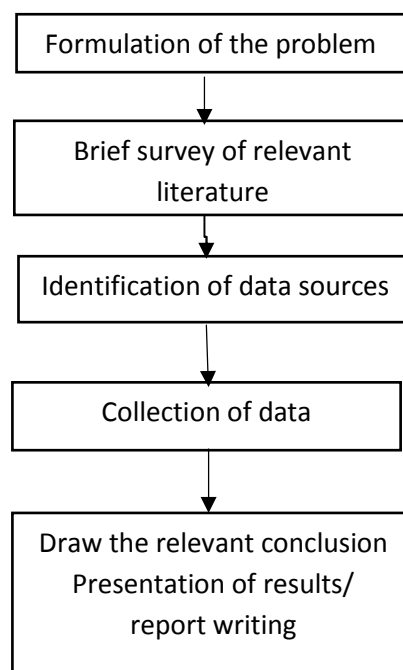
Topics for Secondary Data based Project Work for Class XII

1. Study of supply of credit to the Agricultural Sector/ Industrial Growth
2. Study of poverty in India
3. Study of changing pattern of inequality in India
4. Comparison of Human Development Index between India & a neighbouring country
5. Any other relevant topic from the topics covered

Data Sources for Secondary Data based Project

1. Economic Survey
2. Handbook of Statistics on Indian Economy: <https://rbi.org.in/Scripts/AnnualPublications.aspx>
3. Hand book of Statistics on Indian States
4. RBI Bulletin
5. Annual Survey of Industries
6. Human Development Reports

Flow Chart for Projects Based on Secondary Data



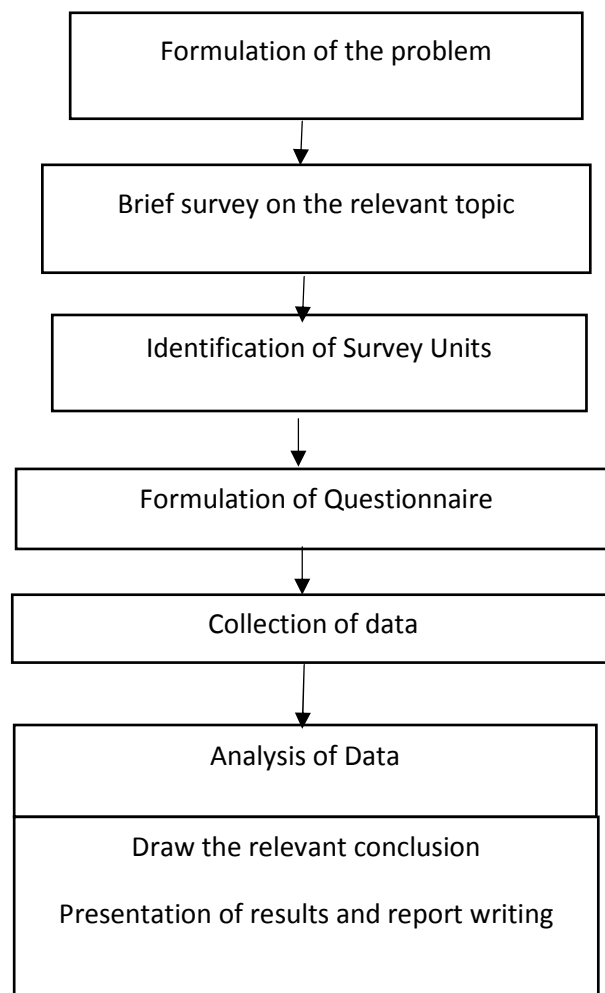
Topics for Primary Survey based Project Work for Class XI

1. Study of consumption pattern of different households
2. Study of savings pattern of different households
3. Study of local vegetable market to identify the extent of completion
4. Provision of public goods by local Gram Panchayat/ Local government
5. Any other relevant topic discussed in the syllabus

Topics for Primary Survey based Project Work for Class XII

1. Study of agricultural land holding pattern in the local area
2. Study of cropping pattern in the local area
3. Study of digitization of transactions among the local population
4. Study of relation between education & income / education & employment based on a local survey
5. Impact of Kanyasree Prakalpa on educational attainment of girl child
6. Any other relevant topic discussed in the syllabus

Flow Chart for Projects Based on Primary Data



Structure of the Project Report

1. Introduction of the issue
2. Description of data (description of data sources in case of secondary data based project, description of how the survey was conducted in case of projects based on primary survey)
3. Analysis of data using the knowledge of Statistics, Graphs & Tables & Interpretation of results
4. Conclusion from the analysis and interpretation
5. Bibliography

Distribution of Contact Hours

Total contact hours available per year	200 hrs
Contact hours for home assignments, tutorials, remedial classes per year	20 hrs
Contact hours available for classes per year	180 hrs
Contact hours for Semester I of Class XI	100 hrs
Contact hours for Semester II of Class XI	80 hrs
Contact hours for Semester III of Class XII	100 hrs
Contact hours for Semester IV of Class XII	80 hrs

Class XI	Hours	Class XII	Hours
Semester I	100 hrs	Semester III	100 hrs
Subject : Microeconomics	50hrs	Subject : Macroeconomics	50 hrs
Unit 1	10 hrs	Unit 1	20 hrs
Unit 2	20hrs	Unit 2	20 hrs
Unit 3	20hrs	Unit 3	10 hrs
Subject : Statistics	50 hrs	Subject : Indian Economic Development	50 hrs
Unit 4	10 hrs	Unit 4	10 hrs
Unit 5	20 hrs	Unit 5	20 hrs
Unit 6	20 hrs	Unit 6	20 hrs
Semester II	80 hrs	Semester IV	80 hrs
Subject : Microeconomics	40 hrs	Subject : Macroeconomics	40 hrs
Unit 1	20 hrs	Unit 1	20 hrs
Unit 2	20 hrs	Unit 2	20 hrs
Subject : Statistics	40 hrs	Subject : Indian Economic Development	40 hrs
Unit 3	17hrs	Unit 3	10 hrs
Unit 4	17 hrs	Unit 4	15 hrs
Unit 5	6 hrs	Unit 5	15hrs

Objectives :

- To understand the meaning, concept and aims of education
- To become aware of different forms and agencies of education and their roles
- To understand the meaning and scope of educational philosophy and its different schools
- To understand the meaning, nature and scope of educational sociology
- To understand the concept of culture, dynamics of social organizations and its role in education

CLASS - XI**SUBJECT : EDUCATION (EDCN)****SEMESTER – I****FULL MARKS : 50****CONTACT HOURS : 100 Hours****THEORY – 40 MARKS & INTERNAL ASSESSMENT – 10 MARKS****COURSE CODE : THEORY****GROUP – A****INTRODUCTION TO EDUCATION**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Meaning Concept & Aims of Education a) Meaning of Education: Meaning, Concept and Definition of Education. Nature and Scope of Education, Narrow and Wider Concept of Education. b) Aims of Education: Individualistic and Socialistic Aims of Education, Specific Aims of Education: Constitutional Values (Democracy and Secularism). c) Factors of Education: Learner, Teacher, Curriculum and Environment.	20	20
UNIT 2	Forms & Agencies of Education a) Forms of Education: Formal, Informal, Non-formal (Meaning, Concept, Characteristics, Need, Role, Limitation). b) Agencies of Education: School, Family, Mass Media, Newspaper, TV, Radio, Cinema, Internet, Library, Religious Institutions, Open School, Open University (Role only). c) Guiding Agencies of Education: NCERT, SCERT, DIET, NCTE, UGC, UNESCO (Full form, Establishment Year and maximum ten functions).	20	
	TOTAL	40 Hours	20

GROUP – B

PHILOSOPHICAL & SOCIOLOGICAL PERSPECTIVE IN EDUCATION

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Education & Philosophy a) Meaning of Educational Philosophy: Needs for Educational Philosophy, Relationship between Education and Philosophy. b) Schools of Indian Philosophy: Astika and Nastika - Naya, Vedanta, Buddhism and Islam - Basic Principles and Educational Implications (With Special Reference to Metaphysics, Epistemology, Axiology). c) Western Schools of Philosophy: Idealism, Realism, Naturalism and Pragmatism (Basic Principles and Educational Implications).	24	20
UNIT 2	Education & Society a) Meaning of Educational Sociology: Needs for Educational Sociology, Relationship between Education and Sociology. b) Social Organization and Social Structure: Folkways, Mores and Social Groups & Social Mobility. c) Culture and Education: Social and Cultural Change, Role of Education regarding these.	16	
	TOTAL	40 Hours	20

INTERNAL ASSESSMENT : 10 Marks (10 Contact Hours)

(Teacher must keep records of the attendance and remarks)

- Demonstration on a Topic** (Teacher will assign a topic from the courses taught and inform students prior to the assessment. Students will demonstrate the topic in front of the class) **5 Marks**
- Group Discussion: (Brain Storming)** (The teacher will assign topics to different student's groups. The group will be based on situation and class size. Topic of group discussion will be informed to the students prior to the assessment.)
- Debate/Discussion on a Topic** (Teacher will assign topic prior to the assessment)
 - The teacher must give remarks on basis of content knowledge, presentation skill, interaction, body language. **5 Marks**

Marks distribution for Internal Assessment

Criteria for Marks	Total Marks 10
Class Attendance	1 Mark
Content Knowledge	1 Mark
Presentation Skill	1 Mark
Interaction	1 Mark
Body Language	1 Mark
Students will write minimum 10 points on the selected topic	5 Marks

Sample Documentation Sheet for School

Sl. No.	Signature of the Student	H.S. Roll No.	H.S. Regi. No.	Awarded Marks

Signature of the Class Teacher

Signature of the HM with Seal

CLASS - XI

SUBJECT : EDUCATION (EDCN)

SEMESTER – II

FULL MARKS : 50

CONTACT HOURS : 80 Hours

THEORY – 40 MARKS & INTERNAL ASSESSMENT – 10 MARKS

COURSE CODE : THEORY

Objectives :

- To comprehend a synoptic view of educational psychology and its different schools
- To understand human growth and development, associated factors and dimensions
- To know the historical development of Indian education along with different committees and discourses
- To know the contributions of Indian social reformers on education

GROUP – C

PSYCHOLOGICAL PERSPECTIVE IN EDUCATION

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Education & Philosophy a) Meaning of Educational Psychology : Needs of Educational Psychology, Relationship between Education and Psychology and Bases on Human Behaviour (Sensation, Perception & Conception). b) Schools of Educational Psychology : Behaviourism, Gestalt and Psycho-analysis. c) Method of Inquiry in Psychology : Observation, Experimentation, Case Study, Survey, Correlation, etc.	20	
UNIT 2	Growth & Development a) Meaning of Growth and Development : Principles of Growth and Development, and its Educational Implication. b) Factors of Development : Heredity and Environment; Role of Education on its. c) Stages of Development : Infancy, Childhood, Adolescence. Dimension of Development : Physical, Mental (Cognitive), Emotional, Social With Reference to Different Levels of Education.	20	20
	TOTAL	40 Hours	20

GROUP – D

Historical Development of Indian Education

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Ancient, Medieval & Pre-Independent Period of Indian Education System a) Ancient Period: Vedic and Buddhist Education System (Synoptic Views). b) Medieval Period: Islamic Education System (Synoptic Views). c) Pre-Independent Period: Charter Act-1813, Macaulay Minute, Woods Dispatch- 1854, Hunter Commission- 1882, Curzon Educational Policy & National Educational Movement-1905, Sadler Commission-1917, Hartog Committee-1929, Sargent Plan- 1944 (Brief Study).	10	20
UNIT 2	Contribution of Indian Social Reformer towards the Development of Education in India a) Raja Ram Mohan Roy b) Iswar Chandra Vidyasagar c) Begum Rokeya d) Savitribai Phule (Brief Biography, Social and Educational Contribution only)	10	
	TOTAL	20 Hours	20

INTERNAL ASSESSMENT : 10 Marks (10 Contact Hours)

(Teacher must keep records of the attendance and remarks)

- Demonstration on a Topic** (Teacher will assign a topic from the courses taught and inform students prior to the assessment. Students will demonstrate the topic in front of the class) **5 Marks**
- Group Discussion: (Brain Storming)** (The teacher will assign topics to different student's groups. The group will be based on situation and class size. Topic of group discussion will be informed to the students prior to the assessment.)
- Debate/Discussion on a Topic** (Teacher will assign topic prior to the assessment)
 - The teacher must give remarks on basis of content knowledge, presentation skill, interaction, body language. **5 Marks**

Marks distribution for Internal Assessment

Criteria for Marks	Total Marks 10
Class Attendance	1 Mark
Content Knowledge	1 Mark
Presentation Skill	1 Mark
Interaction	1 Mark
Body Language	1 Mark
Students will write minimum 10 points on the selected topic	5 Marks

Sample Documentation Sheet for School

Sl. No.	Signature of the Student	H.S. Roll No.	H.S. Regi. No.	Awarded Marks

Signature of the Class Teacher

Signature of the HM with Seal

Project [20 marks] – Marks to be awarded as the cumulative marks of the two Internal assessments awarded in Class XI

CLASS - XII

SUBJECT : EDUCATION (EDCN)

SEMESTER – III

FULL MARKS : 50

CONTACT HOURS : 100 Hours

Theory – 40 Marks & Internal Assessment – 10 Marks

COURSE CODE : THEORY

Objectives :

- To know the post-independence developments of Indian education system
- To be aware of the recent changes in the Indian education system
- To comprehend the challenges in educational opportunities
- To develop an understanding of the contributions made by great educators
- To understand the concept of inclusive education and know the differently abled children
- To get a synoptic view of global perspectives on education
- To understand the role of positive psychology

GROUP – A

EDUCATION IN MODERN INDIA

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Post-Independent Period of Indian Education System a) University Education Commission (1948-49), Secondary Education Commission (1952-53), Indian Education Commission (1964-66) (Major Recommendations). b) National Education Policy 1986 & 2020 – Salient Features. c) Problems of Women Education, SC, ST, OBC, EWS, Equal Opportunity.	24	20
UNIT 2	Great Educators and their Contributions in Education a) Rabindranath Tagore b) Swami Vivekananda c) Mahatma Gandhi d) Jean-Jacques Rousseau e) John Dewey (Aims, Methods, Curriculum, Discipline & Educational Implication)	16	
	TOTAL	40 Hours	20

GROUP – B**RECENT TRENDS & ISSUES IN MODERN EDUCATION**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Inclusive Education a) Education for Differently Abled Children: Meaning, Concept & Classification of Differently Abled (Children with Special Needs). b) Visually Impaired, Hearing Impaired, Autism, Learning Disability, Intellectual Disability (Characteristics & Educational Programme). c) Meaning, Importance, and Barriers of Inclusive Education, Role of Formal & Informal Agencies in Building an Inclusive Society.	20	20
UNIT 2	Education for All a) Education for 21 st Century: Global Vision for Education- Delors Commission (4 Pillers of Education), International Commission on the Futures of Education. b) Education for All: Universalization of Elementary Education in India. c) Role of Positive Psychology in Modern Education.	20	
	TOTAL	40 Hours	20

INTERNAL ASSESSMENT : 10 Marks (10 Contact Hours)

(Teacher must keep records of the attendance and remarks)

- Demonstration on a Topic** (Teacher will assign a topic from the courses taught and inform students prior to the assessment. Students will demonstrate the topic in front of the class) **5 Marks**
- Group Discussion: (Brain Storming)** (The teacher will assign topics to different student's groups. The group will be based on situation and class size. Topic of group discussion will be informed to the students prior to the assessment.)
- Debate/Discussion on a Topic** (Teacher will assign topic prior to the assessment)
 - The teacher must give remarks on basis of content knowledge, presentation skill, interaction, body language. **5 Marks**

Marks distribution for Internal Assessment

Criteria for Marks	Total Marks 10
Class Attendance	1 Mark
Content Knowledge	1 Mark
Presentation Skill	1 Mark
Interaction	1 Mark
Body Language	1 Mark
Students will write minimum 10 points on the selected topic	5 Marks

Sample Documentation Sheet for School

Sl. No.	Signature of the Student	H.S. Roll No.	H.S. Regi. No.	Awarded Marks

Signature of the Class Teacher

Signature of the HM with Seal

CLASS - XII

SUBJECT : EDUCATION (EDCN)

SEMESTER – IV

FULL MARKS : 50

CONTACT HOURS : 80 Hours

Theory – 40 Marks & Internal Assessment – 10 Marks

COURSE CODE : THEORY

Objectives :

- To comprehend a synoptic view of learning mechanism and its different theories
- To understand learning, factors of learning and role of education
- To understand a synoptic view of mental health & well-being
- To develop an understanding of educational technology
- To develop an understanding of the use of computer and internet in education and communication
- To develop an understanding of ICT and e-learning
- To develop the concept of statistics and to develop skill in analyzing descriptive measures

GROUP – C

PSYCHOLOGY OF LEARNING & WELL-BEING

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Learning & Learning Mechanism	24	20
	a) Learning: Meaning, Characteristics, and Factors affecting Learning - Maturation, Motivation, Memory, Imagination, Attention & Interest (Basic Concept).		
	b) Learning Mechanism: Classifications & Basic Characteristics including, Description of Experiment and Educational Implications of • Conditioning (Pavlov, Skinner) • Problem Solving (Thorndike & Gestalt) • Synoptic Views of other Major Learning Approaches, e.g. Brunner, Ausubel, Vygotsky, Bandura.		
	c) Intelligence, Creativity & Personality: Basic Concept.		

UNIT 2	Mental Health & Well-being a) Mental Health: Concept of Health & Mental Health as prescribed by WHO, Means of Identify of Mental Health Problems, Common Mental Health Problems among Adolescence (Anxiety, Stress related, Depression related & Behavioural Problem – Major Symptoms). b) Well-being: Concept of Psychological Well-being, some Strategies for Promotion of Psychological Well-being – Mindfulness, Meditation & other Relaxation Techniques (Evidence Based) c) Life Skills for Promotion of Mental Health & Well-being (Basic Concepts of Ten Core Life Skills as Prescribed by WHO)	16	
	TOTAL	40 Hours	20

GROUP – D

EDUCATIONAL TECHNOLOGY & STATISTICS IN EDUCATION

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Educational Technology a) Educational Technology: Concept, Need & Scope of Educational Technology, Differences between Technology in Education and Technology of Education. b) Components of Educational Technology: Hardware and Software, System Approach (Concept). c) Information & Communication Technology (ICT): Concept & Uses of ICT in Education, Digital Revolution in Education.	10	20
UNIT 2	Statistics in Education a) Statistics in Education: Concept, Applications and Statistical Methods (Data, Frequency Distribution, Graphical Representations) b) Measures of Central Tendency and Standard Deviation (Concept, Applications, and Method of Calculation). c) Correlation: Concept, Types and Methods of Computing Correlation Co-efficient (Product Moment and Rank Difference).	10	
	TOTAL	20 Hours	20

INTERNAL ASSESSMENT : 10 Marks (10 Contact Hours)

(Teacher must keep records of the attendance and remarks)

1. **Demonstration on a Topic** (Teacher will assign a topic from the courses taught and inform students prior to the assessment. Students will demonstrate the topic in front of the class) **5 Marks**
2. **Group Discussion: (Brain Storming)** (The teacher will assign topics to different student's groups. The group will be based on situation and class size. Topic of group discussion will be informed to the students prior to the assessment.)
3. **Debate/Discussion on a Topic** (Teacher will assign topic prior to the assessment)
 - The teacher must give remarks on basis of content knowledge, presentation skill, interaction, body language. **5 Marks**

Project [20 marks] – Marks to be awarded as the cumulative marks of the two Internal assessments awarded in Class XII

Marks distribution for Internal Assessment

Criteria for Marks	Total Marks 10
Class Attendance	1 Mark
Content Knowledge	1 Mark
Presentation Skill	1 Mark
Interaction	1 Mark
Body Language	1 Mark
Students will write minimum 10 points on the selected topic	5 Marks

Sample Documentation Sheet for School

Sl. No.	Signature of the Student	H.S. Roll No.	H.S. Regi. No.	Awarded Marks

Signature of the Class Teacher

Signature of the HM with Seal

SUBJECT : ENGLISH – A

COURSE OVERVIEW :

The short stories and poems selected for XI are a part of popular world literature, which will help students get acquainted with the famous writers and poets of the stories and poems. The themes and ideas dealt with in each story and poem will enable the students to think and appreciate as well as develop their language and vocabulary. Besides conveying social messages, the pieces throw light on Nature (An introduction to the Romantic Movement in English Literature). The complexities of human relationships dealt with in the texts will also lead to a matured understanding and consequent development of the students' intellectual and mental faculties.

COURSE OBJECTIVES :

1. Help the students comprehend the turmoils of human relationships.
2. Help the students appreciate Nature.
3. Introduce students to the prevalent social evils, enabling them to think about the eradication of such evils.
4. Infuse a spirit of patriotism & nationalistic feelings for the motherland as responsible citizens.
5. Overall development of students through appreciation of the discussed themes enhancing their abilities to think, feel and evaluate.
6. An introduction to the different ages of English literature (History of English Literature) will enable the students to carry on with the project allotted to them. This will also help the student who will be pursuing English Honours in their Graduation 1st Year.
7. An introduction to and explanation of Rhetoric (selected Figures of Speech) viz. Simile, Metaphor, Oxymoron, Alliteration, Rhetorical Question, Epigram, Antithesis, Irony, Chiasmus, Epanaphora, Onomatopoeia, Personification, Hyperbole, Transferred Epithet, Litotes, Interrogation, Exclamation will help students understand better while reading the texts.
8. Textual Grammar will help students learn the grammatical rules & sentence structures in detail.
9. Comprehending live and recorded oral presentations on various topics.
10. Participating in group discussions, interviews, and making short oral presentations.
11. Participating in discussions and debates on current national and international affairs, sports, business, etc.
12. Responding in interviews, group discussions, and making meaningful inquiries for travel purposes.

CLASS - XI**SEMESTER – I****SUBJECT : ENGLISH – A (ENGA)****FULL MARKS : 40****CONTACT HOURS : 100 Hours****COURSE CODE : THEORY****MCQ MARKS : 1 MARK PER QUESTION]**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1 : PROSE	1. 'The Last Leaf' by O'Henry	10	05
	2. 'One of these Days' by Gabriel Garcia Marquez	10	05
	TOTAL	20 Hours	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 2 : VERSE	1. 'To Autumn' by John Keats	07	03
	2. 'All the World's a Stage' by William Shakespeare	06	03
	3. 'In the Bazaars of Hyderabad' by Sarojini Naidu	07	04
	TOTAL	20 Hours	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 3 : RHETORIC	Rhetoric from Unit 2: Simile, Metaphor, Oxymoron, Paradox, Alliteration, Antitheses, Irony, Onomatopoeia, Personification, Transferred Epithet	20	05
	TOTAL	20 Hours	05

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 4 : NON- TEXTUAL GRAMMAR	Non-Textual Grammar: Tense, Voice, Direct and Indirect Speech, Phrases and Clauses, Tenses, Articles, Prepositions, Phrasal Verbs	20	01 × 5
	TOTAL	20 Hours	05

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 5 : READING COMPREHENSION	Reading Comprehension (Unseen)	20	01 × 10
	TOTAL	20 Hours	10
	GRAND TOTAL	100	40

CLASS - XI

SEMESTER – II

SUBJECT : ENGLISH – A (ENGA)

FULL MARKS : 40

CONTACT HOURS : 80 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	CONTACT HOURS	MARKS
UNIT 1 : PROSE	1. 'Shooting an Elephant' by George Orwell	02**		01***	06	10
	2. 'The Portrait of a Lady' by Khushwant Singh				06	
	3. 'Araby' by James Joice				08	
	TOTAL				20 Hours	10

**** Any 2 out of 3 Questions of 02 marks. Total marks 2 x2= 04 marks**

*****Any 01 out of 03 questions of 06 marks. Total marks: 06 marks**

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS
UNIT 2 : VERSE	1. ‘I Know Why the Caged Bird Sings’ by Maya Angelou	02**		01***	07	09
	2. ‘Dover Beach’ by Mathew Arnold				08	
	TOTAL				15 Hours	09

**** Any 2 out of 4 Questions of 02 marks. Total marks 2 x2= 04 marks**

*****Any 01 out of 02 questions of 05 marks. Total marks: 05 marks**

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS
UNIT 3 :	Non Textual Prosody		2		20	06
	TOTAL				20 Hours	06

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS
UNIT 4 : NON-TEXTUAL GRAMMAR	Non-textual Grammar: Voice Change, Simple, Complex and Compound Sentences, Finite and Non-Finite Verbs	1 question of 2 marks [1 question to be attempted out of two questions from Transformation of sentences and Correction of errors]	Fill in the blanks as per direction from a paragraph: 01 mark x 3 to be attempted from all topics covered		10	02+03 = 05
	TOTAL				10 Hours	05

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	CONTACT HOURS	MARKS
UNIT 5 :	Essay Writing or Precis Writing				15	10
	TOTAL				15 Hours	10
	GRAND TOTAL				80	40

CLASS - XI

PROJECT WORK / TUTORIAL

FULL MARKS – 20

20 Hours

FORMAT :

- | | |
|------------------------------------|-------------|
| 1) Listening and Speaking Activity | 05+05 marks |
| 2) Writing Activity/ Project | 10 marks |

Instructions :

- Subject teachers will engage students in activities which would assess their listening and speaking skills comprehensively.
- Project Work to be taken up on the following topic in consultation with the concerned subject teacher:
 - Elizabethan Age (All Genres)
- Project/Tutorial may be done throughout both the semesters. However, the student will be evaluated at the end of Semester II.

CLASS - XII

SEMESTER – III

SUBJECT : ENGLISH – A (ENGA)

FULL MARKS : 40**CONTACT HOURS : 100 Hours****COURSE CODE : THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1 : PROSE	1. 'War' by Luigi Pirandello	07	04
	2. 'India through a Traveller's Eyes' by Pearl S. Buck	06	03
	3. 'The Legends of Pensam' by Mamang Dai	07	03
	TOTAL	20 Hours	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 2 : VERSE	1. 'Ode to the West Wind' by P. B. Shelley	10	05
	2. 'My Grandmother's House' by Kamala Das	10	05
	TOTAL	20 Hours	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 3 : DRAMA	'The Glass Menagerie' by Tennessee Williams (Scene I to IV)	30	05
	TOTAL	30 Hours	05

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 4 : TEXTUAL GRAMMAR	Textual Grammar from Unit 1 and Unit 2: Synthesis and Splitting of Sentences; Change of Narration; Correction of errors	10	05
	TOTAL	10 Hours	05

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 5 : READING COMPREHENSION	Reading Comprehension (Unseen)	20	10
	TOTAL	20 Hours	10
	GRAND TOTAL	100	40

*** 5 Questions based on Grammar and Vocabulary Items – 5 Marks****5 Questions based on Understanding and Inference of the Text – 5 Marks****CLASS - XII****SEMESTER – IV****SUBJECT : ENGLISH – A (ENGA)****FULL MARKS : 40****CONTACT HOURS : 80 Hours****COURSE CODE : THEORY****[MCQ : 1 MARK PER QUESTION]**

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS
UNIT 1 : PROSE	1. ‘When Mr. Pirzada Came to Dine’ by Jhumpa Lahiri	02**		01***	05	09
	2. ‘A Clean Well Lighted Place’ by Ernest Hemingway				05	
	TOTAL				10 Hours	09

**** Any 2 out of 4 Questions of 02 marks. And total marks 2 x2= 04 marks**

*****Any 01 out of 02 questions of 05 marks. Total marks: 05 marks**

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS
UNIT 2 : VERSE	1. ‘Last Ride Together’ by Robert Browning	02**		01***	07	09
	2. ‘I cannot live with thee’ by Emily Dickinson				07	
	3. ‘This is a Photograph of Me’ by Margaret Atwood				06	
	TOTAL				20 Hours	09

**** Any 2 out of 4 Questions of 02 marks. And total marks 2 x2= 04 marks**

*****Any 01 out of 03 questions of 05 marks. Total marks: 05 marks**

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS
UNIT 3 : DRAMA	'The Glass Menagerie' by Tennessee Williams (Scene V to VII)			01*	20	05
	TOTAL				20 Hours	05

***Any 01 question to be attempted out of 02 questions. Total Marks - 5**

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	CONTACT HOURS	MARKS

UNIT 4 : NON- TEXTUAL GRAMMAR	Non-textual Grammar: Synthesis and Splitting of Sentences; Change of Narration; Correction of Errors	1 question of 2 marks [1 question to be attempted out of two questions from Synthesis of Sentences and Splitting of Sentences]	Fill in the blanks to report a paragraph or conversation: 01 mark x03 to be attempted		15	02+03 = 05
	TOTAL				15 Hours	05

UNIT No.	TOPICS	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	CONTACT HOURS	MARKS
UNIT 5 : WRITING SKILL	1. Substance Writing			06	7	6
	2. Critical Appreciation			06	8	6
	TOTAL				15 Hours	12
	GRAND TOTAL				80	40

***01 out of Any 02 Questions to be attempted. Total Marks - 10**

CLASS - XII

PROJECT WORK / TUTORIAL

FULL MARKS – 20

20 hours

FORMAT :

- | | |
|--|---------------------|
| 3) Listening and Speaking Activity | 05 marks +05 marks |
| 4) Writing Activity/ Project + Viva Voce | 05 marks + 05 marks |

Instructions:

4. Subject teachers will engage students in activities which would assess their listening and speaking skills comprehensively.
5. Project Work to be taken up on any of the following topics in consultation with the concerned subject teacher:
 - Post War Literature
 - Indian Writing in English

Project/Tutorial may be done throughout both the semesters. However, the student will be evaluated at the end of Semester IV.

Course Overview :

By the time students enter Class XI, it is anticipated that they have attained a reasonable level of language proficiency in English. The primary objective of this course is to further enhance and cultivate higher-order language skills. This proficiency becomes especially crucial as the higher secondary stage serves as a pivotal preparation phase, catering to two significant pathways for students.

Firstly, for a substantial number of students, this stage acts as a preparation for university education which often demand a commendable proficiency in English, making it imperative for students to refine their language skills to meet these academic standards.

Secondly, for another sizable group of students, the higher secondary stage serves as groundwork for entry into the professional realm. In the professional domain as well, effective communication in English is frequently a prerequisite. Hence, this Course is meticulously designed to address the linguistic demands of both academic study and the workplace.

The course structure is strategically crafted to cover a spectrum of language competencies, ensuring that students not only excel in academic contexts but are also well-equipped for effective communication in professional settings.

The Course also introduces diverse literature, including classics and modern texts with themes relevant to contemporary issues. This approach provides students exposure to literature, culture, society, environment, and the broader aspects of humanity. It aims to foster a comprehensive understanding of the world through engaging literary exploration in the context of today's global landscape.

The annual course duration spans 200 contact hours, with 100 hours allocated to Semester I & III, 80 hours to Semester II & IV and the remaining 20 hours dedicated to Home Assignments, Projects and Tutorial / Remedial Classes. By embracing a comprehensive approach, this Course seeks to prepare students for the diverse linguistic challenges they may encounter in their future endeavours, whether in university studies or the professional world.

Course Objectives :

The general objectives include developing skills in listening, speaking, reading, and writing. Specific competencies include:

Listening and Comprehension:

- Comprehending live and recorded oral presentations on various topics.
- Participating in group discussions, interviews, and making short oral presentations.

Speaking Skills :

- Participating in discussions and debates on current national and international affairs, sports, business, etc.
- Responding in interviews, group discussions, and making meaningful inquiries for travel purposes.

Reading and Comprehension :

- Perceiving the overall meaning and organization of texts.
- Identifying central points, supporting details, and building communicative competence in different lexicons.
- Promoting advanced language skills for reasoning and drawing inferences.
- Reading and comprehending extended texts in various genres.
- Developing study skills such as skimming and scanning for main ideas and details.
- Understanding the writer's purpose, tone, and distinguishing between literal and figurative language.
- Comprehending technical language in computer-related fields.

Writing Skills :

- Writing expository/argumentative essays, formal/informal letters.
- Using contextual clues to infer meanings of unfamiliar vocabulary.
- Producing unified paragraphs with adequate details and support.
- Writing workplace-related items such as summaries, and reports.
- Writing Commercial Documents like posters and advertisements.

Grammar :

- Developing an advanced understanding and usage of various tenses, surpassing the levels acquired in previous classes, enabling them to express nuanced temporal relationships in English sentences.
- Constructing and deconstructing sentences, including diverse types (simple, compound, complex) and clauses, showcasing their ability to use varied sentence structures for effective communication.
- Emphasis on the use of passive forms in scientific and innovative writings.
- Usage of modal auxiliaries based on semantic considerations

Methods and Techniques :

- Emphasis on self-learning, reducing dependence on teachers.
- Multi-skill, learner-centred, activity-based approach.
- Incorporating silent reading, role-play, dramatization, group discussions, and other interactive activities.
- Encouraging independent interpretation of texts.
- Overall, the curriculum aims to develop well-rounded language skills for academic and professional success, fostering independent thinking and lifelong learning.

CLASS - XI

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – I

COURSE STRUCTURE

FULL MARKS : 40

CONTACT HOURS : 100 HOURS

UNIT	TOPICS	CONTACT HOURS	MARKS
I	Prose	15	10
II	Verse	15	10
III	Rapid Reader	20	10
IV	Textual Grammar	25	05
V	Reading Comprehension	25	05

CLASS - XI

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – II

COURSE STRUCTURE

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

UNIT	TOPICS	CONTACT HOURS	MARKS
I	Prose	20	10
II	Verse	10	10
III	Rapid Reader	20	05
IV	Non-Textual Grammar	15	05
V	Writing	15	10

ACTIVITY / PROJECT :

FULL MARKS : 20

CONTACT HOURS : 20 HOURS

ACTIVITY/ PROJECT	CONTACT HOURS	MARKS
Assessment of Listening & Speaking (ALS)	10	Listening – 5 Marks Speaking – 5 Marks
Project	10	10 Marks

CLASS - XI

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – I

MARKS :40

CONTACT HOURS : 100 HOURS

(MCQ 1 MARK PER QUESTION)

UNIT 1 : Prose

Sl. No.	TOPICS	CONTACT HOURS	MARKS
1	'An Astrologer's Day' – R. K. Narayan	05	04
2	'The Swami and Mother-Worship' from <i>The Master as I Saw Him</i> by Sister Nivedita	05	03
3	'Amarnath' from <i>The Master as I Saw Him</i> by Sister Nivedita	05	03
4	'The Model Millionaire' by Oscar Wilde	05	03

Schools will select **any three** prose (One text of 04 marks and two other texts of 03 marks) topics from the four available options and guide the students accordingly. Students will be informed of the chosen texts by the School at the beginning of the semester.

UNIT 2 : Verse

Sl. No.	TOPICS	CONTACT HOURS	MARKS
1	'Composed Upon Westminster Bridge' – William Wordsworth	05	04
2	'The Bangle Sellers' – Sarojini Naidu	05	03
3	'The Second Coming' – W.B. Yeats	05	03

UNIT 3 : Rapid Reader

Sl. No.	TOPICS	CONTACT HOURS	MARKS
1	'Macbeth' from A Textbook of English(B) Rapid Reader & A Book on ESP & Project by WBCHSE	08	04
2	'Othello' from A Textbook of English(B) Rapid Reader & A Book on ESP & Project by WBCHSE	06	03
3	'As You Like It' from A Textbook of English(B) Rapid Reader & A Book on ESP & Project by WBCHSE	06	03
4	'The Story of an Hour' by Kate Chopin	08	04
5	'The Open Window' by Saki	06	03

Schools will select **any three** Rapid Reader texts (One text of 04 marks and two other texts of 03 marks) from the five available options and guide the students accordingly. Students will be informed of the chosen texts by the School at the beginning of the semester.

UNIT 4 : Textual Grammar

Sl. No.	TOPICS	CONTACT HOURS	MARKS
1	Textual Grammar from Unit 1 and Unit 2: Tense, Voice, Indirect Speech, Phrasal Verbs, Preposition, Clause, Participle, Gerund and Transformation of sentences	25	01 x 5

UNIT 5 : Comprehension (Unseen)

Unit	TOPICS	Contact Hours	Marks
1	Reading Comprehension (Unseen)	25	01 x 5

CLASS - XI

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – II

Total Theory Marks : 40

CONTACT HOURS : 80 HOURS

UNIT 1 : Prose

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	Total Marks	Total Contact Hours
1	'The Garden Party' – Katherine Mansfield	02		01	10	05
2	'Alias Jimmy Valentine' – O' Henry					05
3	'Of Studies' - Francis Bacon					04
4	'Nobel Lecture' - Mother Teresa					06
5	'The Fly' by Katherine Mansfield					05

Schools will select **any three** prose topics from the five available options and guide the students accordingly. Students will be informed of the chosen texts by the School at the beginning of the semester.

UNIT 2 : Verse

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	Total Marks	Total Contact Hours
1	'My last Duchess' – Robert Browning	02		01	10	05
2	'And Still I Rise' – Maya Angelou					05

**** Any 2 out of 4 Questions of 02 marks. And total marks 2 x2= 04 marks**

*****Any 01 out of 02 questions of 06 marks. Total marks: 06 marks**

UNIT 3 : Rapid Reader

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	Total Marks	Total Contact Hours
1	'The Ghost Brahman' from <i>Folk-Tales of Bengal</i> by Lal Behari Dey			1 question to be attempted out of 3 questions	05	07
2	'A Ghostly Wife' from <i>Folk-Tales of Bengal</i> by Lal Behari Dey					07
3	'The Man Who Wished to be Perfect' from <i>Folk-Tales of Bengal</i> by Lal Behari Dey					06
4	'At School (1)'-Extract from Netaji's Life and Writings – Part-I – Autobiography of Subhas Chandra Bose, Chapter - 4					07
5	'The Parrot's Tale' by Rabindranath Tagore					06

Schools will select **any three** Rapid Reader texts from the five available options and guide the students accordingly. Students will be informed of the chosen texts by the School at the beginning of the semester.

UNIT 4 : Non-Textual Grammar

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	Total Marks	Total Contact Hours
	Non-textual Grammar: Tense, Voice, Phrasal Verbs, Preposition, Clause, Participle, Gerund, Transformation of Sentences	1 question of 2 marks [1 question to be attempted out of two questions from Transformation of sentences or Correction of errors]	Fill in the blanks as per direction: 0.5 marks x 6 to be attempted from all topics covered		02+03 = 05	15

UNIT 5 : Writing Skill

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5/10 marks)	Total Marks	Total Contact Hours
1	Paragraph Writing [150 words] OR Formal Letter Writing [75 words] & Event Report Writing [75 words]			1 x 10 OR 1 x 05 + 1 x 05	10	15 hours

PROJECT / TUTORIAL**FULL MARKS : 20****CONTACT HOURS : 20 HOURS****FORMAT:**

- 1) Listening and Speaking Activity 05+05 marks
- 2) Writing Activity 10 marks

Subject teachers will engage students in activities which would test their listening and speaking skills comprehensively.

Instructions:

1. For writing activity concerned teacher will select relevant topic from:
 - a) Environmental Issues
 - b) Society and Culture
 - c) Eminent Personalities
2. Project/Tutorial may be done throughout both the semesters. However the student will be evaluated at the end of Semester II.

CLASS - XII

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – III

COURSE STRUCTURE

FULL MARKS : 40

CONTACT HOURS : 100 HOURS

UNIT	TOPICS	CONTACT HOURS	MARKS
I	Prose	20	10
II	Verse	15	10
III	Drama	20	05
IV	Textual Grammar	20	05
V	Reading Comprehension	25	10

CLASS - XII

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – IV

COURSE STRUCTURE

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

UNIT	TOPICS	CONTACT HOURS	MARKS
I	Prose	10	10
II	Verse	10	10
III	Drama	20	05
IV	Non-Textual Grammar	20	05
V	Writing	20	10

ACTIVITY / PROJECT :

FULL MARKS : 20

CONTACT HOURS : 20 HOURS

ACTIVITY/ PROJECT	CONTACT HOURS	MARKS
Assessment of Listening & Speaking (ALS)	10	Listening – 5 Marks Speaking – 5 Marks
Project	10	10 Marks

CLASS - XII

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – III

MARKS : 40

CONTACT HOURS : 100 HOURS

(MCQ : 1 MARK PER QUESTION)

UNIT 1 : Prose

Sl. No.	Topic	Contact Hours	Marks
1	'The Night Train at Deoli' by Ruskin Bond	07	04
2	'Strong Roots'(Extract from the chapter Orientation from Wings of Fire) by APJ Abdul Kalam	06	03
3	'The Bet' by Anton Chekhov	07	03

UNIT 2 : Verse

Sl. No.	Topic	Contact Hours	Marks
1	'Our Casuarina Tree' by Toru Dutt	07	05
2	'Ulysses' by Alfred Lord Tennyson	08	05

UNIT 3 : Drama

Sl. No.	Topic	Contact Hours	Marks
1	'Riders to the Sea' by J. M. Synge	20	05

- Students will go through the same drama in Sem 3 for MCQs and in Sem 4 for Descriptive Type Questions respectively.

UNIT 4 : Textual Grammar

Sl. No.	Topic	Contact Hours	Marks
1	Textual Grammar from Unit 1 and Unit 2: Synthesis and Splitting of Sentences; Change of Narration; Correction of errors	20	05

UNIT 5 : Reading Comprehension

Sl. No.	Topic	Contact Hours	Marks
1	Reading Comprehension (Unseen)	25	10*

*** 5 Questions based on Grammar and Vocabulary Items – 5 Marks**

5 Questions based on Understanding and Inference of the Text – 5 Marks

CLASS - XII

SUBJECT : ENGLISH – B (ENGB)

SEMESTER – IV

TOTAL THEORY MARKS : 40

CONTACT HOURS : 80 HOURS

UNIT 1 : Prose

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	Total Marks	Total Contact Hours
1	'Alice in Wonderland – Down the Rabbit Hole' by Lewis Carroll	02**		01***	10	05
2	'Three Questions' by Leo Tolstoy					05

**** Any 2 out of 4 Questions of 02 marks. And total marks 2 x2= 04 marks**

***** Any 01 out of 02 questions of 06 marks. Total marks: 06 marks**

UNIT 2 : Verse

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (6 marks)	Total Marks	Total Contact Hours
1	'Hawk Roosting' by Ted Hughes	02**		01***	10	03
2	'That time of year' - Sonnet 73 by William Shakespeare					03
3	'The Greenhouse Effect' by Carl Dennis					04

**** Any 2 out of 4 Questions of 02 marks. And total marks 2 x2= 04 marks**

***** Any 01 out of 03 questions of 06 marks. Total marks: 06 marks**

UNIT 3 : Drama

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	Total Marks	Total Contact Hours
1	'Riders to the Sea' by J. M. Synge			01*	05	20

*Any 01 question to be attempted out of 02 questions. Total Marks – 5

Students will go through the same drama in Sem 3 for MCQs and in Sem 4 for Descriptive Type Questions respectively.

UNIT 4 : Non-Textual Grammar

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (5 marks)	Total Marks	Total Contact Hours
1	Non-textual Grammar: Synthesis and Splitting of Sentences; Change of Narration; Correction of Errors	1 question of 2 marks [1 question to be attempted out of two questions from Synthesis of Sentences and Splitting of Sentences]	Fill in the blanks to report a paragraph or conversation: 01 marks x03 to be attempted		02+03 = 05	20

UNIT 5 : Writing

Sl. No.	Topic	Short Answer Type Questions (2 marks)	Short Answer Type Questions (3 marks)	Descriptive Type Questions (10 marks)	Total Marks	Total Contact Hours
1	Essay Writing (150 words) Or Drafting email (75 words) & Blog Writing (75 words)			01*	01 x 10 = 10 Or 01 x 5 + 01 x 5	20

Word Limit for Text Oriented Questions :

- For 02 marks questions students are expected to write answers within 20-30 words in all the semesters.
- For 05 or 06 marks students are expected to write answers within 100-120 words in all the semesters.

PROJECT / TUTORIAL**FULL MARKS : 20****CONTACT HOURS : 20 HOURS****FORMAT :**

- 3) Listening and Speaking Activity 05+05 marks
- 4) Writing Activity/ Project 10 marks

Instructions :

- Subject teachers will engage students in activities which would assess their listening and speaking skills comprehensively.
- For writing activity concerned teacher will select a relevant topic from any of the following themes:
 - Health and Fitness
 - Film and Drama
 - Technology and Innovation
 - Local Festivals and Traditions
- Project/Tutorial may be done throughout both the semesters. However, the student will be evaluated at the end of Semester IV.

Word Limit for Text Oriented Questions :

- For 02 marks questions students are expected to write answers within 20- 30 words in all the semesters.
- For 05 or 06 marks students are expected to write answers within 100-120 words in all the semesters.

CLASS-XI

SUBJECT : ENVIRONMENTAL SCIENCE (EVSC)

SEMESTER-I

FULL MARKS:35

CONTACT HOURS :80 HOURS

COURSE CODE : THEORY

Chapter	Subtopics	Contact Hours	Marks
1.Environment Through Ages	1.1 Origin of Earth • The processes and events leading to the formation of Earth. 1.2 Evolution of Earth • Lithospheric Evolution: Development and transformation of Earth's crust. • Atmospheric Evolution: Changes in Earth's atmosphere over time. • Hydrospheric Evolution: Formation and progression of Earth's water systems. • Origin and Evolution of Life • Development of life on Earth before the emergence of humans. • Origin of Human Beings • The appearance and development of humans on Earth. 1.3 Human Interaction with the Environment • Concept of Natural Resources • Definition and Types: Various categories of natural resources and their significance. • Exploitation and Degradation: Human use of natural resources and the resulting environmental impacts.	16	8
2. Concept of Ecology	2.1 Basic Concepts and Definitions • Key terms: Ecology, landscape, habitat, ecological niche, eco-zones, biosphere, ecosystems, ecosystem stability, ecotone, edge effect. • Branches of Ecology: Autecology (individual species) and Synecology (communities). • Overview of major terrestrial biomes. 2.2 Types of Ecosystems • Categories: Terrestrial: Forests, grasslands, deserts; Aquatic: Lentic (still water), lotic (flowing water), estuarine, marine, and wetlands.	32	15

	• Structure and Function: Biotic (living) and abiotic (non-living) components. • Metabolism: Primary production and energy flow models. 2.3 Ecosystem Connections • Energy Flow: Food chains, food webs, Lindeman's 10 Percent Energy Law. • Characteristics: Decomposition, ecological efficiencies, ecological pyramids (number, biomass, energy). 2.4 Biogeochemical Cycles and Nutrient Cycles • Key Cycles: Oxygen, carbon, nitrogen, phosphorus, sulfur. • Nutrient Dynamics: Models, inputs, biotic accumulation, and losses.		
3. Biodiversity and Conservation	3.1 Biodiversity • Concept: Biodiversity, Definition, and importance of speciation in biodiversity. • Types: Species, genetic, ecosystem, and domestic diversity. • Importance: Biodiversity's role in ecosystem function. • Highlights: Hotspots of biodiversity, India's mega biodiversity. 3.2 Loss of Biodiversity • Causes and Effects: Factors driving biodiversity loss and their impacts. • IUCN Red List: Categories (e.g., Extinct, Critically Endangered, Vulnerable). Red Data Book. • Extinction: Chronospecies extinction, true extinction. • Trade : Wildlife trade 3.3 Biodiversity Conservation Strategies • In-situ: Biosphere reserves, national parks, sanctuaries. • Ex-situ: Aquariums, botanical gardens, zoos, cryopreservation, DNA banks. • Significance: Importance of biodiversity conservation. 3.4 Biodiversity Assessment • Parameters: Frequency, density, abundance, relative density. • Indices: Shannon-Wiener diversity index, Simpson's index, evenness index • Nutrient Dynamics: Models, inputs, biotic accumulation, and losses.	32	12

CLASS-XI

SUBJECT : ENVIRONMENTAL SCIENCE (EVSC)

SEMESTER-II

FULL MARKS:35

CONTACT HOURS : 80 HOURS

COURSE CODE : THEORY

Chapter	Subtopics	Contact Hours	Marks
4. Environment and sustainability	4.1 Sustainable Development • Definition and origin of sustainable development. • Relationship between environment and development. • Importance for present and future quality of life. • Brief overview of the Brundtland Commission Report. 4.2 Science of Sustainability • Concepts: Sustainability, carrying capacity, deep and shallow ecology, sustainability index. • Population growth vs. resource availability. • Sustainable consumption. 4.3 Sustainable Agriculture • Concept, need, and action plan for sustainable agriculture. • Green Revolution: Impact of fertilizers, pesticides, mechanization. • New practices: Animal husbandry, livestock management, aquaculture. 4.4 Urban Environmental Challenges and Sustainability • Urban Heat Island (UHI) and its impact on city climates. • Urban Stress and its effects on human health and quality of life. • Mitigation strategies for urban environmental issues. 4.5 Global Sustainability Challenges and Solutions • SDGs and global sustainability efforts • Circular economy and sustainable waste practices • Water management and governance • Fossil fuels vs. renewable energy	30	13

	• Technological innovations for sustainability • Life cycle assessment and eco-labelling		
5. Environmental Physics	5.1. Basic Concepts of Light and Matter • Spectroscopy: Absorption and emission spectra, Beer-Lambert law, light transmission, and scattering. • Laws of thermodynamics: Entropy, enthalpy, and free energy. • Heat transfer: Conduction, convection, and radiation. 5.2 Atmospheric Physics • Albedo, solar constant, and Earth's heat budget. • Radiation and subsidence inversion, mixing depth, and pollutant dispersal. 5.3 Physics of Climate • Temperature, humidity (absolute, relative, specific), dew point, pressure, wind, and precipitation. • Climatological norms, Earth's conveyor belt, ocean circulation, and ocean-atmosphere interactions. • Cyclone and anticyclone formation, climate classification.	25	10
6. Environmental Chemistry	6.1 Introduction to Environmental Chemistry • Definition & Scope: Overview of environmental chemistry and its importance. • Applications of Environmental Chemistry 6.2 Air Chemistry • Atmosphere composition and stratification. • Photochemical reactions: NO_x, SO_x, O₃, O₂, CO₂, CO, ions, and radicals. • Issues: Acid rain, ozone depletion, particulate matter, aerosols. 6.3 Water Chemistry • Physical and chemical properties of water. • Dissolved gases (O₂, CO₂) and Henry's Law. • Heavy metals (As, Hg, Pb), acidity, alkalinity, and water hardness. 6.4 Soil Chemistry • Soil composition: Inorganic and organic components. • Cation/anion exchange, and key nutrients (N, P, K) in soil.	25	12

CLASS-XII

SUBJECT : ENVIRONMENTAL SCIENCE (EVSC)

SEMESTER-III

FULL MARKS:35

CONTACT HOURS :80 HOURS

COURSE CODE : THEORY

Chapter	Subtopics	Contact Hours	Marks
7.Environmental microbiology and biotechnology	7.1 Introduction to Microbiology • Definition & Scope: Study of microorganisms and their roles. • Types of Microorganisms: Bacteria, fungi, algae, protozoa, viruses. • Microbial World: Habitats of microorganisms and their ecological roles. 7.2 Microbial Ecology • Symbiotic Relationships: Mutualism, commensalism, parasitism. • Microbial Communities: Presence in soil, water, and air. • Nutrient Cycles: Microbes' role in nitrogen, carbon, sulfur, and phosphorus cycles. 7.3 Microorganisms and Human Health • Impact on Health: Disease-causing microorganisms (e.g., E. coli, Salmonellosis). • Immunology: Antigen-antibody interactions, vaccines. 7.4 Biotechnology for Environmental Sustainability • Introduction to Biotechnology: Definition, branches, and tools/techniques. • Genetic Engineering: GMOs with examples. • Sustainable Agriculture: Biofertilizers, organic farming, vermicomposting, integrated pest management. • Concept of Bioremediation with their types. • Microbes for wastewater treatment and pollutant cleanup. • Biofuels: Concepts and types of biofuels based on generations. • Biosensors: Role in environmental monitoring. • Ethics & Biosafety: Guidelines and protocols for safe biotechnology use	30	12
8. Environmental Health and Toxicology	8.1 Environmental Health • Concept, Principle, and Components: Understanding environmental factors affecting health. • Epidemiological Concepts: Measurement of	30	12

	mortality, morbidity, screening, and surveillance. • Public Health & Environmental Stewardship: Promoting health through environmental protection. 8.2 Community and Health • Health Education & Communication: Introduction to health programs and family planning in India. 8.3 Occupational Health: Health issues in various occupations (e.g., Anthracosis, Silicosis, Asbestosis). 8.4 Concept of Toxicology • Toxicants & Xenobiotics: Types of toxic substances, exposure routes, and their effects. Toxicokinetic and Toxicodynamic. • Acute & Chronic Toxicity: Understanding dose-response (LD50, LC50). • Sublethal Concentration: NOEL, MATC for safe levels. • Bioassay: Types and methodologies for toxicity testing. • Biomarkers, Bioaccumulation, Bioconcentration, and Biomagnification: Key concepts in tracking toxic substance effects.		
9. Environmental legislation and policy	9.1 Environmental Conservation in British India & Independent India: • Indian Penal Code 1860, Van Mahotsava (1950), National Forest Policies (1952, 1988), National Water Policy (2002), National Environment Policy (2006). 9.2 Legal Provisions for Environmental Management: • Indian Forest Act (1927), Wildlife (Protection) Act (1972), Water (Pollution Control) Act (1974), Water Cess Act (1977), Forests (Conservation) Act (1980), Air (Pollution Control) Act (1981), Environment (Protection) Act (1986), Motor Vehicle Act (1988), Public Liability Insurance Act (1991), Noise Pollution Rules (2000), Biological Diversity Act (2002). 9.3 International Environmental Agreements: • Ramsar Convention (1971), Stockholm Conference (1972), Montreal Protocol (1987), UN Conference on Environment and Development (1992), Kyoto Protocol (1997), Convention on Climate Change, Carbon Credit and Trading, Clean Development Mechanism (CDM), World Summit on Sustainable Development (2002), Paris Agreement (2015).	20	11

CLASS-XII

SUBJECT : ENVIRONMENTAL SCIENCE (EVSC)

SEMESTER-IV

FULL MARKS:35

CONTACT HOURS :80 HOURS

COURSE CODE : THEORY

Chapter	Subtopics	Contact Hours	Marks
10.Environmental pollution control and Green Technology	10.1 Introduction to Environmental Pollution • Pollutants: Definition and types. 10.2 Air Pollution • Pollutants: Major primary and secondary pollutants. • Effects: Health impacts, photochemical smog, industrial smog, temperature inversions, greenhouse effect, and global warming. • Control: Gaseous absorption, adsorption, cyclone separators, ESP. • Case Studies: Black carbon aerosol in the Himalayas, Bhopal Gas Tragedy. • NAAQS & AQI: Standards and calculations. 10.3 Water Pollution • Sources: Surface, ground, and ocean water pollution. • Concepts: DO, BOD, COD, Eutrophication. • Health Impact: Waterborne diseases (Diarrhea, Typhoid). • Control: Water quality standards, STP, ETP, WHO guidelines. • Case Studies: Ganga Action Plan (GAP), Minamata Disaster. 10.4 Soil Pollution • Causes and Effects: Impact on environment, vegetation, and life forms. • Control: Soil reclamation and pollution control measures. 10.5 Noise Pollution • Sources & Effects: Measurement, current situation in India, prevention, and control. 10.6 Radioactive Pollution • Types & Sources: Hazards and disposal methods. • Case Study: Chernobyl disaster, 1986. 10.7 Analytical Methods with principles and application • Techniques: UV-VIS Spectrophotometry, Atomic Absorption Spectrophotometry, Electrophoresis, Chromatography, and Microscopy—properties, types, and applications.	30	13

	10.8 Green Technologies with principles and application • Technologies: CFLs, motion detection lighting, programmable thermostats, carbon capture and storage (CCS), Flue Gas Desulfurization (FGD), and solvent recovery systems.		
11.EIA and Environmental management	11.1 Concept of Environmental Management • Need for Environmental Management: Importance of managing environmental resources for sustainable development. • Environmental Aspects: Social, economic, and moral approaches to environmental management. 11.2 Waste Management • Solid Waste: Disposal, recycling, and treatment methods. • Liquid Waste: Management of wastewater, and treatment technologies. • Biomedical Waste: Proper disposal and treatment of healthcare-related waste. • Hazardous Waste: Handling, disposal, and risk mitigation. 11.3 3R Management: • Reduce, Reuse, Recycle Key principles for waste minimization and resource efficiency. 11.4 Environmental Impact Assessment (EIA) • Introduction to EIA: Definition, objectives, and significance of conducting EIA for development projects. • Types of EIA: o Rapid EIA: Quick assessment for urgent projects. o Comprehensive EIA: In-depth evaluation of large-scale projects. o Strategic EIA: Focused on policies, plans, and programs. • Methodologies of EIA: General procedures used to evaluate the environmental impacts of projects. • Fundamentals of the Latest EIA Notification (Draft) 2020: Overview of the new guidelines and changes in the draft notification. • Case Studies of EIA: o Hydropower Projects: Environmental concerns and EIA studies for hydroelectric plants. o Thermal Power Projects: Environmental impact assessment for coal and gas-fired power plants.	30	12
12.Environmental statistics	12.1• Statistical Analysis Tools: Brief principle of Sampling, probability theory, distributions (Normal, Lognormal, Binomial, Poisson, t, F). Basic concepts of mean, median, mode, standard error, standard deviation, correlation, regression, hypothesis testing (t-test, Chi-square) with examples. 12.2 • Data Presentation: Basic concepts of Frequency, histograms, pie charts, and pictograms. 12.3 • Environmental Modelling: Principles of Linear simple and multiple regression models, validation, and forecasting.	20	10

	12.4 • Air Pollution Dispersion: Principles of Box model for air pollution modelling and prediction.		
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CLASS XI

SUBJECT: ENVIRONMENTAL SCIENCE (EVSC)

COURSE CODE: PRACTICAL

FULL MARKS: 30

CONTACT HOURS: 40

Evaluation Scheme	Marks
One experiment from Group A	7
One experiment from Group B	7
Spotting (Total three) from Group C. One from each set.	(3X2)=6
Practical Record Book +Viva Voce	(3+2)=5
One Investigatory field study from Group D + viva voce on field study	(3+2)=5

A. List of experiments:

1. Determination of total hardness of water sample provided by EDTA method.
2. Determination of total alkalinity of water sample provided by phenolphthalein/methyl orange.
3. Characterization of water samples: Turbidity, Color, Odor, Temperature, pH,
4. To test the presence of chloride, nitrates, carbonates and bicarbonates in the water sample provided.

B. List of experiments:

5. To measure the climate temperature by Six's maximum and minimum Thermometer : Correctly read the maximum and minimum temperature from Six's Thermometer and record them in a proper table.
6. To measure the humidity by hygrometer : Note down the dry bulb and wet bulb temperature reading in a proper table and find out the Relative Humidity of the atmosphere with the help of relative humidity chart.
7. To measure atmospheric pressure: Correctly read the atmospheric pressure from the

Fortin's Barometer with the help of Vernier scale and record them in a proper table. Calculate the atmospheric pressure.

8. To Study the effect of light on the growth of plants (Moong /Pea/ barley/ wheat) provided (Grown in two pots in light and dark conditions). Make the following observations using your experimental data and record in a tabular form as given below.

Number of seeds germinated in each pot, Color of seedling and leaves ,Height of seedlings, Length of internodes, Number of leaves produced per plant (seedlings)

C. Study and observe the following (Spotting):

(Set-I): Identification of specimen (photograph/museum specimen) and comment on their importance: Sarpagangha, Yam, Sami, Cinchona, Neem

(Set-II): Identification of specimen (Photograph) and comment on their cause of endangeredness : Red panda, Snow leopard, Cheetah, Koala, one horned rhino , Ginkgo biloba.

(Set-III): Identification of specimen (photograph/museum specimen) and comment on their impact on biodiversity of India: *Lantana sp*, *Parthenium sp*, *Tilapia sp*.

D. Field study

1. Field study in ecology using both qualitative and quantitative studies (Checklist/Quadrat /Transect) from any one of the following bio-geographical area (plain/coastal/ forest/ Hills) with report submission.
2. Identification of flora with highlight on economic significance and importance to ecological balance
3. Identification of fauna with highlight on economic significance and importance to ecological balance
4. Choice of five plants of your neighbourhood and description in terms of height and leaf characteristics.
5. To describe: a) climate of an urban area b) yearly variation in suspended particulate matter in the same area.

CLASS XII

SUBJECT: ENVIRONMENTAL SCIENCE (ENVSC)

COURSE CODE: PRACTICAL

FULL MARKS: 30

CONTACT HOURS: 40

Evaluation Scheme	Marks
One experiment from Group A	7
One experiment from Group B	7
Spotting (Total three) from Group C. One from each set.	(3X2)=6
Practical Record Book +Viva Voce	(3+2)=5
One Investigatory field study from Group D + viva voce on field study	(3+2)=5

A. List of experiments:

1. Determination of LC50 and LD50 with data provided.
2. Determination of Water Holding Capacity of soil sample provided (at least two samples).
3. Determination of soil moisture content.
4. Determination of soil pH of at least soil sample.

B. List of experiments:

5. Determination of Dust fall rate.
6. Estimation of Noise level (dB(A)).
7. Studies on goodness of fit for regression model with given data set on any environmental issues.
8. Simple staining using curd.

C. Study and observe the following (Spotting):

(Set-I):Identification of pest in agriculture (photographs/ museum specimen) and comment on their economic importance: *Leptocorisa sp* , *Scirpophaga incertulus* , *Gryllinae sp* , *Amphipyra pyramidoides*.

(Set-II): Identification of Microorganisms in environmental sample (Permanent slide / photograph) and comment on their economic importance: *Escherichia coli*, *Bacillus sp*, *Staphylococcus aureus* , *Plasmodium sp* , *Aspergillus sp*

(Set-III): Identification of bio-indicators (Photograph/ museum specimen/permanent slides):

Microbes: *Escherichia coli*, *Lichens*

Plants: *Mosses*, *Water hyacinth*

Animals: *Dragonfly larva*, *Earthworm*

D. Field study

1. Preparation of an Environmental Management Plan on the basis of given scenario.
2. Review on local environmental issues and making design of action plan.
3. Review on social responsibility of environmental management and report submission.
4. Report on global/ local environmental pollution issues.
5. To segregate domestic waste into bio-degradable and non-biodegradable components.
6. To make an audit of the electrical energy consumption by various household appliances.

Objectives of H.S. Geography Course

Objective 1 : Familiarize with basic concepts and core contents of Geography and examine man environment relationship.

Objective 2 : Description and interpretation of spatial pattern of features on thematic maps with location, time and space.

Objective 3 : Understanding the physico-cultural relationship with respect to different environmental adaptations.

Objective 4 : Apply geographical knowledge and skills of inquiry to emerging situations and problems at different levels, i.e. local, regional, national and international.

Objective 5 : Develop geographical skills to collect primary data from field survey and preparation of field based project report either manually or using computer based techniques whichever possible.

Learning Outcomes of H.S. Geography Course

LO1: Understanding the Geosystems : Students will accrue a comprehensive understanding of the basics of man-environment interactions at global, regional, and local scales.

LO2:Synthesizing Environmental and Sustainability issues : Students will be able to synthesize social and environmental issues and grasp the know-hows of sustainable development.

LO3:Acquiring Skills of Geographical Analysis : Students will acquire fundamentals of geospatial skills and geographical instrument handling capabilities to analyze geographical problems.

CLASS - XI

SUBJECT : GEOGRAPHY (GEGR)

SEMESTER – I

FULL MARKS : 35

CONTACT HOURS : 80 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF PHYSICAL GEOGRAPHY	34	15
Unit 1	Geography as a discipline : Definition,nature and classification of Geography; Scope and content of Physical Geography	03	
Unit 2	I. Origin of Earth : Classical theories (Kant, Jean and Jeffrey); Modern theory (Big Bang theory) II. Interior of the Earth : Sources of information regarding the Earth's interior; Application of Seismology for analyzing Earth's interior; Layering of the earth	09	
Unit 3	Geomorphic Processes : Endogenic processes – Definition, concept, and types; Vulcanicity(definition,concept,causes, and types of vulcanicity; associated landforms; global distribution of volcanoes);Earthquake (concept and definition of earthquake related aspects; causes, and consequences; measuring instruments and scale; global earthquake prone zones with special reference to India; Seaquake and Tsunami) Exogenic processes – Definition, concept, and types	14	
Unit 4	Weather and Climate : Composition of the Atmosphere : gaseous, liquid, and solid components of atmosphere ; Layering structure of the Atmosphere : on the basis of thermal characteristics and chemical components ; Importance of ozonosphere ; causes and consequences of ozone depletion.	08	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF HUMAN GEOGRAPHY	26	12
Unit 1	Human Geography : Scope and content of Human Geography ; concept of man-environmental relationship.	04	
Unit 2	Economic Geography : Classification of Economic activities – Primary, Secondary, Tertiary, Quaternary, and Quinary sectors with characteristics and examples Primary Activities – Definition, characteristics of Hunting and gathering ; Pastoralism ; Agriculture (Subsistence, Commercial, Mixed, Market gardening, and Dairy farming); Favourable geographical conditions and Production-wise distribution of important cash crop of any one globally leading country (except India) — Coffee, sugarcane, and cotton ; Mining : types and environmental problems.	22	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: GEOGRAPHY OF INDIA	20	8
Unit 1	India as a country : Geographical location; Size; Administrative setup; General information of neighbouring countries	02	
Unit 2	India – Physiography : Physiographic divisions of India with special reference to Tectonic provinces (Peninsular; Extra-peninsular; Indo-Gangetic; Coastal plains and Islands)	10	
Unit 3	India – Drainage systems : Drainage systems of India with reference to flow directions and perennality (The Himalayan drainage system; The Peninsular drainage system); Uses and sharing of river water (Ganga, Indus, Krishna, Godavari and Kaveri)	08	

CLASS - XI

SUBJECT : GEOGRAPHY (GEGR)

SEMESTER – II

FULL MARKS : 35

CONTACT HOURS : 60 HOURS

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF PHYSICAL GEOGRAPHY	30	15
Unit 1	Concept of Isostasy : Concept of Isostatic anomalies; Theories of Airy and Pratt; Isostatic Adjustments; Cymatogeny	04	
Unit 2	Geomorphic processes : Endogenic Processes – Definition, mechanism, structural elements and resultant landforms of Fold and Fault : Types of fold on the basis of axis and axis plane : types of fault on basis of origin. Exogenic processes – Weathering (definition, types, and resultant features); Soil forming process and factors; Soil profile development; Soil erosion; Soil conservation and management	12	
Unit 3	Weather and Climate : Climatic Elements - Solar radiation; Heat balance; Temperature distribution (horizontal and vertical); Controlling factors of temperature distribution; Inversion of temperature (definition and types) Atmospheric Circulation - Controlling factors of atmospheric motion; Tri-cellular model; Planetary winds; Zonal winds (Surface – Walker circulation; Upper Air – Jet stream)	10	
Unit 4	Hydrosphere : Modes and occurrence of water on Earth; Global hydrological cycle ; Definition, Types (surface and sub-surface) and controlling factors of run-off ; Drainage basin as a hydrological unit	04	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF HUMAN GEOGRAPHY	11	12
Unit 1	Secondary Activities – Industry : Classification of Industries; Factors responsible for industrial location; Distribution and favourable condition for the development of the following industries of any one leading country of world (except India) - a) Agro-based: food processing industries b) Sea-based: commercial marine fishing c) Forest-based: paper industries d) Mineral-based: Metallic (Iron and steel); Non-metallic (Petrochemical) e) Manufacturing-based: Automobile	6	
Unit 2	Tertiary Activities : Definition and Classification : Trade, Transport, Service, Communication and Tourism.	03	
Unit 3	Quaternary Activities : Nature and characteristics of Information and Communication Technology (ICT) based industries; Research and Development (R&D) based industries	01	
Unit 4	Quinary Activities : Roles of specialists; decision-makers; consultants, policy formulators	01	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: GEOGRAPHY OF INDIA	19	8
Unit 1	Indian Climate : Controlling factors of Indian climate; Nature of Indian monsoon; Seasonal variability of weather; Monsoon and Indian Economy; Impact of ENSO Phenomena and global warming on Indian climate	08	
Unit 2	Forests of India : Types of forests; Ecological and economic significance; Programmes and policies of forest management in India	04	
Unit 3	Natural Hazards and Disasters of India : Definition and classification of hazards and disasters ; National disaster management policy, 2019 ; Hazards prone zones of India with special reference of West Bengal.	07	

CLASS : XI

SUBJECT : GEOGRAPHY (GEGR)

COURSE CODE : PRACTICAL

FULL MARKS : 30

CONTACT HOURS : 40 HOURS

Unit No.	Unit Content**	Contact Hours	Marks
Unit 1	Introduction to Maps : Definition; components; types, importance and uses	02	02
Unit 2	Map scale: Concept and types of map scales; Graphical scale (Concept of Linear; Comparative; Diagonal; and Vernier scales; Construction of Linear scale)	09	04
Unit 3	Map Projection : Concept of map projection; Classification; Mathematical construction and properties of following projections – a. Polar Zenithal Stereographic b. Simple conical with one standard parallel c. Mercator's Projection	10	05
Unit 4	Interpretation of Topographical Maps : Study of Open series topographical maps (1:50000 scale) Preferably of a plateau region; Identification of topographical features using cross-section drawings. Identification of break of slopes from cross-section drawing and preparation of Broad Physiographic Divisions Map. Typical features identification (Drainage; Natural Vegetation; Transport and Communication; Settlement); Establishment of relationship between different physical and cultural elements using Transect Chart (Schematic method)	12	06
Unit 5	Interpretation of Indian Daily Weather Maps : Daily Weather Maps of January and July months under following Heads - Pressure condition; Wind condition; Sky condition (cloudiness and precipitation)	05	04
Unit 6	Preparation and Presentation of Poster[†] : Poster related to any one prominent hazard/ disaster (causes, consequences, preparedness and management) with respect to West Bengal	02	04(2+2)
Unit 7	Laboratory Notebook* and Viva voce	-	05(3+2)

* The laboratory notebook should contain A3 sized (42 cm × 29.7 cm) white pages in landscape mode.

[†] Poster related to selected topic should be of A1 (59.4 cm × 84.1 cm, approximately). Technique of preparing the hardcopy poster can be manual/ digital/ blended. The poster should carry signatures of teachers responsible for supervising the poster preparation.

** Each topic of laboratory notebook should contain the following sub-heads: statement of the problem, objectives, materials and methods (with data source), calculations, drawings/ representation (if any), analysis and interpretation.

Tutorial+ Remedial+ Assignments: 6+10+4= 20 Hours(Semester: I+II)

CLASS - XII

SUBJECT : GEOGRAPHY (GEGR)

SEMESTER – III

FULL MARKS : 35

CONTACT HOURS : 79 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF PHYSICAL GEOGRAPHY	34	15
Unit 1	Geotectonic : Continental Drift Theory according to Alfred Wegner — Concept, evidence, force of drifting and criticism	03	
Unit 2	Geomorphic Process : Exogenic processes – Mass wasting (Definition and types) ; Glacial Landform (Types of Glacier, erosional and depositional processes with associated landforms) ; Work of Ground water and associated landforms (Erosional processes, landforms developed by Ground water — A. Erosional landforms : Sinkholes, dolines, caves, cavern and karst lakes. B. Depositional landforms : Stalactites, stalagmites and cave pillars).	09	
Unit 3	Climate : Water in the atmosphere (Condensation – formation and types, Precipitation – formation and types)	06	
Unit 4	Hydrosphere : Topography of Ocean floor — Major divisions, General characteristics of Ocean floor ; Controlling factors and distribution of temperature, salinity and density of ocean water ; Importance of Marine resources — Minerals, Food and Power ; Classification of Ocean deposits according to origin and location.	10	
Unit 5	Biosphere : Nature, concept and types of ecosystem; Concept of trophic levels; Food chain and food web	06	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF HUMAN GEOGRAPHY	15	10
Unit 1	Population: Concept of Demography; Distribution and Density of population (worldwide) ; Determinants and measures of Population Growth — Birth rate, Death rate, Migration (Definition and types of Migration) ; Concept of Age-Sex pyramid.	07	
Unit 2	Settlement: Types and patterns of settlement; Factors determining the types of rural settlement; Functional classification of urban settlement; Problems of rural and urban settlements	08	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: GEOGRAPHY OF INDIA	30	10
Unit 1	Population of India : Distribution, density, growth, and composition of population	03	
Unit 2	Water Resources : Water conservation and management; Watershed management and rainwater harvesting; Nature and types of irrigation systems; Sustainable use of water resources	04	
Unit 3	Mineral and Energy Resources : Types, distribution and uses – Mineral resource (Iron Ore, Manganese, Bauxite, Copper, Mica); Conventional energy resource (Coal, Petroleum, Natural Gas, Nuclear Energy); Non-conventional energy resource (Solar, Wind, Tidal, Geothermal, Biogas energy, OTEC)	10	
Unit 4	Agriculture : Importance of agriculture in Indian economy : Basic concept of Crop calendar, Crop rotation, Crop combination and cropping intensity ; Concept of Green, White and Blue revolution in India ; Production-wise distribution of important cash crops (Jute, Cotton, Sugarcane and Tea)	08	
Unit 5	Transport and Communication : Importance of different modes of Transport; Types and importance of personal and mass communication systems	02	
Unit 6	Trade and Economy : Concept, Classification ; Bases and Importance ; Role of WTO in International Trade.	03	

CLASS - XII

SUBJECT : GEOGRAPHY (GEGR)

SEMESTER – IV

FULL MARKS : 35

CONTACT HOURS : 61 HOURS

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF PHYSICAL GEOGRAPHY	34	15
Unit 1	Geotectonic : Sea floor spreading — Definition, evidences and causes ; Concept and classification of Plate, types of Plate boundaries and associated landforms.	06	
Unit 2	Geomorphic Processes : Exogenic processes - Fluvial landforms; Coastal landforms; Aeolian landforms; Combined works of exogenic processes and associated landforms	10	

Unit 3	Cycle of erosion : The role of WM Davis; Davis's concept of landform evolution; Rejuvenation and resultant landforms.	03	
Unit 4	Atmosphere : Atmospheric Disturbances — Definition, characteristics and mechanism of Cyclone (Tropical and Temperate zones) and Anti-cyclones ; Climatic classification after Koppen ; Causes and evidences of climatic change.	05	
Unit 5	Hydrosphere : Movement of ocean currents and associated environmental effects	05	
Unit 6	Biosphere : Biodiversity - Definition, types, significance; Factors responsible for biodiversity depletion; Strategies and programs of biodiversity conservation	05	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: FUNDAMENTALS OF HUMAN GEOGRAPHY	12	10
Unit 1	Population : Impact of migration on global demographic changes; Malthusian theory of population growth; Concept of optimum, under, and over population; Demographic transition model by Thompson and Notestein; Need of population control measures and population policy	09	
Unit 2	Human Development : Concept; definition; measurement (Human Development Index, Concept of human poverty index); and approaches (Welfare approach; Amartya Sen's Capability approach)	03	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	SUB-TOPIC: GEOGRAPHY OF INDIA	15	10
Unit 1	Distribution causes of development, problems and prospects : a. Agro-based – Food processing b. Sea-based – Commercial marine fishing c. Forest-based –Paper industry d. Mineral-based –Metallic (Iron and steel); Non-metallic (Petrochemical) e. Manufacturing-based –Automobile f. Information and communication technology	06	

Unit 2	Human Settlement and Development in Indian context : Human settlement types in India; Land resources and agricultural land use — Concept of land use and land cover, land use pattern of India, quality and size of cultivable land holdings, methods to reduce fragmentation of land holdings; Sustainable development in Indian context	04	
Unit 3	Geographical Perspectives on Selected Issues and Problems : a. Water pollution in Ganga Basin – Causes, consequences, and management b. Air pollution in Kolkata and National Capital Region – Causes, consequences, and management c. Arsenic pollution in southern West Bengal – Causes, consequences, and management d. Human-wildlife conflict in Dooars and Sundarbans regions – Causes, consequences, and management e. Land degradation in Jangalmahal– Causes, consequences, and management	05	

CLASS : XII

SUBJECT : GEOGRAPHY (GEGR)

COURSE CODE : PRACTICAL

FULL MARKS : 30

CONTACT HOURS : 40 HOURS

Unit No.	Unit Content	Contact Hours	Marks
Unit 1	Data Processing Using Statistical Techniques : a. Types and sources of data b. Tabulation and processing of data; Construction of data array; Frequency distribution table; Histogram, Frequency polygon; and Ogives c. Measures of central tendency - Mean; Median; Mode d. Measures of dispersion – Absolute measures (range, mean deviation, standard deviation); Relative measure (co-efficient of variation)	10	05
Unit 2	Cartograms and Thematic Mapping : a. Bar graphs – Simple; Multiple; Compound b. Divided proportional circles c. Flow chart d. Ombrathermic diagram e. Thematic maps – Dot, Isopleths and Choropleth	10	05

Unit No.	Unit Content	Contact Hours	Marks
Unit 3	Application of Geographical Instruments : a. Angular measurement using Prismatic Compass (forward and backward bearings) b. Pebble diameter measurement using Slide Caliper (major, minor, and intermediate axes) c. Measurement of strike direction and dip amount using Clinometer Compass d. Six's maximum & minimum thermometer	12	05
Unit 4	Remote Sensing and Geographic Information System (GIS) : a. Remote sensing - Concept of remote sensing; Definition; Source of energy; Types of sensors; Types of image acquisition platforms; Modern application b. GIS – Introduction to GIS; Hardware and software requirements; Data formats – vector and raster; Concept of data input, editing, analysis, manipulation, and representation	04	05
Unit 5	Preparation and Presentation of Field Report : Report based on structured survey schedule on (any one) – a. Household survey b. Market survey c. Traffic survey Note: Students of school in rural area will visit any type of Urban Centre or unit / Students of school in urban area will visit a rural area for Household survey (household survey only)	04	05 (3+2)
Unit 6	Laboratory Notebook* and Viva Voce	-	05 (3+2)

* The laboratory notebook should contain A3 sized white pages (42 cm ×29.7 cm) in landscape mode

Tutorial+ Remedial+ Assignments: 6+10+4= 20 Hours(Semester: III+IV)

Guidelines for Preparation of Field Report

Every student needs to participate in fieldwork and prepare a filed report according to the following guidelines:

A. Household Survey :

- Each student will prepare a report based on primary data collected from household survey with the help of properly formulated and structured survey schedule having open ended, double choice, and multiple choice questions.
- Student will select either a rural area/ mouza (for school students from urban area) or an urban (at least a Class-VI Town as categorized by Census of India) area/ municipal ward (for school students from rural area) for the study, with primary objective of demographic and socio-economic information.
- Each student should survey at least five households selected through simple random sampling without replacement method.
- The report should be handwritten in A4 sized white pages in candidate's own words. The word limit should be of 500 words excluding tables, figures, maps, photographs, and appendices.
- A copy of the field report duly signed by the concerned teacher will be submitted on the date of examination.
- The field report should contain the following sections – Introduction, Selection of Study Area, Objectives of the Study, Methodology, Results and Discussion/ Major Inferences, Conclusions.

B. Market Survey :

1. Each student will prepare a report based on primary data collected from market survey with the help of properly formulated and structured survey schedule having open ended, double choice, and multiple choice questions.
2. Each student should survey at least five shops selected through simple random sampling without replacement method.
3. The report should be handwritten in A4 sized white pages in candidate's own words. The word limit should be of 500 words excluding tables, figures, maps, photographs, and appendices.
4. A copy of the field report duly signed by the concerned teacher will be submitted on the date of examination.
5. The field report should contain the following sections – Introduction, Selection of Study Area, Objectives of the Study, Methodology, Results and Discussion/ Major Inferences, Conclusions.

C. Traffic Survey :

1. Each student will prepare a report based on primary data collected from road traffic survey with the help of properly formulated and structured inventory, specifically having frequency distribution tables.
2. Each student should survey at an important three-point or four-point junction, nearest to the school.
3. The survey should be conducted in two sessions (forenoon and afternoon) of one hour duration.
4. The report should be handwritten in A4 sized white pages in candidate's own words. The word limit should be of 500 words excluding tables, figures, maps, photographs, and appendices.
5. A copy of the field report duly signed by the concerned teacher will be submitted on the date of examination.
6. The field report should contain the following sections – Introduction, Selection of Study Area, Objectives of the Study, Methodology, Results and Discussion/ Major Inferences, Conclusions.

HIST**SUBJECT HISTORY****CLASS - XI****SUBJECT : HISTORY (HIST)****SEMESTER—I****FULLMARKS : 40****CONTACTHOURS : 90HOURS****COURSECODE : THEORY**

UNITNo.	TOPICS	CONTACT HOURS	MARKS
<u>Unit1:</u> Learning History	<u>Understanding History:</u> Pre-History, Proto-History.Early Sources and their nature. Forms of Recorded History. Facts and interpretation. Indo- Persian tradition of History writing in Medieval India. Concept of time in History. Linear and cyclical periodisation of History and chronology.	24	10
<u>Unit2:</u> E mpires	Empires Across the three Continents 100BCE TO 1300CE Introducing the age of Empires. Dynamics of the Roman Empire. Implications of the Contact of the Romans with the sub-continental empires—the importance of slavery in the economy- Cultural transformation and impact on the slave economy	33	15
<u>Unit3:</u> ComparativeStudi es	<u>Concept of Governance</u> 3.1. Citystates : Classical Governments 3.2. Monarchies from <i>Janapadas</i> to <i>Mahajanapadas</i> (Chiefdoms to kingdoms) 3.3. Empires. Definition, Difference with Monarchy- Comparative history of Empires a) The Mauryan empire and the Macedonian empire b) Roman Empire and the Gupta Empire c) The Mughal Empire and the Ottoman Empire	33	15

CLASS-XI

SUBJECT : HISTORY (HIST)

SEMESTER—II

FULLMARKS : 40

CONTACTHOURS : 70HOURS

COURSECODE : THEORY

UNITNo.	TOPICS	CONTAW HOURS	MARKS
<u>Unit 4:</u> NatureofState	<u>State and its apparatus</u> 4.1. Nature of state,The ideal prototype a) The Indian context-Kautilya, the Arthasastra and the state craft ; Ziauddin Barani: <i>Fatwa-i-Jahandari</i> and the nature of the state under the Delhi sultans b) The European context : Greek and the Roman world. Thomas Cromwell and the new Monarchy: intellectual basis of the early modern state. 4.2. Apparatus of Governance a. Persian Satraps b. Chinese Mandarins c. Delhi Sultans : <i>Iqtadars</i> d. Mughal <i>Mansabdars</i>	25	15
<u>Unit 5:</u> Changing Traditions	The Crusades – Changing Cultural Traditions- To appreciate the history of Cultural transformations with reference to paintings, art & architecture of the period. Renaissance periods. Debate on Renaissance —positive and negative impact. Roman Catholic Church & Protestant movements.	20	10
<u>Unit 6:</u> Expanding Horizons	Origins of Modern Science-From Witchcraft to Social emancipation- Astrology to Astronomy, Towards a solar centric universe. Geographical Explorations and new geographical knowledge - Technological advancements. Agricultural, Military and Shipbuilding technology- Printing Revolution in Western Europe (With reference to the contributions of China, Japan and the Arab World to the art of Printing)	25	15

CLASS-XI

SUBJECT : HISTORY (HIST)

COURSE CODE : PROJECT

FULL MARKS : 20

1.	Slavery and its impact on the contemporary world
2.	Map work : On a world map point out the places which were under the Roman Empire
3.	Mapwork On an outline map of India locate and name the sixteen <i>mahajanapadas</i>
4.	Contributions of (any one) a. Kautilya b. Zia-ud-din Barani c. Thomas Cromwell
5.	Renaissance and its impact on the Human Life
6.	Printing Revolution and its impact on the spread of education

[Note:40Hours reserved for Project, Remedial classes, Tutorials and Home Assignments.]

CLASS-XII

SUBJECT: HISTORY (HIST)

SEMESTER-III

FULLMARKS : 40

CONTACTHOURS : 90HOURS

COURSECODE : THEORY

UNITNo.	TOPICS	CONTACT HOURS	MARKS
<u>Unit1:</u> 1) Throughtheeyes of the travelers. 2) Cultural amalgamation 3)Imperialcapital	1. Society and Polity of India (Between 10th and 17th Centuries) with reference to travelers' accounts with special reference on the following travelers : (1) Al Biruni-Uzbekistan, (2) Marco Polo-Italy, (3) Ibn Battuta-Morocco, (4) Abdur Razzak - Iran (5) Francois Bernier-France Administrative Structure - Economies Life—Social and Religious Life—Literature, Art and Architecture 2. Bhakti and Sufi Movement The Bhakti Cult— -Bhakti Saints — Kabir, Guru Nanak, Chaitanya Dev and Meerabai — Impact of Bhakti Ideology — Origins of Sufism—Sufi Orders in India— Impact of Sufism etc. 3 Vijayanagar, Bahmani and Bengal : Iliyas Shahi and Hussain Shahi Dynasties; Origins— Administrative Structure— Society and Economy — Religion and Culture— Artand Architecture— Literature— The Bahmani Successor States	40	20
<u>Unit2:</u> 1) Colonialism and Imperialismin the19 th and20 th Centuries 2) The levers of Colonial Control	1. Brief overview of 17 th and 18 th Century Colonialization in Asia and the New World : - Brief overview of Colonization in Asia & the New World- - Economic Dynamics of Imperialism and Colonialism; from Mercantile Capital to Industrial & Finance Capital - Political and Economic Domination and Subordination of the Colonies— - Racism and Its Impact on Colonial Societies— Neo-imperialism 2. The Levers of Colonial Control : India-as a 'colonized' state— The instruments of control : Law/Legislature/ Bureaucracy/Police/Army- The subordination of the colonial economy	50	20

CLASS-XI

SUBJECT:HISTORY(HIST)

SEMESTER—IV

FULL MARKS : 40

CONTACT HOURS: 70 HOURS

COURSECODE : THEORY

UN ITNo.	TOPICS	CONTACTHOURS	MARKS
<u>Unit 3:</u> Rebels and Raj	1. Revolt of 1857 and its representations : Causes of the Revolt—Character—Causes of Failure — The impact of the Revolt	70	07
<u>Unit 4:</u> Nationalism and Separatism	1. Aligarh Movement, Birth of Muslim League, Hindu Mahasabha : The Aligarh Movement — Sayyid Ahmed Khan—The 'two nation theory'- Birth of the Muslim League — The Hindu Mahasabha — Politics of Separatism and the Pathway to Pakistan 2. Emergence of Organised Nationalist Politics – Pre-1917 and Post-1917, till Independence : Establishment of the Indian National Congress in 1885 — The Moderate phase of Nationalist Politics – The Surat Split (1907) and the birth of Extremism in Indian Politics – Role of Tilak, Pal, Aurobindo Ghosh and Lajpat Rai in extremist politics – Revolutionary movements in Bengal, Punjab and Maharastra. 3. Impact of First World War on India : Economic Impact of First World War on India – Emergence of Gandhi – Concept of Satyagraha : Champaran, Kheda and Ahmedabad – Non-Cooperation Movement (1921-1922) – Civil Disobedience Movement (1930). 4. Prelude to Independence : The Quit India Movement (1942) – Subhas Chandra Bose and the INA—Post-War upsurge in India: Naval Mutiny –The transfer of Power – Cabinet Mission – Mountbatten's negotiations – Mountbatten award – Partition and Independence.		19

UNIT No.	TOPICS	CONTACT HOURS	MARKS
<u>Unit 5 :</u> Framing of the Constitution	The beginning of a New Era, India after 75 years of Independence : Making of the Indian Constitution — Salient Features of the Constitution with emphasis on social and economic empowerment and women empowerment—Five Year Plans – Nehru and Non-Alignment Movement — Liberalization and the Opening up of the Indian Economy in the 1990s		07
<u>Unit 6 :</u> Nation Formation and related aspects	The 1971 Indo-Pak War Formation of Bangladesh and its impact on India.		07

CLASS-XII

SUBJECT : HISTORY (HIST)

COURSE CODE : PROJECT

FULL MARKS : 20

1. Globalization
2. Women 's Right Movement
3. Recent Political Crisis (International)
4. New Social Movement
5. A visit to any museum (Local or otherwise) in West Bengal
6. A visit to any place of historical significance in West Bengal
7. Anti-Corruption Movement
8. Environmental Movement
9. World Peace Movement

[Note: 40 Hours reserved for Project, Remedial classes, Tutorials and Home Assignments.]

Course Objectives

The Mathematics curriculum has undergone periodic revisions in response to the field's expansion and the changing demands of society. The senior secondary stage serves as a springboard for students to pursue professional programs in engineering, physical and biological science, commerce, or computer applications, or to pursue higher education in mathematics. In order to address the evolving demands of all student categories, the current revised syllabus has been created. More focus has been placed on the application of certain principles, drawing inspiration for the issues from real-world scenarios and other academic disciplines.

The following are the main goals of teaching mathematics to senior school students:

- To develop general interest in Mathematics as a discipline.
- To gain critical insight and knowledge of fundamental terminology, concepts, principles, symbols, and skills, especially through motivation and visualization, as well as mastery of underlying procedures and abilities.
- To experience the logic flowing while demonstrating an outcome or resolving an issue.
- To use the gained information and abilities to solve issues, using many approaches where feasible.
- To familiarize students with the various applications of mathematics in everyday life.
- Developing a sense of appreciation and respect for notable mathematicians and their contributions to mathematics is important.
- To cultivate an optimistic outlook in order to reason, evaluate, and speak coherently.
- To cultivate curiosity for the topic by taking part in competitions related to it.

Course Outcomes

At the end of the course the students are expected to develop expertise in various areas of the subject and gain critical insights into the background dynamics of the problem solving process.

The following are the major course outcomes. A student is expected to:

- Develop problem solving skills and apply mathematical concepts to real life situations.
- Cultivate critical thinking and analytical skills in mathematical context.
- Collaborate with peers to solve complex mathematical problems.
- Make predictions and draw conclusions based on statistical data.
- Get a preliminary idea of using technology, like calculators in problem solving.
- Effectively communicate mathematical ideas and solutions both verbally and in writing.
- Present mathematical arguments and justifications.
- Prepare for standardized examinations based on the curriculum.
- Understand the relevance of mathematics in real-world applications.

CLASS - XI**SUBJECT : MATHEMATICS (MATH)****SEMESTER – I****FULL MARKS : 40****CONTACT HOURS : 100 HOURS****COURSE CODE : THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT-I	SETS AND FUNCTIONS	45	15
	1. Sets Sets and their representations, Empty set, Finite and Infinite sets, Equal sets, Subsets, Subsets of the set of real numbers especially intervals (with notations). Power set. Universal set. Venn diagrams. Union and Intersection of sets. Difference of sets. Complement of a set. Properties of Complement sets.	15	4
	2. Relations and Functions Ordered pairs. Cartesian product of sets, Number of elements in the Cartesian product of two finite sets. Cartesian product of the set of reals with itself (up to $R \times R \times R$). Definition of relation, pictorial diagrams, domain, co-domain and range of a relation. Function as a special kind of relation from one set to another. Pictorial representation of a function, domain, co-domain and range of a function. Real valued functions, domain and range of these functions, constant, identity, polynomial, rational, modulus, exponential, logarithmic, signum and greatest integer functions with their graphs. sum, difference, product and quotients of functions.	15	4
	3. Trigonometric Functions Positive and negative angles, Measuring angles in radians and in degrees and conversion from one measure to another. Definition of trigonometric functions with the help of unit circle. Truth of the identity $\sin^2 x + \cos^2 x = 1$, for all x . Signs of trigonometric functions, domain, range and sketch their graphs. Expressing $\sin(x \pm y)$ and $\cos(x \pm y)$ in terms of $\sin x$, $\cos x$, $\sin y$ and $\cos y$.	15	7

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	Deducing identities like the following : $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}, \quad \cot(x \pm y) = \frac{\cot x \cot y \mp 1}{\cot y \pm \cot x}$ $\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$ $\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2},$ $\sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2},$ $\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}$ Identities related to $\sin 2x$, $\cos 2x$, $\tan 2x$, $\sin 3x$, $\cos 3x$ and $\tan 3x$. General solutions of trigonometric equations of the type $\sin \theta = \sin \alpha$, $\cos \theta = \cos \alpha$ and $\tan \theta = \tan \alpha$.		
Unit-II	ALGEBRA	30	15
	1. Complex Numbers and Quadratic Equations Need for complex numbers, especially $\sqrt{-1}$, to be motivated by inability to solve some of the quadratic equations. Algebraic properties of complex numbers. Argand plane, polar representation of complex numbers, modulus, argument. solution of quadratic equation in complex number system.	13	6
	2. Linear Inequalities Linear inequalities. Algebraic solutions of linear inequalities in one variable and modulus function and their representation on the number line. Graphical solution of linear inequalities in two variables.	5	4
	3. Permutations and Combinations Fundamental principle of counting. Factorial n ($n!$). Permutations and combinations, derivation of formulae for ${}^n P_r$ and ${}^n C_r$ and their connections, simple applications.	12	5
Unit-III	CALCULUS	25	10
	1. Limits and Derivatives Intuitive idea of limit. Limits of polynomials and rational functions, trigonometric, exponential and logarithmic functions. Derivative introduced as rate of change both as that of distance function and geometrically. Definition of derivative, relate it to slope of tangent of the curve, derivative of sum, difference, product and quotient of functions. Derivatives of polynomial and trigonometric functions.		

CLASS - XI
SUBJECT: MATHEMATICS (MATH)
SEMESTER – II

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit-I	ALGEBRA	35	15
	1. Principle of Mathematical Induction Process of the proof by induction motivating the application of method by looking at natural numbers as the least inductive subset of real numbers. The principle of mathematical induction and simple applications.	7	3
	2. Binomial theorem History, Statement and proof of the binomial theorem for positive integral indices. Pascal's Triangle, General and middle term in Binomial expansion, Simple applications.	13	6
	3. Sequence and series Sequence and series. Arithmetic Progression (A.P.), Arithmetic Mean (A.M.), Geometric Progression (G.P.), Geometric Mean (G.M,) relation between A.M. & G.M., Arithmetic-Geometric Progression Series (AGP series), infinite G.P. and its sum, sum to n terms of the special series $\sum x$, $\sum x^2$ and $\sum x^3$	15	6
Unit-II	COORDINATE GEOMETRY (2D)	30	15
	1. Straight lines Brief recall of two dimensional geometry from earlier classes. Slope of a line and angle between two lines. Various forms of equations of a line: Parallel to Axis, Point-slope form, slope intercept form, two point form, intercept form, distance of a point from a line.	10	5
	2. Conic sections Sections of a Cone: circle, ellipse, parabola, hyperbola, a point, a straight line and a pair of intersecting lines as a degenerated case of conic section; Standard equation of circle, general equation of circle, Standard equations and simple properties of Parabola, Ellipse and Hyperbola.	20	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit-III	STATISTICS AND PROBABILITY	15	10
	1. Statistics Measures of dispersion: Range, mean deviation, variance and standard deviation of ungrouped/ grouped data	5	3
	2. Probability Random experiments, outcomes, Sample spaces (set representation), Events: Occurrence of events, 'not', 'and' and 'or' events, exhaustive events, mutually exclusive events, Axiomatic (set theoretic) probability, connections with other theories of earlier classes. Probability of an event, probability of 'not', 'and' and 'or' events.	10	7

[Note:20 Hours reserved for Remedial classes, Tutorials and Home Assignments.]

Course : Project for Class XI

Full Marks: 20

Projects should be conducted regularly throughout the year. A project notebook is to be prepared by each and every student where all the below mentioned activities should be recorded. There should be a project assessment once a year (once in Class XI and once in Class XII) where the student will be asked to do one of the activities and write it in his/her script provided for the purpose. The student should carry his/her project notebook during the assessment. A viva should also be conducted during the assessment to test the knowledge of the student regarding the project activity.

List of Projects for Class XI

Sl. No.	Topics	Activities
1	Sequence and Series	To illustrate that the arithmetic mean of two different positive numbers is always greater than the geometric mean.
2	Complex Number	To interpret geometrically the meaning of $i = \sqrt{-1}$ and its integral powers.
3	Trigonometric Functions	To illustrate the values of sine and cosine functions for different angles which are multiples of $\frac{\pi}{2}$ and π .
4	Theory of Sets	To show that the total number of subsets of a given set with 'n' number of elements is 2^n .

5	Theory of Sets	Theoretic Operations using Venn Diagrams.
6	Relations and Functions	To verify that for two sets A and B , $n(A \times B) = pq$ and the total number of relations from A and B is 2^{pq} , where $n(A) = p$ and $n(B) = q$.
7	Limits and Derivatives	To find analytically $\lim_{x \rightarrow c} f(x) = \frac{x^2 - c^2}{x - c}$
8	Probability	To write the sample space, when a coin is tossed once, two times, three times.
9	Conic Sections	To recognize different types of conics and its parts.
10	Permutations and Combinations	To find out the number of permutations and combinations from a set of 3 different objects taking 2 at a time.

Marks Division for the Project Assessment

<u>Sl. No.</u>	<u>Item</u>	<u>Marks</u>
1	Project Notebook	10
2	Doing and writing a project during the project assessment	5
3	Viva	5
	Total	20

CLASS - XII**SUBJECT : MATHEMATICS (MATH)****SEMESTER – III****FULL MARKS : 40****CONTACT HOURS : 100 HOURS****COURSE CODE : THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT-I	RELATIONS AND FUNCTIONS	20	7
	1. Relations and Functions Types of relations: Reflexive, symmetric, transitive and equivalence relations. One-to-one and onto functions, composite functions, inverse of a function.	10	4
	2. Inverse Trigonometric Functions Definition, range, domain, principal value branches. Graphs of inverse trigonometric functions. Elementary properties of inverse trigonometric functions.	10	3
UNIT- II	ALGEBRA	25	10
	1. Matrices Concept, notation, order, equality, types of matrices, zero matrix, identity matrix, transpose of a matrix, symmetric and skew-symmetric matrices. Addition, multiplication and scalar multiplication of matrices; properties of addition, multiplication and scalar multiplication. Simple properties of addition, multiplication and scalar multiplication. Non-commutativity of multiplication of matrices. Existence of non-zero matrices whose product is a zero matrix (restrict to square matrices of order 2). Invertible matrices and proof of the uniqueness of inverse (if it exists). (Here all matrices will have real entries).	15	6
	2. Determinants Determinant of a square matrix (upto 3×3 matrices), properties of determinants, minors, cofactors and application of determinants in finding the area of a triangle. Adjoint and inverse of a square matrix. Consistency, inconsistency and number of solutions of system of linear equations by examples. Solutions of system of linear equations in two or three variables (having unique solution) using inverse of a matrix.	10	4

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT-III	CALCULUS	38	15
	1. Continuity and Differentiability Concept of Continuity and differentiability, derivative of composite functions, chain rule, derivatives of inverse trigonometric functions, derivative of implicit functions, concept of exponential and logarithmic functions, Derivatives of logarithmic and exponential functions, Logarithmic differentiation, derivative of functions expressed in parametric forms. Second order derivatives.	20	8
	2. Application of Derivatives Application of derivatives, Rate of change of quantities, increasing and decreasing functions, tangents and normals, maxima and minima (first derivative test motivated geometrically and second derivative test given as a provable tool). Simple problems on basic principles and real life situations.	18	7
UNIT-IV	PROBABILITY	17	8
	Conditional Probability, Multiplication theorem on probability, independent events, total probability, Bayes' theorem, Random variable and its probability distribution. Mean and variance of a random variable.		

CLASS - XII
SUBJECT : MATHEMATICS (MATH)
SEMESTER – IV

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit-I	VECTORS AND THREE-DIMENSIONAL GEOMETRY	30	15
	1. Vectors Vectors and scalars, magnitude and direction of a vector. Direction cosines and direction ratios of a vector. Types of vectors (equal, unit, zero, parallel and collinear vectors), position vector of a point, negative of a vector, components of a vector, addition of vectors, multiplication of a vector by a scalar, position vector of a point dividing a line segment in a given ratio. Definition, Geometrical Interpretation, properties and application of scalar (dot) product of vectors, vector (cross) product of vectors.	10	5
	2. Three-Dimensional Geometry Introduction to 3D geometry, Coordinate axes and coordinate planes in 3D. Coordinates of a point, distance between two points, Direction cosines and direction ratios of a line joining points. Cartesian equation and vector equation of a line, skew lines, shortest distance between two lines. Angle between two lines.	20	10
Unit-II	CALCULUS	40	20
	1. Integrals Integration as inverse process of differentiation. Integration of a variety of functions by substitution, by partial fractions and by parts. Evaluation of simple integrals of the following types and problems based on them. $\int \frac{dx}{x^2 \pm a^2} \int \frac{dx}{\sqrt{x^2 \pm a^2}} \int \frac{dx}{\sqrt{a^2 - x^2}} \int \frac{dx}{ax^2 + bx + c} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$ $\int \frac{px+q}{ax^2+bx+c} dx \int \frac{px+q}{\sqrt{ax^2+bx+c}} dx \int \sqrt{a^2 \pm x^2} dx \int \sqrt{x^2 - a^2} dx$ $\int \sqrt{ax^2 + bx + c} dx$ Fundamental theorem of Calculus (without proof). Basic properties of definite integrals and evaluation of definite integrals.	20	9

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	2. Applications of the Integrals Applications in finding the area under simple curves, especially lines, circles/parabolas/ ellipses (in standard form only)	10	6
	3. Differential Equations Definition, order and degree, general and particular solutions of a differential equation. Solution of differential equations by method of separation of variables, solutions of homogeneous differential equations of first order and first degree. Solutions of linear differential equation of the type: $\frac{dy}{dx} + py = q$ where p and q are functions of x or constants $\frac{dx}{dy} + px = q$ where p and q are functions of y or constants	10	5
Unit-III	LINEAR PROGRAMMING	10	5
	Introduction, related terminology such as constraints, objective function, optimization, graphical method of solution for problems in two variables, feasible and infeasible regions (bounded or unbounded), feasible and infeasible solutions, optimal feasible solutions (up to three non-trivial constraints).		

[Note:20 Hours reserved for Remedial classes, Tutorials and Home Assignments.]

Course : Project for Class XII

Full Marks : 20

Projects should be conducted regularly throughout the year. A project notebook is to be prepared by each and every student where all the below mentioned activities should be recorded. There should be a project assessment once a year (once in Class XI and once in Class XII) where the student will be asked to do one of the activities and write it in his/her script provided for the purpose. The student should carry his/her project notebook during the assessment. A viva should also be conducted during the assessment to test the knowledge of the student regarding the project activity.

List of Projects for Class XII

<u>Sl. No.</u>	<u>Topics</u>	<u>Activities</u>
1.	Relations and Function	To verify that the relation R in the set L of all straight lines in a plane, defined by $R = \{(l, m): l \parallel m\}$ is an equivalence relation,
2.	Relations and Function	To demonstrate a function which is one-one but not onto.
3.	Relations and Function	To demonstrate a function which is not one-one but onto.
4.	Differential Calculus	To find analytically the limit of a function $f(x)$ at $x = c$ and also to check the continuity of the function at that point.
5.	Differential Calculus	To verify that amongst all the rectangles of the same perimeter, the square has the maximum area.
6.	Differential Calculus	To understand the concepts of absolute maximum and minimum values of a function in a given closed interval through its graph.
7.	Three-Dimensional Geometry	To explain the concept of octant by three mutually perpendicular planes in space.
8.	Three-Dimensional Geometry	To measure the shortest distance between two skew lines and verify it analytically.
9.	Probability	To explain the computation of conditional probability of a given event A , when event B has already occurred, through an example of throwing a pair of dice.
10.	Linear Inequalities	To verify that a given inequality of the form $ax + by + c < 0$, $a, b > 0$, $c < 0$ represents only one of the two half planes.

Marks division for the Project Assessment

<u>Sl. No.</u>	<u>Item</u>	<u>Marks</u>
1.	Project Notebook	10
2.	Doing and Writing a project during the project assessment	05
3.	Viva	05
	Total	20

Aims and Objectives of the Proposed Curriculum of Philosophy for Higher Secondary level :**General Objectives :**

- To prepare young minds to become constructive thinkers and responsible citizens.
- To impart an education that enables them to be capable of meaningfully contributing to a society that understands the environment and imbibes the ideals of sustainable development
- To impart an education that is not a mere collection of facts but is a tool that helps the student to find his or her status in society and life in general
- To impart an education that helps students in their character-building so that they embody empathetic social concerns and
- To ensure parity with the key ideals of NEP-2020, SEP-2023 as well as the CBSE curriculum
- To generate informative and intuitive knowledge as well as ensure skill development and analytical ability.
- To develop a mind oriented towards research and innovation.
- To develop social and ethical values in students
- To prepare students for the College /University entrance examination for higher education as well as different competitive exams

Subject-specific Objectives :

- Develop a foundational philosophical understanding of key issues relating to epistemology, metaphysics, logic and ethics and take informed actions after identifying the assumptions that frame our thinking and actions
- Develop a foundational understanding of Classical Indian Philosophy as well as Western Philosophy
- Develop interest and basic knowledge of some key modern Indian philosophers
- Develop critical thinking abilities

CLASS - XI

SUBJECT : PHILOSOPHY (PHIL)

SEMESTER – I

FULL MARKS : 40

CONTACT HOURS : 100 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Introduction to Indian Philosophy (24 marks)	1. দর্শন শব্দের অর্থ (The meaning of the term <i>Darśana</i>)	3	2
	2. ভারতীয় দর্শন সম্প্রদায়গুলির শ্রেণীবিভাগঃ সংক্ষেপে (Classification of Indian Philosophical System: in brief)	8	3
	3. জ্ঞানলাভের ছয়টি পদ্ধতি – প্রমাণ সম্পর্কে ভারতীয় তত্ত্বসমূহ (Six ways of knowing – Indian theories of <i>pramāṇa</i>)	10	3
	4. চার্বাক কর্তৃক অনুমান প্রমাণ খণ্ডন (Refutation of inference by <i>Cārvāka</i>)	5	3
	5. ন্যায়সম্মত প্রত্যক্ষের লক্ষণ (<i>Nyāya</i> definition of <i>pratyakṣa</i> or perception)	6	3
	6. নির্বিকল্পক ও সবিকল্পক প্রত্যক্ষের পার্থক্য (Distinction between determinate & indeterminate perception)	6	3
	7. ন্যায় দর্শনের অনুমান সম্পর্কে বক্তব্য (<i>Nyāya</i> theory of inference):		
	• অনুমিতির লক্ষণ (Definition of <i>anumiti</i>)	5	2
	• পক্ষ, সাধ্য ও হেতুর ধারণা (The concepts of <i>pakṣa</i> , <i>sādhya</i> , <i>hetu</i>)	9	2
	• ব্যাপ্তির স্বরূপ (The nature of <i>vyāpti</i>)	6	3

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 2 Introduction to Western Philosophy (16 marks)	1. দর্শনের স্বরূপ ও শাখাসমূহ (Nature and branches of Philosophy)	10	6
	2. জ্ঞানের উৎসসমূহ (Sources of Knowledge) a. অভিজ্ঞতাবাদ -লক, বার্কলে, হিউম (Empiricism -- Locke, Berkeley, Hume)	16	5
	b. বুদ্ধিবাদ – ডেকার্ট, স্পিনোজা, লাইবনিজ (Rationalism -- Descartes, Leibnitz, Spinoza)	16	5

CLASS - XI

SUBJECT : PHILOSOPHY (PHIL)

SEMESTER – II

FULL MARKS : 40

CONTACT HOURS : 80 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Introduction to Logic (24 marks)	1. যুক্তিবিজ্ঞানের প্রকৃতি – অবরোহ এবং আরোহ (The nature of Logic – Deductive and Inductive)	4	3
	2. পদ, বাক্য, বচন, পদের ব্যাপ্যতা, সত্যতা ও বৈধতা (Terms, sentences, propositions, distribution of terms, truth & validity)	6	3
	3. বাক্য থেকে বচনে রূপান্তর (Changing sentences into logical forms)	8	2
	4. বচনের বিরোধিতা (Opposition of proposition)	4	4
	5. অমাধ্যম অনুমান : আবর্তন, বিবর্তন, সমবিবর্তন (Immediate Inferences: Conversion, Obversion, Contraposition)	8	3
	6. নিরপেক্ষ ন্যায় : মূর্তি ও সংস্থান এবং বৈধতা বিচার (Categorical Syllogism: Figure & Mood)& validity testing	10	6
	7. যৌগিক যুক্তিঃ প্রাকল্পিক ও বৈকল্পিক (Compound arguments: Hypothetical and Disjunctive)	6	3

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 2 Introduction to Ethics (16 marks)	1. ভারতীয় নীতিবিদ্যা (Indian Ethics): a. পুরুষার্থ (Purusartha) b. শ্রীমদ্ভগবৎ গীতা – নিকাম কর্মের ধারণা (Srimadbhagavad Gita – The concept of Niskama Karma) c. বৌদ্ধদের অষ্টাঙ্গিক মার্গের ধারণা (Buddhist concept of Eight-fold Margas) d. চার্বাক সুখবাদ (Carvaka Hedonism) 2. পাশ্চাত্য নীতিবিদ্যা (Western Ethics): a. নৈতিক প্রত্যয়সমূহ : নৈতিক, অনৈতিক, নীতি-বহির্ভূত; উদ্দেশ্য ও অভিপ্রায়; ঠিক ও ভুল; ভাল ও মন্দ; ন্যায়বিচার (Ethical concepts: moral, immoral, non-moral; motive and intention; right and wrong; good and bad; justice) b. নৈতিক তত্ত্বসমূহ (Theories of Ethics): Utilitarianism - Bantham and Mill, Kant's categorical imperative	4 4 6 4 8 8	(a+b+c)=6 (d) 2 8

PROJECT

Suggested Topics :

1. Distinction between Ordinary and Extra-ordinary Contact (Laukika and Alaukika Sannikarsa)
2. Svarthanumiti and Parathanumiti
3. Realism and Its Different Forms
4. Idealism – Subjective Idealism of Berkeley

Project Guideline :

Project is a piece of planned work or an activity that is finished over a period and intended to achieve a particular purpose following some steps.

1. Topic or Title of the study
2. Purpose of the study
3. Method of the study
4. Analysis and findings of the study
5. Conclusions

Reference

Preliminary pages should contain-

- i. Acknowledgement
- ii. Certificate
- iii. Content page

CLASS - XII

SUBJECT : PHILOSOPHY (PHIL)

SEMESTER – III

FULL MARKS : 40

CONTACT HOURS : 100 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Introduction to Metaphysics: Indian & Western (20 marks)	1. Western Metaphysics : - Substance: Descartes, Spinoza & Leibnitz – Locke, Berkeley & Hume - Causality: Regularity Theory & Entailment Theory - Mind-Body Problem: Interactionism & Parallelism 2. Indian Metaphysics : - Vedanta - Basic concepts of Vedanta. - Brahman and Maya according to Sankaracharya	16 12 10	1x12=12 1x8=8
Unit – 2 Ethics & Social and Political Philosophy (20 marks)	Practical Ethics - Suicide and Euthanasia - Environmental Ethics : Nature and Scope Social and Political Philosophy Basic Concept: Society, Community, Association, Institutions, State, Law	15 12 20	1x10=10 1x10 =10

CLASS - XII

SUBJECT : PHILOSOPHY (PHIL)

SEMESTER – IV

FULL MARKS : 40

CONTACT HOURS : 80 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1 Western Logic (22 marks)	1. Symbolic Logic		
	• Truth-functions – Variables, Constants, Truth-functional propositions: Negation, Conjunction, Disjunction, Material Implication, Material Equivalence	6	6
	• Boolean Interpretation of categorical propositions	6	2
	• Venn Diagram of categorical propositions	8	2
	• Truth-Table method of determining forms of proposition : Tautology, Self-Contradictory, Contingent	8	6
	2. Inductive Logic		
Unit – 2 Contemporary Indian Thoughts (18 marks)	• Nature of Induction	6	6
	• Causality	6	
	• Mill's Method of Experimental Enquiry	10	
	• Rabindranath Tagore's Humanism	10	6
	• Swami Vivekananda's Karma yoga	10	6
	• Mahatma Gandhi's Ahimsa	10	6

PROJECT

Suggested Topics :

1. Vaishesika Categories (padarthas)
2. Ontological Arguments for the Existence of God (Western Philosophy)
3. The Idea of Democracy
4. Science and Hypothesis

Project Guideline

Project is a piece of planned work or an activity that is finished over a period and intended to achieve a particular purpose following some steps.

1. Topic or Title of the study
2. Purpose of the study
3. Method of the study
4. Analysis and findings of the study
5. Conclusions

Reference

Preliminary pages should contain-

- i. Acknowledgement
- ii. Certificate
- iii. Content page

Preamble :

This Higher Secondary level Physics syllabus has been framed in such a way that it can serve as a bridge between the general physical science course taught at the secondary level and the discipline based curriculum followed at higher education. Effort has been given to make this crucial transition as smooth as possible.

The syllabus is divided into **Units** spread over two year's duration. The Units are logically so arranged that the students can gradually learn the different topics of Physics with higher degree of difficulty. Conventional topics as well as modern concepts have been included in the syllabus so that the students can cope up with the present day needs of the society committed to the use of Physics and technology. Both breadth and depth wise the syllabus is comparable with the national as well as international standards. At the same time emphasis has been given to reduce the syllabus load by eliminating overlapping contents within the subject or with other subjects.

Hope the students will enjoy learning Physics at this stage and will develop passion for the subject.

Outcome :

- The students will learn the basic physics laws and will develop conceptual understanding of the physical processes.
- Students will be able to understand and analysis the real-life events from physics point of view.
- Develop problem solving ability, experimental ability and analytical skills.
- Sufficient conceptual background of physics will be created to make the students competent to meet the requirements of academic and professional courses after the higher secondary stage.
- Interest will be developed for pursuing career in Physics.
- Inculcate scientific aptitude in the learners.

Course Structure :

Class	Semester	Contact Hours				Marks		Credit	
		Theory	Remedial Tutorial	Practical	Total	Theory	Practical	Theory	Practical
11	I	70	10	30 + 20 (50)	110	35	30		
	II	60	10		90	35			
12	III	70	10	30 + 20 (50)	110	35	30		
	IV	60	10		90	35			

CLASS - XI

SUBJECT : PHYSICS (PHYS)

SEMESTER – I

FULL MARKS : 35

CONTACT HOURS : 70 Hours

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
1	PHYSICAL WORLD AND MEASUREMENT Physics - scope and excitement, nature of physical Law, physics technology and society. Need for measurement, units of measurement, length, mass and time measurement, accuracy and precision of measuring instruments, error in measurement, rounding off and order of magnitude, significant figures. Dimensions of physical quantities, dimensional analysis and its applications.	6	3

UNIT No.	TOPICS	CONTACT HOURS	MARKS
2	KINEMATICS SUB TOPIC : MOTION IN A ONE DIMENSION AND TWO DIMENSION Frame of reference (inertial and non-inertial frames). Motion in a straight line, position - time graph, speed and velocity. Elementary concepts of differentiation and integration for describing motion. Uniformly accelerated motion. Graphical analysis: position - time and velocity - time graph and calculation of relevant quantities Relations for uniformly accelerated motion (using graphical and calculus method). SUB TOPIC : MOTION IN A PLANE Scalar and vector quantities, position and displacement vectors, general vectors and their notations, equality of vectors, multiplication of vectors by a real number, addition and subtraction of vectors. Relative velocity. Unit vector, resolution of a vector in a plane - rectangular and non - rectangular components. Scalar and vector product. Motion in a plane. Cases of uniform velocity and uniform acceleration - projectile motion.	24	12
3	LAWS OF MOTION Intuitive concept of force. Inertia, Newton's first law of motion. Momentum and Newton's second law of motion, impulse and concept of impulsive force, Newton's third law of motion and its examples. Law of Conservation of Linear Momentum and its application, concept of free body diagram and its application (simple cases). Equilibrium of concurrent forces. Static and kinetic friction, laws of friction, ideas of coefficient of friction, angle of friction and angle of repose. Rolling friction. Dynamics of uniform circular motion, centripetal force, and example of circular motion (motion of a cyclist, vehicle on level circular road, vehicle on bank road). Concept of centrifugal force.	16	8

UNIT No.	TOPICS	CONTACT HOURS	MARKS
4	WORK , ENERGY AND POWER Work done by a constant force and variable force, kinetic energy. Work - energy theorem, power. Notion of potential energy, potential energy of a spring, conservative forces, conservation of mechanical energy (kinetic and potential energies). Non-conservative forces. Motion in a vertical circle. Elastic and inelastic collisions in one and two dimensions.	10	5
5	MOTION OF SYSTEM OF PARTICLES AND RIGID BODY Centre of mass of a two - particle system. Momentum conservation and motion of centre of mass. Centre of mass of a rigid body (examples of simple geometrical bodies). Moment of a force, torque, angular momentum, conservation of angular momentum with examples. Equilibrium of rigid bodies, rigid body rotation and equations of rotational motion, comparison of linear and rotational motion, moment of inertia, radius of gyration. Values of moment of inertia for simple geometrical objects (no derivation). Statement of parallel and perpendicular axis theorem and their applications.	14	7

FOR SEMESTER – I

- CONTACT HOURS FOR THEORY PART – 70 HOURS
 - CONTACT HOURS FOR PRACTICAL PART – 30 HOURS
 - CONTACT HOURS FOR REMEDIAL CLASSES AND TUTORIAL – 10 HOURS
- SO TOTAL CONTACT HOURS FOR 1st SEMESTER IS 110 HOURS.**

CLASS - XI
SUBJECT : PHYSICS (PHYS)
SEMESTER – II

FULL MARKS : 35

CONTACT HOURS : 60 HOURS

COURSE CODE : THEORY

UNIT No.	TOPICS	CONTACT HOURS	MARKS
6	GRAVITATION The universal law of gravitation. Acceleration due to gravity and its variation with altitude, depth and rotation of earth. Kepler's laws of planetary motion. Gravitational potential energy, Gravitational potential. Escape velocity, Orbital velocity of a satellite. Geostationary satellite.	9	5
7	PROPERTIES OF BULK MATTER SUB TOPIC : MECHANICAL PROPERTIES OF SOLIDS Elastic behavior, stress - strain relationship. Hooke's law, Young's modulus(Y), bulk modulus(K), shear modulus of rigidity(η), Poisson's ratio(σ), relation between Y, K, η, σ (no derivation). Elastic energy for stretched string and extended spring. SUB TOPIC: MECHANICAL PROPERTIES OF FLUIDS Streamline and turbulent flow, Critical velocity. Viscosity, Newton's law of viscosity, Stoke's law, terminal velocity, Reynolds' number. Bernoulli's theorem and its applications. Surface energy and surface tension, angle of contact, excess of pressure, application of surface tension, ideas to drops, bubbles. Capillary rise and fall (no derivation, only analytical treatment). SUB TOPIC : THERMAL PROPERTIES OF MATTER Heat, temperature, thermal expansion of solids, liquids, and gases. Anomalous expansion of water and its effects. Specific heat capacity, principle of calorimetry, change of state, latent heat capacity. Heat transfer: conduction, convection and radiation, black body radiation, Kirchhoff's law, absorptive and emissive powers, thermal conductivity. Newton's law of cooling, Wien's displacement law, Stefan's law and Boltzmann's correction.	17	10

UNIT No.	TOPICS	CONTACT HOURS	MARKS
8	THERMODYNAMICS Thermal equilibrium and definition of temperature, Zeroth law of thermodynamics. Heat, work and internal energy, First law of thermodynamics, C_p and C_v and determination of their relation. Isothermal and Adiabatic processes. P-V diagram, calculation of external work done in different cases. Second law of thermodynamics, reversible and irreversible processes. Heat engine, Calculation of efficiency of Carnot engine only, efficiency of refrigerator (only qualitative idea).	9	5
9	KINETIC THEORY OF GASES Assumptions for the kinetic theory of gases, RMS speed of gas molecules, degrees of freedom. Concept of pressure, kinetic energy and temperature in the light of kinetic theory, ideas of gas laws in the light of kinetic theory of gases. Law of equipartition of energy (statement only) and application regarding of specific heats of the gases. Concept of mean free path, Avogadro's number.	8	5
10	OSCILLATION AND WAVES SUB TOPIC : OSCILLATION Periodic motion-period, frequency, displacement as a function of time, Periodic functions. Simple harmonic motion (S.H.M) and its equation, phase, oscillation of a spring - restoring force and force constant, combination of springs, energy in S.H.M - kinetic and potential energies. Simple pendulum, loaded spring - derivation of expression for time period. Free, damped and forced oscillations, resonance (qualitative ideas only). SUB TOPIC : WAVES Wave Motion: longitudinal and transverse waves, speed of travelling wave motion. Velocity of sound in gaseous medium - Newton's law and Laplace's correction. Displacement relation for a progressive wave. Principle of superposition of waves. Formation of Stationary waves, reflection of waves in string and organ pipes: fundamental mode and harmonics. Formation of beats. Doppler effect of sound.	17	10

FOR SEMESTER II

- CONTACT HOURS FOR THEORY PART – 60 HOURS
 - CONTACT HOURS FOR PRACTICAL PART – 20 HOURS
 - CONTACT HOURS FOR REMEDIAL CLASSES AND TUTORIAL – 10 HOURS
- SO TOTAL CONTACT HOURS FOR 2nd SEMESTER IS 90 HOURS.

CLASS : XI

COURSE CODE: PRACTICAL

SUBJECT: PHYSICS (PHYS)

FULL MARKS: 30

CONTACT HOURS: 50 HOURS (30+ 20)

PRACTICAL WORKS + VIVA (16+4) = 20 MARKS

Given below is a list of required experiments.

In each experiment students are expected to record their observations in tabular form with unit at the column head.

Students should plot an appropriate graph where required, work out the necessary calculation and arrive at the result.

SECTION : A

No. OF EXPT.	TOPICS
1	To measure diameter of a small spherical / diameter and length of a cylindrical body using slide calipers, hence calculate its volume with proper formula
2	To measure the internal diameter and depth of a beaker using slide calipers and hence find its volume.
3	To measure diameter of a given thin wire using screw gauge
4	To determine the volume of an irregular but uniform thickness lamina using screw gauge and graph paper.
5	To determine the radius of curvature of a given spherical surface by a spherometer.
6	Consider equilibrium of three concurrent coplanar forces. To verify the parallelogram Law of forces and to determine weight of a body.
7	To study the force of limiting friction for a wooden block placed on horizontal plane surface and to study its relationship with normal reaction. To determine the coefficient of friction.
8	To study the downward force acting along the inclined plane on a roller due to gravitational pull of earth and to study its relationship with angle of inclination(θ) by plotting graph between force and $\sin \theta$.

SECTION : B

No. OF EXPT.	TOPICS
1	To study the acceleration due to gravity by measuring variation in time period (T) with effective length (L) of a simple pendulum, plot graphs of L - T and L - T ² . Determine the effective length of second pendulum from L - T ² graph
2	To study the force constant of a spring and to study variation in time period of oscillation with mass (m) of a body suspended by a spring. To find the spring constant by plotting a graph of m – T ²
3	To study the force constant of a helical spring by plotting graph between load and extension.
4	To study the variation in volume with pressure for a sample of air at constant temperature by plotting graphs between P – (1 / V) and between P - V
5	To study the fall in temperature of a body (like hot water) with time, by plotting a cooling curve.
6	To study the surface tension of water by capillary rise method.
7	To study the coefficient of viscosity of a given viscous liquid by measuring the terminal velocity of given spherical body.
8	To study the speed of sound of in air at room temperature using resonance column apparatus by two resonance positions.
9	To study the frequency of a tuning fork using resonance column apparatus is by two resonance positions, where the data of the speed of sound in air medium at room temperature will be supplied
10	To study the relationship between frequency and length of a given wire under constant tension using sonometer

The students have to do one practical each from section A and section B in the examination.

PROJECT WORK (MARKS – 7)

All candidates will be required to do one project involving physics related topic/topics of their theory syllabus under the guidance of the Physics teacher.

Candidate should take under any one of the following types of projects:

- 1) Theoretical project
- 2) Working model

Candidates are to prepare a technical report formally written including title, abstract, some theoretical discussion, experiment set up, observations with tables of data collected, graph / chart (if any), analysis and discussion of result, deduction, conclusion etc. The report should be kept simple but neat and elegant.

No extra credit shall be given for typewritten material or decorative cover etc.

Suggested heading of project file for theory based project

Title of the project
Introduction
Contains
Analysis / material aid (graph, Data, Structure, diagram etc)
Conclusions/ comments

Suggested heading of project file for model based projects

Title of the project
Model construction
Principle used, concise project report
Conclusion / comments

PRACTICAL FILE – (MARKS – 3)

Teachers are required to assess the students on the basis of Physics practical file maintained by them during the academic year. Generally students are not expected to write the procedure of the experiments. The students will write the working formula, draw the figure or circuit diagram, collection of data in proper tabular form, results and few ideas of precautions associated with the experiments.

PRACTICAL MARKS SCHEME

EXPERIMENT						SUB TOTAL (16)	VIVA (4)	L.N.B (3)	PROJECT (7)	TOTAL (30)
No. 1(2+5+1=8)			No. 2 (2+5+1=8)							
THEORY	RECORD	RESULT	THEORY	RECORD	RESULT					

CLASS - XII**SUBJECT : PHYSICS (PHYS)****SEMESTER – III****FULL MARKS : 35****CONTACT HOURS : 70 Hours****COURSE CODE : THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
1	ELECTROSTATICS SUB TOPICS: ELECTRIC CHARGES AND FIELDS Electric charges, conservation of charge. Coulomb's law - force between two point charges, forces between multiple charges, superposition principle and continuous uniform distribution of charges. Electric field: electric field due to a point charge, electric field lines. Electric dipole, electric field due to a dipole (at a point on its axis, at a point on its perpendicular bisector, at any point), torque on a dipole in uniform electric field. Electric flux, statement of Gauss's theorem and its application to find the field due to infinitely long straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell (field inside and outside) SUB TOPICS: ELECTROSTATIC POTENTIAL AND CAPACITANCE Electric potential, potential difference, relation between electric field intensity and potential, electric potential : due to a point charge, a dipole and system of point charges, equipotential surface and its properties, electrical potential - energy of a system of two point charges and of electric dipole in electrostatic field. Conductors and insulators, free charges and bound charges inside a conductor. Dielectrics and electric polarization. Capacitors and capacitance, combination of capacitors in series and in parallel. Capacitance of parallel plate capacitors with or without dielectric medium between the plates. Capacitances of solid and hollow spherical capacitors. Energy stored in a capacitor. Example of capacitors in our daily life (only qualitative idea).	18	8

UNIT No.	TOPICS	CONTACT HOURS	MARKS
2	CURRENT ELECTRICITY Electric current, flow of electric charge in a metallic conductor. Drift velocity, mobility and their relation with electric current. Ohm's law, electrical resistance, resistivity and conductivity. V-I characteristics for ohmic resistance, temperature dependence of resistance. Series, parallel and mixed grouping of resistances. Internal resistance of a cell, potential difference and emf of a cell, combination of cells in series and in parallel and in mixed grouping. Parallel combination of two cells of unequal emfs, series combination of n cells of unequal emfs. Kirchhoff's law and simple applications. Wheatstone bridge principle, Metre Bridge principle (end error correction not required). Potentiometer: principle and its applications to measure the potential difference and for comparing emfs of two cells and measurement of internal resistance of a cell.	16	8
3	MAGNETIC EFFECTS OF CURRENT AND MAGNETISM SUB TOPICS: MOVING CHARGE AND MAGNETIC FIELD Concept of magnetic field, Oersted's experiment. Biot - Savart law, calculation of magnetic field for linear and circular current carrying conductors and its simple applications. Ampere's circuital law and its application to infinitely long straight wire and straight solenoid. Force on a moving charge in a uniform magnetic and electric fields - Lorentz force. Motion of a charged particle in a perpendicular magnetic field (Cyclotron frequency). Force on a current carrying conductor in a uniform magnetic field. Force between two parallel current carrying conductors - definition of ampere. Torque experienced by a current carrying loop in uniform magnetic field, moving coil galvanometer -its current sensitivity. Conversion of galvanometer into ammeter and voltmeter. SUB TOPICS: MAGNETISM AND MATTER Current loop as a magnetic dipole and its magnetic dipole moment. Magnetic dipole moment of a revolving electron. Magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis.	16	8

UNIT No.	TOPICS	CONTACT HOURS	MARKS
	Torque on a magnetic dipole (bar magnet) in a uniform magnetic field, magnetic field lines. Magnetic properties of a material: magnetic permeability, magnetic susceptibility, intensity of magnetization, magnetic retentivity and coercivity. Hysteresis: B - H loop and its significance, (only qualitative idea) Earth's magnetic field and magnetic elements. Dia, Para and Ferro - magnetic substances with examples. Electromagnets and factor affecting their strengths.		
4	ELECTROMAGNETIC INDUCTION AND ALTERNATING CURRENT SUB TOPICS : ELECTROMAGNETIC INDUCTION Electromagnetic induction, concept of magnetic flux. Faraday's laws, induced emf and current, Lenz's law, Eddy current. Concept of self and mutual inductance, self-inductance of a solenoid and mutual inductance of two coaxial solenoids (qualitative ideas). SUB TOPICS : ALTERNATING CURRENT Alternating current, peak and RMS values of alternating current/voltage, reactance and impedance. Concept of phasor diagram, only resistive circuit, only inductive circuit, only capacitive circuit, LR circuit, CR circuit, and LCR series circuit, resonance, LC oscillator (qualitative idea only). Power in AC circuit, power factor in AC circuit, wattless current. AC generator and transformer.	15	8
5	ELECTROMAGNETIC WAVES Basic idea of displacement current, electromagnetic waves and their characteristics (qualitative ideas only). Transverse nature of electromagnetic waves. Electromagnetic spectrum (radio waves, infrared, visible, ultraviolet, X-rays, Gamma Rays) including elementary facts about their uses.	5	3

FOR SEMESTER – III

- CONTACT HOURS FOR THEORY PART – 70 HOURS
- CONTACT HOURS FOR PRACTICAL PART – 30 HOURS
- CONTACT HOURS FOR REMEDIAL CLASSES AND TUTORIAL – 10 HOURS

SO TOTAL CONTACT HOURS FOR 3RD SEMESTER IS 110 HOURS.

SUBJECT CODE : PHYSICS (PHYS)**SEMESTER – IV****FULL MARKS : 35****CONTACT HOURS : 60 HOURS****COURSE CODE : THEORY**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
6	OPTICS SUB TOPICS : RAY OPTICS AND OPTICAL INSTRUMENTS Reflection of light, spherical mirrors, mirror formula, refraction of light, total internal reflection and its applications, optical fibers. Refraction at spherical surfaces, lenses, thin lens formula. Lens -Maker's Formula. Displacement method to find the position of image (conjugate points), magnification power of a lens. Combination of thin lenses in contact, combination of lens and mirrors. Refraction and dispersion of light through a Prism. Scattering of light - blue colour of the sky and reddish appearance of the sun at sunrise and sunset. Optical instruments: human eye, image formation and accommodation, correction of eye defects (myopia and hypermetropia) only qualitative Ideas. Microscopes and astronomical telescopes (reflecting and refracting) and their magnifying powers. SUB TOPICS : WAVE OPTICS Wave front and Huygens' principle, reflection and refraction of plane wave at a plane surface using Huygens' principle. Interference: interference of monochromatic light by double slits –Young's experiment, conditions for sustained interference of light - coherent sources, condition of maxima and minima in the term of path difference and phase difference, expression for the fringe width. Diffraction: Fraunhofer's diffraction due to single slit, width of central maximum. Resolving power of microscope and astronomical telescope. Polarization, plane polarized light. Brewster's law, uses of plane polarized light and polaroid.	25	14
7	DUAL NATURE OF RADIATION AND MATTER Dual nature of radiation. Photoelectric effect. Hertz and Lenard's observations, Einstein's Photoelectric equation - particle nature of light. Matter waves - wave nature of particles, de Broglie relation and its simple applications.	7	4

UNIT No.	TOPICS	CONTACT HOURS	MARKS
8	ATOMS AND NUCLEI SUB TOPICS : ATOMS Alpha - particle scattering experiment, Rutherford's model of atom, Bohr model of hydrogen like atoms, energy levels, hydrogen spectrum. Elementary theory of X -ray production, continuous and characteristic X-ray(their origin and properties only),Moseley's law. SUB TOPICS : NUCLEI Composition and size of nucleus, atomic mass, isotope, isobar, isotone. Radioactivity: alpha, beta and gamma particles / rays and their properties, radioactive decay law. Mass - energy relation, mass defect, binding energy per nucleon and its variation with mass number, Nuclear fission and fusion.	10	6
9	ELETRONIC DEVICES Thermal emission of electrons and only the basic concepts of vacuum diode and triodes. Energy bands in solids: conductors, insulators and semiconductors (qualitative idea only) Intrinsic and extrinsic semiconductors, band diagram. P- N junction diode, forward and reverse bias, I - V characteristics of junction diode (nonlinear concept). Special type of diodes: LED, photodiode, solar cell and Zener diode with their characteristics. Zener diode as a voltage regulator. Junction transistor, npn and pnp transistor, transistor action, characteristics of a transistor, transistor as an amplifier (common emitter configuration). Transistor as a switch. Elementary idea of analogue and digital signals. Concepts of decimal and binary numbers. Logic gates : OR,AND,NOT,NAND,NOR (Symbols, input, output Boolean equations, truth table, qualitative explanation). Simple cases of combination of gates.	15	8
10	COMMUNICATION SYSTEM Elements of a communication system (Block diagram only), concepts of amplitude and frequency modulation. Band width of signals (speech, TV and digital data). Band width of transmission medium. Propagation of electromagnetic waves in the atmosphere, sky wave and space wave propagation (qualitative idea only).	5	3

FOR SEMESTER – IV

- CONTACT HOURS FOR THEORY PART – 60 HOURS
- CONTACT HOURS FOR PRACTICAL PART – 20 HOURS
- CONTACT HOURS FOR REMEDIAL CLASSES AND TUTORIAL – 10 HOURS

SO TOTAL CONTACT HOURS FOR 4TH SEMESTER IS 90 HOURS.

CLASS: XII

SUBJECT : PHYSICS (PHYS)

COURSE CODE : PRACTICAL

FULL MARKS : 30

CONTACT HOURS : 50 HOURS (30+ 20)

PRACTICAL WORKS + VIVA (16+4) = 20 MARKS

The experiments for laboratory work are from two groups :

- 1) Experiment based on current electricity and Magnetism
- 2) Experiments based on ray optics and Semiconductor devices.

The main skill required in group 1 is understanding the circuit diagram and making connections, polarity of cells, meters, their ranges, zero error, least count, concept of magnetic lines of force and neutral point.

The main skill required in group 2 is to remove parallax between a needle and a real image of another needle. Basic circuit idea and knowledge of characteristic graphs of Semiconductor devices.

A graph is a convenient and effective way of representing result of measurement so it is an important part of the experiments. (Where it is applicable).

All the calculations should be rounded off up to proper decimal places or significant figures.

SECTION : A

No. OF EXPT.	TOPICS
1	Verify Ohm's law for a given unknown resistance (a 100 cm uniform wire) by plotting a graph of potential difference versus current. Calculate the resistance and hence resistance per cm of the wire from the slope of the graph.
2	Using a Metre Bridge determine the resistance of about 100 cm of uniform wire. If its length and diameter are supplied, calculate the specific resistance of the material of the wire.
3	To verify the law of series combination of resistances, using two resistances (2 ohm / 3 ohm range) in the Metre Bridge circuit.
4	To verify the law of parallel combination of resistances using two resistances (2 ohm / 3 ohm range) in Metre Bridge circuit.
5	To compare emfs of two cells using potentiometer circuit.

6	To determine the internal resistance of a cell using potentiometer circuit.
7	To determine resistance of a galvanometer by half deflection method and to find its figure of merit.
8	To convert a given Galvanometer (of known resistance and figure of merit) into an ammeter and voltmeter of desired range and verify same.
9	Draw the lines of forces for a magnet placing its north pole towards the geographic north. Also determine the position of neutral points on two sides of the magnet.

SECTION : B

No. OF EXPT.	TOPICS
1	To find the value of image distance (v) for different values of object distance (u) of a concave mirror. By drawing $1/v - 1/u$ graph determine the focal length of the mirror.
2	To find the focal length of a convex mirror, using a convex lens.
3	To find the focal length of a convex lens by plotting graphs between $u - v$ and between $1/u - 1/v$
4	To find the focal length of a concave lens using a convex lens.
5	To determine the angle of minimum deviation for a given prism by plotting a graph between angle of incidence and angle of deviation
6	To determine the refractive index of a glass slab using a travelling microscope.
7	To draw I - V characteristic curves of a P - N junction diode in forward and reverse bias.
8	To draw the reverse bias characteristics of Zener diode and to determine its breakdown voltage
9	To study the characteristic of common emitter configuration of NPN or PNP transistor and to find dynamic resistances and amplification factor.

The students have to do one practical each from section A and section B in the examination.

PROJECT WORK (MARKS – 7)

All candidates will be required to do one project involving physics related topic/topics of their theory syllabus under the guidance of the Physics teacher.

Candidate should take under any one of the following types of projects:

1) Theoretical project

2) Working model

Candidates are to prepare a technical report formally written including title, abstract, some theoretical discussion, experiment set up, observations with tables of data collected, graph / chart (if any), analysis and discussion of result, deduction, conclusion etc. The report should be kept simple but neat and elegant.

No extra credit shall be given for typewritten material or decorative cover etc.

Suggested heading of project file for theory based project

Title of the project
Introduction
Contains
Analysis / material aid (graph, Data, Structure, diagram etc)
Conclusions/ comments

Suggested heading of project file for model based projects

Title of the project
Model construction
Principle used, concise project report
Conclusion / comments

PRACTICAL FILE – (MARKS – 3)

Teachers are required to assess the students on the basis of Physics practical file maintained by them during the academic year. Generally students are not expected to write the procedure of the experiments. The students will write the working formula, draw the figure or circuit diagram, collection of data in proper tabular form, results and few ideas of precautions associated with the experiments.

PRACTICAL MARKS SCHEME

EXPERIMENT						SUB TOTAL (16)	VIVA (4)	L.N.B (3)	PROJECT (7)	TOTAL (30)
No. 1(2+5+1=8)			No. 2 (2+5+1=8)							
THEORY	RECORD	RESULT	THEORY	RECORD	RESULT					

Overview and Objective :

Political Science as a discipline deals with understanding the social structures and methods used to understand a government or a State. It also deals with the historical, philosophical, constitutional, and legal foundation of the political society. It further provides scope to identify the political values and ideas, governing institutions and their policy making processes. The subject enhances the ability to address the functions and processes of government and politics both at the National and International level.

The present WBCHSE curriculum of Political Science is framed in a systematic manner to facilitate students to have an understanding of political ideas, ideologies, institutions, policies, processes, and behaviour. The contents enrich student's writing, communication, data analysis skills and also develop knowledge about current and past political events across the world.

CLASS - XI**SUBJECT : POLITICAL SCIENCE (POLS)****SEMESTER – I****TOPIC : 1. POLITICAL SCIENCE : THEORIES AND CONCEPTS****2. INDIAN CONSTITUTION****FULL MARKS : 40****CONTACT HOURS : 100 Hours****COURSE CODE : THEORY****Sub Topic :**

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Nature and Scope of Political Science as a Discipline Definition Nature of Political Science : Scientific approach of the Subject on Governance and Politics Scope of Political Science	15	04
UNIT 2	State: Definition and Characteristics Definition of State Characteristics of State (Please note that "Theories regarding origin of the state" is not included)	10	05

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 3	Citizenship Meaning and Definition of Citizenship Methods of Acquisition of Citizenship Reasons of Loss of Citizenship Citizenship in the Constitution of India	15	07
UNIT 4	Understanding Constitutions: Definition and types Definition of Constitution Types of Constitution- Written, Unwritten, Rigid, Flexible- Merits and Demerits	20	06
UNIT 5	Making and the Philosophy of the Constitution Functions of the Constitution Framing of the Constitution : Demand for the establishment of a Constituent Assembly, Composition of the Assembly, Methods of deliberation, Role of Dr. B.R. Ambedkar Preamble – ideals and significance Incorporating features of the constitutions of the different countries of the world	25	08
UNIT 6	Salient Features of the Constitution of India Largest Constitution Written Constitution Partly Rigid and Partly Flexible Parliamentary System Federalism- Quasi-Federal Structure Fundamental Rights and Duties and Directive Principles Independence of the Judiciary Single Citizenship Anti-Defection Law Reservation Minority Rights Secularism Universal Adult Suffrage Supremacy of the Constitution	15	10
	TOTAL	100 Hours	40

CLASS - XI

SUBJECT : POLITICAL SCIENCE (POLS)

SEMESTER – II

TOPIC : 1. POLITICAL SCIENCE: THEORIES AND CONCEPTS

2. INDIAN CONSTITUTION

FULL MARKS : 40

CONTACT HOURS : 80 Hours

COURSE CODE : THEORY

Sub Topic :

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	Key Concepts of Political Theory : Law – Source and Classification Liberty – Meaning, Classification, Safeguards Equality- Meaning and Nature, Different forms (Social, political, legal, economic, culture, gender equality) Justice – Meaning, Forms Separation of Powers - Definition, Arguments for and against the notion	15	08
UNIT 2	Nation and Nationalism (with emphasis on internationalism) Meaning of Nation and Nationalism Elements of Nationality National Self- Determination Tagore’s views of Nationalism(with emphasis on internationalism)	10	06
UNIT 3	Types of Government Definition and Basic Features – Democracy, Authoritarianism, Totalitarianism	15	08
UNIT 4	Contemporary Indian Political Thought : Selected thinkers Mahatma Gandhi : <i>Satyagraha, Non-Violence and Trusteeship</i> Vivekananda: <i>Socio-Political reforms</i> Netaji Subhas Chandra Bose: <i>Freedom and Nationalism</i> Maulana Abul Kalam Azad: <i>Freedom and Education</i>	15	06

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 5	Fundamental Rights : Meaning and Types Directive Principles Fundamental Duties of Indian Citizens	10	06
UNIT 6	Election and Representation : First Past the Post-System Election Commission of India - Composition and Functions	15	06
	TOTAL	80 Hours	40

[Note : 20 Hours reserved for Remedial classes, Tutorials and Home Assignments.]

CLASS : XI

SUBJECT : POLITICAL SCIENCE (POLS)

COURSE CODE : PROJECT

FULL MARKS : 20

Sub Topic :

1.	Success and Problems of the Application of Universal Adult Franchise (local area studies)
2.	Right to Education and its Actual Implementation- Problem of Dropouts (Local Area Studies)
3.	Role of Media in Democracy
4.	Fundamental Rights in the Indian Constitution and Two Case Studies Indicating Violation of any of these Fundamental Rights

CLASS - XII

SUBJECT : POLITICAL SCIENCE (POLS)

SEMESTER – III

TOPIC : 1. CONTEMPORARY WORLD POLITICS

2. INDIAN GOVERNMENT AND CONTEMPORARY POLITICS

FULL MARKS : 40

CONTACT HOURS : 100 Hours

COURSE CODE : THEORY

Sub Topic :

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	International Relations in the Post-Second World War Period : Cold War Era – Brief Discussion on Bipolarity and End of Bipolarity (outline of ideological conflict with emphasis on NATO and Warsaw Pact) Non-Aligned Movement : Origin, Aims and Objectives; Leaders of NAM and present relevance of NAM	20	08
UNIT 2	International Organizations and Institutions • United Nations- Aims and Principles • General Assembly – Composition and Functions • Security Council -Composition and Functions • Secretary General- Power and Functions • Agencies- UNICEF, UNESCO, WHO • World Bank and International Monetary Fund (IMF)	20	08
UNIT 3	Security in the Contemporary World: • Traditional Security – Internal and External • Non-traditional Security – Human Security, Terrorism, Migration, Poverty, Epidemics Environmental Security (Brief Discussions on the topics)	10	07
UNIT 4	Challenges of Nation-Building • Basic Concept-Partition of India and the Refugee Problem • Integration of Princely States – with brief reference to Jammu and Kashmir, Manipur and Hyderabad • Linguistic Re-organization of States – Background and Brief Discussion (To also include a Table on the Newly Created States in India-Original State, New State, Date of Creation, Criteria of Creation)	20	05

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 5	Political Parties and Party System - Definition and Features of Political Parties - Functions of Political Parties - National and Regional Parties-Definition and Criteria of Formation	20	05
UNIT 6	India's Foreign Policy - Objectives of India's foreign policy with her Neighbours in the contemporary period (Year 2000 onwards)-Sri Lanka, Pakistan, Bangladesh, Nepal, Bhutan, Maldives - India's relations with the Major powers in the post-Cold War period with reference to the USA, Russia and China - Features of India's Nuclear Policy-Brief Analysis	10	07
	TOTAL	100 Hours	40

CLASS - XII

SUBJECT : POLITICAL SCIENCE (POLS) SEMESTER – IV

**TOPIC : 1. CONTEMPORARY WORLD POLITICS
2. INDIAN GOVERNMENT AND CONTEMPORARY POLITICS**

FULL MARKS : 40

CONTACT HOURS : 80 Hours

COURSE CODE : THEORY

Sub Topic :

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 1	International Relations (IR) -Key Concepts and Political Doctrines - Evolution of IR as a Discipline - Definition and Basic Features - Realism, Idealism, Liberalism, Marxism	18	08
UNIT 2	Major Regional Organisations - European Union - SAARC - ASEAN	10	06
UNIT 3	Globalisation - Definition and Features - Economic, Political and Cultural Consequences, - India and Globalisation : An Analysis	08	06

UNIT No.	TOPICS	CONTACT HOURS	MARKS
UNIT 4	Organs of the Indian Government: - Executive in India - Powers and Functions (President, PM, Governor, Chief Minister, Bureaucracy as a Permanent Executive) - Legislature in India - Composition and Functions (Lok Sabha, Rajya Sabha, Vidhan Sabha) ; Speaker of Lok Sabha – (Power and Functions) - Judiciary in India – Composition and Jurisdiction (Supreme Court, High Court, PILs, Consumer Courts)	24	08
UNIT 5	Contemporary Civil Society Movements in India - Civil Society Movements : Concept and Importance Brief Outline of : - Narmada Bachao Andolan - Chipko Movement	08	06
UNIT 6	Constitutional Amendments and Local Self-Government (as per original syllabus) Methods of Amendments (Article 368) 73 rd Amendment -Rural Local Self-Governemnt – 3 tier panchayat system- Composition, Functions and Source of Income 74 th Amendment -Urban Local Self-Government- Municipality and Corporation – Composition and Functions and Source of Income	12	06
	TOTAL	80 Hours	40

[Note:20 Hours reserved for Remedial classes, Tutorials and Home Assignments.]

CLASS : XII

SUBJECT : POLITICAL SCIENCE (POLS)

COURSE CODE : PROJECT

FULL MARKS : 20

Sub Topic :

1.	Survey and observation on the functioning of Gram Panchayat/Panchayat Samiti /Zilla Parishad/ Municipality/ Borough Committee/ Ward Committee etc.
2.	Women's Participation in Local Level Politics
3.	Major Initiatives to Empower Women- <i>Kanyashree (2013)/ Beti Bachao, Beti Padoo (2015)</i>
4.	Indian Foreign Policy- Analyses of India's relation with <u>any one</u> major power/ South Asian Neighbour
5.	Globalisation– Its impact on India

Course Overview

Psychology has been introduced as an elective subject in Higher Secondary school education as a social science aiming wholesome development of the students. The discipline Psychology explores the human mental processes and life experiences shaped by biology and rooted in a definite socio-historic and cultural perspective. The syllabus focuses on teaching fundamental principles, explanations and methods of Psychology. Introduction of the discipline at this academic level aims to create a critical reasoning, rational thinking and deeper understanding of human experiences with an application based orientation in resolving human problems.

Course Objective

- * The aim is to orient the students to learn about human behaviour, life experiences in the socio-cultural context.
- * To develop in the students an attitude to understand and apply the knowledge to solve day to day life problems.
- * To help the students to become socially aware, receptive, self- reflective and to facilitate their personal growth.
- * To enhance the ability of adjustment of the students with the environment.

Career Prospect

- * This discipline helps the student to build a critical and rational thinking about human psychological processes and socio-cultural context. Thus they can engage themselves in academic and research career.
- * The students are also being trained to apply the knowledge in practical field. Thus they can engage themselves later as psychologists for curative, rehabilitation and organizational purposes.

CLASS - XI

SUBJECT : PSYCHOLOGY (PSYC)

SEMESTER – I

FULL MARKS : 35

CONTACT HOURS : 70 Hours

COURSE CODE : THEORY

Sub-topics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1	What is Psychology i) Introduction ii) Definition and scope of Psychology Psychology as a discipline and its evolution Psychology as natural and social science iii) Understanding mind and behaviour iv) Psychology in India v) Branches in Psychology vi) Relevance of psychology in everyday life	10 hours contact hours for sub-topics (1+1 + 3+ 2+2+1)	07
Unit – 2	Methods of enquiry in Psychology i) Introduction ii) Steps in conducting scientific research iii) Methods of Psychology with advantages and limitations, Quantitative and qualitative methods, Observation, . Experimentation, . Correlational research, . Survey research, . Interview . Case study iv) Psychological testing, v) Ethical Issues	10 hours contact hours for sub-topics (1+1+4+2+2)	07

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 3	Bases of Human Behaviour i) Introduction ii) Evolutionary perspective iii) Biological bases of behaviour, Structure and functions of neuron iv) Classification of Nervous system - Structure and functions v) Endocrine system and its relation with behaviour and experience	20 hours contact hours for sub-topics (1+2+3+8+6)	07
Unit – 4	Sensory, Attentional and Perceptual processes i) Introduction ii) Types of senses - Visual, auditory, olfactory, tactile, gustatory Special senses Organic senses - visceral, motor Other senses - kinaesthetic, vestibular iii) Attentional process Determinants of attention, shift, fluctuation, oscillation, selective and sustained attention iv) Perception Perceptual Process, Principles of perception, . Perception of space, depth, distance, motion, . monocular and binocular cues v) Illusion, types of illusions vi) Social and cultural influences on perception	20 hours contact hours for sub-topics (1+6+2+7+2+2)	07
Unit – 5	Thinking i) Introduction ii) Nature of thinking Building blocks of thought iii) Culture and thinking iv) The process of thinking Problem solving, Reasoning, Decision making v) Nature and process of creative thinking vi) Thought and language	10 hours contact hours for sub-topics (1+1+1+3+2+2)	07

CLASS - XI

SUBJECT : PSYCHOLOGY (PSYC)

SEMESTER – II

FULL MARKS : 35

CONTACT HOURS : 70 Hours

COURSE CODE : THEORY

Sub-topics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 6	Human Development i) Introduction ii) Meaning of development Concept of growth, maturation and evolution iii) Lifespan perspective on development iv) Factors influencing development(Heredity & Environment) v) Overview of developmental stages - . Prenatal development, . Infancy(Motor, Cognitive, Socio-emotional), . Childhood, (Motor, Cognitive, Socio-emotional) . Adolescence and its challenges, (Physical, Moral, Cognitive) vi) Adulthood and Old age	15 hours contact hours for sub-topics (1+2+1+3+6+2)	Consult Question Pattern to be published by the Council
Unit – 7	Learning i) Introduction ii) Nature of Learning iii) Paradigms of learning . Classical conditioning - Pavlov . Instrumental conditioning - Skinner . Observational learning . Cognitive learning . Verbal learning Skill learning iv) Transfer of learning v) Factors facilitating learning vi) Learning disabilities	20 hours contact hours for sub-topics (1+2+12+2+2+1)	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 8	Human Memory i) Introduction ii) Nature of memory iii) Information processing approach: the stage model iv) Memory systems: Sensory, short term and long term memories v) Types of long term memory - declarative, procedural, episodic and semantic vi) Levels of processing vii) Forgetting - concept, nature, causes viii) Theories of forgetting - .Trace, . decay, . Interference and retrieval failure ix) Enhancing memory	15 hours contact hours for sub-topics (1+1+2+2+2+1+2+3+1)	
Unit – 9	Motivation i) Introduction ii) Nature of motivation iii) Types of motives - Biological, psychological, social iv) Maslow's hierarchy of needs v) Extrinsic and intrinsic motivation – Differences with examples	10 hours contact hours for sub-topics (1+1+3+3+2)	
Unit– 10	Emotion i) Introduction ii) Nature of emotions iii) Expressions of emotions - Emotional expression and labelling Culture and emotion iv) Managing negative emotions vi) Enhancing positive emotions	10 hours contact hours for sub-topics (1+1+4+2+2)	

CLASS - XI

SUBJECT : PSYCHOLOGY (PSYC)

SEMESTER – II

FULL MARKS : 30

CONTACT HOURS : 40 Hours

COURSE CODE : PRACTICAL

Sub-topics

1	Compare the rate of perceptual reversibility between two respondents using visual stimulus (Rubin's Cross/human profile-Flower vase) and interpret the result.	7 hours
2	Compare the responses of three respondents to optical illusory figures (Muller- Lyer) and interpret the result	7 hours
3	Compare the rate of learning of two respondents varying modality (auditory & visuals) and interpret the result	8 hours
4	Determine the effect of meaning on memorization capacity of the respondent	8 hours
	PROJECT 1) Biological bases of Behaviour (Neuron, spinal cord, Brain and endocrine glands) - File work on above mentioned topic explaining structure and functions with diagrams.	10 hours

[**Note** : 20 Hours reserved for Remedial classes, Tutorials and Home Assignments.]

Distribution of Marks in Practical (Total =30)

Topics	Marks
Practical Problem	10
Project	10
Practical File	5
Viva voce	5

Distribution of Marks in Theory (Total 35+35=70)**Semester – I**

Type of Question	Allotted Marks	Marks Pattern
MCQ	35	1 x 35
		Total = 35

Semester – II

Types of Question	Allotted Marks	Marks Pattern
SAQ	6	2 x 3
LAQ	9	3 x 3
LAQ	8	4 x 2
LAQ	12	6 X 2
		Total = 35

CLASS - XII

SUBJECT : PSYCHOLOGY (PSYC)

SEMESTER – III

FULL MARKS : 35

CONTACT HOURS : 70 Hours

COURSE CODE : THEORY

Sub-topics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 1	Variations in Psychological attributes i) Introduction ii) Individual differences in human functioning iii) Assessment of psychological attributes - different domains- intelligence, aptitude, interest, personality, values iv) Methods of assessments - Psychological test, interview, observation, case study, self report v) Intelligence - Concept vi) Theories of intelligence - .Spearman's two-factor theory, . Guilford's structure of intellect model, . Gardner's Multiple intelligences, . . Sternberg's Triarchic theory of intelligence, . Pass Model by Das vii) . Individual differences in intelligence with special reference to normal probability curve - . Concept of IQ and categories of intellectual ability with special reference to Weschler. viii) Classification of Intelligence test ix) . Emotional intelligence - Concept by Salovey and Mayer, Goleman x) Creativity	20 hours contact hours for sub-topics (1+1+3+4+2+4+2+2+1)	09

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 2	Social Influence and Group Processes i) Introduction ii) Group - Concept and characteristics iii) Types of group iv) Why do people join groups v) Formation of group - Factors and stages of group formation vi) Influence of group on individual behaviour - . Social Facilitation, .social Loafing, . Group Polarization vii) Process of social influences - .Conformity, . Compliance, .Obedience	15 hours contact hours for sub-topics (1+2+2+1+2+4+3)	09
Unit – 3	Identity Development i) Introduction ii) Identity - concept iii) Identity development - Erikson's theory of psycho-social development. iv) Identity crisis - Theory of Erikson and Marcia v) Factors influencing identity development	10 hours contact hours for sub-topics (1+1 + 3+ 2+2+1)	08
Unit – 4	Meeting life challenges i) Introduction ii) Nature, types and sources of Stress iii) Effect of Stress . General adaptation syndrome . Stress and immune system . Lifestyle iv) Coping with stress - . Stress management techniques v) Promoting positive health and wellbeing - . Life skills, . positive health	25 hours contact hours for sub topics (2+6+6+2+4)	09

CLASS - XII

SUBJECT : PSYCHOLOGY (PSYC)

SEMESTER – IV

FULL MARKS : 35

CONTACT HOURS : 70 Hours

COURSE CODE : THEORY

Sub Topics

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 5	Attitude and Social Cognition i) Introduction ii) Explaining social behaviour - .Social Cognition, . Schemas and . Different pro-social behaviour iii) Nature and components of attitude: A-B-C components iv) Attitude formation and change - . Attitude formation, .Attitude change, . Attitude -behaviour relationship v) Prejudice, Stereotype and Discrimination vi) Strategies for handling prejudice	12 hours contact hours for sub-topics (1+3+2+4+3+1)	Consult Question Pattern to be published by the Council
Unit – 6	Self and Personality i) Introduction ii) Self and Personality - .concepts of Self - Personal and Social self iii) Cognitive and Behavioural aspects of self - . Self concept, . Self esteem, .Self efficacy, . Self regulation iv) Culture and Self v) Concept and characteristics of personality vi) Major approaches to the study of personality - . Type approaches, .Trait approaches, . Psychodynamic approach, . Behavioural approach, . Humanistic approach vii) Assessment of Personality - .Self report Measures, Projective Techniques, Behavioural Analysis	15 hours contact hours for sub-topics (1+2+3+1+2+4+2)	

UNIT No.	TOPICS	CONTACT HOURS	MARKS
Unit – 7	Psychological Disorders i) Introduction ii) Concept of Abnormality and Psychological Disorder . Classification of Psychological Disorders (ICD 11 & DSM V) iii) Factors underlying abnormal behaviour - . Biological, . Psychological and . Socio-economic iv) Major Psychological Disorders - . Anxiety Disorders, . Obsessive -compulsive and Related Disorders, . Somatic symptoms and related disorders, . Dissociative disorders, . Depressive disorders, . Bipolar and related disorders, . Schizophrenia spectrum and other psychotic disorders, . Neuro-developmental disorders, . Disruptive, Impulse control and Conduct disorders . Substance related and Addictive disorders	25 hours contact hours for sub-topics (1+3+6+15)	
Unit – 8	Therapeutic approaches i) Introduction ii) Nature and process of Psychotherapy iii) Therapeutic Relationship iv) Types of Therapies - . Psychodynamic Therapy . Behaviour Therapy . Cognitive Therapy . Humanistic - Existential Therapy . Alternative Therapies - Kriya Yoga v) Ethics in Psychotherapy vi) Rehabilitation	18 hours contact hours for sub-topics (1+2 + 2+ 10+2+1)	

CLASS - XII

SUBJECT : PSYCHOLOGY (PSYC)

SEMESTER – IV

FULL MARKS : 30

CONTACT HOURS : 40 Hours

COURSE CODE : PRACTICAL

Sub-topic

1	Students will administer and interpret any one Intelligence Test from the following. Intelligence Tests -Alexander Pass along test/ Dearborn Form Board Test	10 hours
2	Students will administer and interpret Academic Stress Scale [Educational Stress Scale for Adolescents -(ESSA) by Sun, Dunne, Xiang-yu Hou and Xu] This test should be administered on two participants to compare and interpret the results.	10 hours
3	Students will administer and interpret any one Personality Test from the following. Personality tests - Kundu Neurotic Personality Inventory-KNPI/ Kundu Introversion Extroversion Inventory -KIEI	10 hours
	PROJECT Development of a comparative case study using General health questionnaire of male and female participants.	10 hours

[**Note** : 20 Hours reserved for Remedial classes, Tutorials and Home Assignments.]

Distribution of Marks in Practical (Total= 30)

Topics	Marks
Practical Problem	10
Project	10
Practical File	5
Viva Voce	5

Distribution of Marks in Theory (Total 35+35=70)**Semester – III**

Type of Question	Allotted Marks	Marks Pattern
MCQ	35	1 x 35
		Total = 35

Semester – IV

Types of Question	Allotted Marks	Marks Pattern
SAQ	6	2 x 3
LAQ	9	3 x 3
LAQ	8	4 x 2
LAQ	12	6 X 2
		Total = 35

COURSE OVERVIEW:

Sanskrit has been an enlightening language since the dawn of Indian intellectual endeavour. It has the indomitable power to disseminate the pearls of wisdom along with its adorable potential to impel human spirit to undertake a rapturous journey to the abode of truth. In the contemporary times when the Artificial Intelligence (AI) is gaining a commanding influence on earth and almost taking over the human intelligence, this wonderful language has the ability to strengthen human intelligence and raise to a higher degree of perfection. Moreover, the spirit of Indian knowledge systems and culture is enshrined in this divine language. The vastness of Sanskrit literature, in innumerable fields of diversities, at once awakens awe and wonder. It encompasses varied disciplines like literature, grammar, linguistics, philosophy, mathematics, astronomy, *yoga*, *āyurveda*, law and ethics, polity, economics, sociology, fine arts, natural science and technology. Sanskrit is not merely a carrier of thought but is the cradle of profound ideas, throbbing emotions and ethical values permeated in Indian culture. Therefore, cultivation of this language is not only an intellectual pursuit but is a dynamic-force in building a self-reliant, self-enabled, prosperous and awakened nation.

The new syllabus of the Sanskrit (SNSK) course, prepared for the students of Higher Secondary classes under the umbrella of the West Bengal Council of Higher Secondary Education, is offering a scope for acquiring a profound knowledge about several genres of Sanskrit literary heritage.

The new syllabus of the Sanskrit (SNSK) course has introduced texts and/or narratives from the *Rāmāyaṇa*, the *Mahābhārata* (including *Śrīmadbhagavadgītā*), the Buddhist *Avadānaśataka*, the *Carakasamhitā*, the works of Kālidāsa, Bhāsa, Bhavabhūti and two authors of modern Sanskrit literature also. Sanskrit grammar is always an essential part of Sanskrit learning. Therefore, the new syllabus also includes a few topics of Sanskrit grammar. A student of Sanskrit requires to develop a holistic idea about the Sanskrit literary heritage. So, a basic outline of the history of Sanskrit literature will be taught in this course.

A student having a basic knowledge of any of the Indian vernacular languages may opt for this course.

It aims to encourage the students of Sanskrit to continue with their higher studies and researches on several domains of Indian knowledge system.

OBJECTIVES OF THE COURSE:

The new syllabus of the Sanskrit (SNSK) course, prepared for the students of Higher Secondary classes under the umbrella of the West Bengal Council of Higher Secondary Education, aims to offer an intensive knowledge about several genres of Sanskrit literary heritage.

The following are the main objectives of the course:

- Develop the skill of reading and comprehending Sanskrit texts.
- Make the students acquainted with the fundamentals of Sanskrit grammar, which would help them analysing the grammatical applications of Sanskrit texts.
- Introduce the salient features of Vedic, Epic, Purāṇic and Classical Sanskrit literatures including the texts on Āyurveda and other scientific and technical literature through the lessons on History of Sanskrit Literature.
- Offer a general idea about the socio-cultural, socio-political and socio-economic conditions of the ancient, mediaeval and even contemporary India to boost their management skill and to inspire them to choose the areas of interdisciplinary studies and researches in future.
- Introduce the basics of health-awareness, human values and self-management through certain texts.
- Encourage the students of Sanskrit to continue with their higher studies and researches on several domains of Indian knowledge system.
- Inspire the students to utilise their acquired knowledge of ethics, values and self-management for building a self-reliant, self-enabled, prosperous and awakened nation.

CLASS - XI**SUBJECT : SANSKRIT (SNSK)****SEMESTER - I****FULL MARKS : 40****CONTACT HOURS : 100 Hours****COURSE CODE: THEORY**

UNIT NO.	TOPICS	CONTACT HOURS	MARKS [QUESTION TYPE: MCQ]
Part I	संस्कृत साहित्य (Sanskrit Literature)	40	15
Unit – I	गद्य (Prose): उपमन्युकथा (बैयासिक-महाभारतेर आदिपर्वेर् निर्वाचित अंश अवलम्बने) [Upamanyukathā (narrative based on select portion from Ādiparvan of Vaiyāsika-Mahābhāratam)]		05 [1 × 5=5]
Unit – II	पद्य (Poetry/Verse): वर्षावर्णनम् (वाल्मीकि-रामायणेर् किष्किन्ध्याकाण्डेर् अंशविशेष) [Varṣāvarṇanam (select portion from Kiṣkindhyākāṇḍam of Vālmīki- Rāmāyaṇam)]		05 [1 × 5=5]
Unit – III	दृश्यकव्य (Drama): कृपणः कपाली (श्रीजीव न्यायतीर्थ-कृत 'चिपिटकचर्चणम्'-एर् निर्वाचित अंश) [Kṛpaṇaḥ Kapālī (select portion from Cipīṭakacarvaṇam of Srijeeb Nyayatirtha)]		05 [1 × 5=5]
Part II	संस्कृत व्याकरणं उ ससंस्कृत साहित्येर् इतिहास (Sanskrit Grammar & History of Sanskrit Literature)	60 (40+20)	25
Unit – IV	व्याकरण (Grammar):	40	15
	५. सङ्कि [Sandhi] - ➤ स्वरसङ्कि [Vowel sandhi] - सवर्णदीर्घ, गुण उ वृद्धि (savarnadīrgha, guṇa and vr̥ddhi) ➤ व्यञ्जनसङ्कि [Consonant sandhi] - शूढ, श्रूढ उ जश्ढ (ścutva, śrutva and jaśtva)		1 × 3=3 1 × 3=3

	২. শব্দরূপ [Declension] - ➤ অজন্ত শব্দ - বালক, লতা, ফল, কবি, মতি, বারি, নদী [এবং এইগুলির সমতুল অন্যান্য শব্দ] (Words ending with vowel: <i>bālaka, latā, phala, kavi, mati, vāri, nadī</i> and similar words) ➤ সংখ্যাবাচক শব্দ - এক, দ্বি (তিন লিঙ্গে) (Numerals: <i>eka, dvi</i> in three genders) ➤ সর্বনাম শব্দ - অস্মদ, যুস্মদ (Pronouns: <i>asmad, yuṣmad</i>)		[1 × 3=3]
	৩. ধাতুরূপ [Conjugation] - লট্, লঙ্ ও লৃট্- এই তিন লকারে (in three tenses: present, past, future) ➤ পরস্মৈপদী - √ভূ, √গম, √কৃ, √পূজ্ (<i>Parasmaipadī</i> - √<i>bhū</i>, √<i>gam</i>, √<i>kr</i>, √<i>pūj</i>)		[1 × 3=3]
	৪. প্রত্যয় [Suffix] - ➤ ক্ত, ক্তবতু, ক্ত্বা, ল্যপ্, ক্তিন্ (<i>ktā, ktavatu, ktvā, lyap, ktin</i>)		[1 × 3=3]
Unit – V	বৈদিক, জাতীয় মহাকাব্য ও লৌকিক সংস্কৃত সাহিত্যের ইতিহাস (History of Vedic, Epic & Classical Sanskrit Literature):	20	10
	১. বৈদিক সাহিত্য (সংক্ষিপ্ত পরিচয়) [Brief Introduction to Vedic Literature] - ➤ ঋগ্বেদ, সামবেদ (<i>Rgveda, Sāmaveda</i>)		[1 × 4=4]
	২. বাল্মীকি-রামায়ণ (রচয়িতা ও বিষয়বস্তুর সংক্ষিপ্ত পরিচয়) [Brief Introduction to the author and subject-matter of <i>Valmīki-Rāmāyaṇa</i>]		[1 × 3=3]
	৩. সংস্কৃত গল্প সাহিত্য (রচয়িতা ও বিষয়বস্তুর সংক্ষিপ্ত পরিচয়) [Brief Introduction to the author and subject-matter of Sanskrit Narrative Literature] -		[1 × 3=3]
	➤ পঞ্চতন্ত্র, হিতোপদেশ, কথাসরিৎসাগর (<i>Pañcatantra, Hitopadeśa, Kathāsaritsāgara</i>)		

CLASS - XI**SUBJECT : SANSKRIT (SNSK)****SEMESTER - II****FULL MARKS : 40****CONTACT HOURS : 80 HOURS****COURSE CODE: THEORY**

UNIT NO.	TOPICS	CONTACT HOURS	MARKS [QUESTION TYPE: SAQ INCLUDING DQ]
Part I	संस्कृत साहित्य (Sanskrit Literature)	40	20
Unit – I	गद्य (Prose): प्रतिज्ञासाधनम् (पण्डित अम्बिकादत्त व्यास रचित 'शिवराजविजयम्' এর অংশবিশেষ) [Pratijñāsāadhanam (select portion from Śivarājavijayam of Ambikadatta Vyasa)]	40 (20+20)	20 - SAQ: 06 Marks 3 questions out of 4 (covering all the given texts of this part) each containing 2 marks to be answered in Sanskrit [সংক্ষিপ্ত উত্তরের প্রশ্ন: চারটি (এই অংশে প্রদত্ত সকল গ্রন্থ অবলম্বনে) ২ নং-এর প্রশ্নের মধ্যে যেকোনো তিনটির সংস্কৃত-তে উত্তর করতে হবে]. [2×3=6] - SAQ: 04 Marks 2 questions out of 3 (1+1+1) each containing 2 marks to be answered in Sanskrit/Bengali/ English/ Hindi [সংক্ষিপ্ত উত্তরের প্রশ্ন: তিনটি (১+১+১) ২ নং-এর প্রশ্নের মধ্যে যেকোনো দুটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [2×2=4] - DQ: 10 Marks 2 questions out of 3 (1+1+1) each containing 5 marks to be answered in Sanskrit/Bengali/ English/ Hindi [ব্যখ্যামূলক উত্তরের প্রশ্ন: তিনটি (১+১+১) ৫ নং-এর প্রশ্নের মধ্যে যেকোনো দুটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [5×2=10]
Unit – II	পদ্য (Poetry/Verse): ঋতুচর্যা ('চরকসংহিতা'র নির্বাচিত অংশ) [Rtucaryā (select portion from Carakasamhitā)]		
Unit – III	দৃশ্যকাব্য (Drama): দানবীরঃ কর্ণঃ (মহাকবি-ভাস-রচিত 'কর্ণভারম্' রূপকের অংশ) [Dānavīrah Karṇaḥ (select portion from Bhāsa's Karṇabhāram)]		
Part II	संस्कृत व्याकरण ও সংস্কৃত সাহিত্যের ইতিহাস (Sanskrit Grammar & History of Sanskrit Literature)	40 (20+20)	20
Unit – IV	व्याकरण (Grammar): ১. সন্ধি [Sandhi] -	20	10 SAQ: 10 Marks

	➤ স্বরসন্ধি [Vowel sandhi] - যণ, অয়াদি (<i>yan, ay etcetera</i>) ➤ ব্যঞ্জনসন্ধি [Consonant sandhi] - অনুস্বার, পরসবর্ণ (<i>anusvāra, parasavarṇa</i>) ➤ বিসর্গসন্ধি [Visarga-sandhi] - উত্, রুত্, লোপ, বিসর্গস্থানে স্, শ্, ষ্ (<i>utva, rutva, deletion, and replacement of visarga by s, ś, ṣ</i>) ২. শব্দরূপ [Declension] - ➤ অজন্ত শব্দ - শিশু, ধেনু, মধু, মাতৃ, পিতৃ ও সমতুল অন্যান্য শব্দ (Words ending with vowel: <i>śiśu, dhenu, madhu, mātṛ, pitṛ</i> and similar words) ➤ হলন্ত শব্দ - রাজন্, ভবৎ, কর্মন্, পথিন্, গুণিন্, দিশ্, আত্মন্ [এবং এইগুলির সমতুল অন্যান্য শব্দ] (Words ending with consonant: <i>rājan, bhavat, karman, pathin, guṇin, diś, ātman</i> and similar words) ➤ সংখ্যাচক শব্দ - ত্রি, চতুঃ (তিন লিঙ্গে) (Numerals: <i>tri, catuḥ</i> in three genders) ➤ সর্বনাম শব্দ - সর্ব, তদ, ইদম্, কিম্ (তিন লিঙ্গে) (Pronouns: <i>sarva, tat, idam, kim</i> in three genders) ৩. ধাতুরূপ [Conjugation] - লট্, লঙ্ ও লৃট্- এই তিন লকারে (in three tenses: present, past, future) ➤ পরস্মৈপদী - √দা, √স্থা, √শ্র্, √দৃশ্ (<i>Parasmaipadī - √dā, √sthā, √śru, √dṛś</i>) ৪. প্রত্যয় [Suffix] - ➤ তুমুন্, শতৃ, শানচ্, কৃত্য প্রত্যয় (<i>tumun, śatr, śānac, kṛtya</i> suffixes)		5 questions out of 6/7 (covering all the topics of this unit) each containing 2 marks to be answered as per given instructions [সংক্ষিপ্ত উত্তরের প্রশ্ন: (এই অংশের সকল বিষয় অবলম্বনে প্রদত্ত) ছয়/সাতটি ২ নং-এর প্রশ্নের মধ্যে যেকোনো পাঁচটির প্রদত্ত নির্দেশানুসারে উত্তর করতে হবে]. [2×5=10]
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Unit – V	বৈদিক, জাতীয় মহাকাব্য ও লৌকিক সংস্কৃত সাহিত্যের ইতিহাস (History of Vedic, Epic & Classical Sanskrit Literature): ১. বৈদিক সাহিত্য (সংক্ষিপ্ত পরিচয়) [Brief Introduction to Vedic Literature] - ➤ যজুর্বেদ, অথর্ববেদ ও বেদাঙ্গ [Yajurveda, Atharvaveda and Vedāṅgas] ২. বৈয়াসিক-মহাভারত (রচয়িতা ও বিষয়বস্তুর সংক্ষিপ্ত পরিচয়) [Brief Introduction to the author and subject-matter of <i>Vaiyāsika-Mahābhārata</i>] ৩. সংস্কৃত গীতিকাব্য (রচয়িতা ও বিষয়বস্তুর সংক্ষিপ্ত পরিচয়) [Brief Introduction to the authors and subject-matters of Sanskrit lyrical poetries] - ➤ গীতগোবিন্দ ও মেঘদূত (<i>Gītāgovinda</i> and <i>Meghadūta</i>)	20	10 - SAQ: 6 Marks 3 questions out of 4 (covering all the topics of this unit) each containing 2 marks to be answered in Sanskrit/Bengali/ English/ Hindi [সংক্ষিপ্ত উত্তরের প্রশ্ন: (এই অংশের সকল বিষয় অবলম্বনে প্রদত্ত) চারটি ২ নং-এর প্রশ্নের মধ্যে যেকোনো তিনটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [2×3=6] - DQ: 4 Marks 1 question out of 2 each containing 4 marks to be answered in Sanskrit/Bengali/ English/ Hindi [ব্যাখ্যামূলক উত্তরের প্রশ্ন: দুটি ৪ নং-এর প্রশ্নের মধ্যে যেকোনো একটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [4×1=4]
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CLASS: XI

SUBJECT: SANSKRIT (SNSK)

COURSE CODE: PRACTICAL/PROJECT

FULL MARKS : 20

CONTACT HOURS : 20

	TOPICS (PROJECT)	CONTACT HOURS	MARKS
১.	প্রাচীন ভারতের জনপদ, পাহাড়, নদ-নদী থেকে একটি বিষয় [Provinces, Mountains and Rivers of ancient India – any one topic]	20	20 Pictorial and Informative Project work in Sanskrit language with Devanāgarī script on any two topics [ছবি ও তথ্য সহ সংস্কৃত ভাষায় দেবনাগরীলিপিতে যে কোনো দুটি বিষয় অবলম্বনে প্রকল্প-কর্ম] [10 + 10 = 20]
২.	প্রথম ও দ্বিতীয় সেমিস্টার-এর পাঠ্যান্তর্গত বিষয়সমূহ (সংস্কৃত সাহিত্য ও সাহিত্যের ইতিহাস) থেকে একটি বিষয় [Any one topic from the syllabus of Semester I or II (Sanskrit Literature and History of Sanskrit Literature)]		
৩.	অনুচ্ছেদ রচনা (প্রকৃতি-বিষয়ক, জীবনী-বিষয়ক, নীতি/শিক্ষা-বিষয়ক) – একটি [Paragraph writing (Nature, Biography, Ethics/Morality/Education) – any one topic]		

CLASS - XII

SUBJECT : SANSKRIT (SNSK)

SEMESTER – III

FULL MARKS : 40

CONTACT HOURS : 100 Hours

COURSECODE : THEORY

UNIT NO.	TOPICS	CONTACT HOURS	MARKS [QUESTION TYPE : MCQ]
Part I	संस्कृत साहित्य (Sanskrit Literature)	40	15
Unit - I	गद्य (Prose) : राजवाहनचरितम् (दशकुमारचरितम्) –एर अंशविशेष) [Rajvahanacharitam (Select portion of Daskumaracharitam)]		05 [1 × 5 = 5]
Unit - II	पद्य (Poetry/Verse) : योगः कर्मसु कौशलम् ('श्रीमद्भगवद्गीता' –र निर्वाचित अंश [Yogah karmasu kaushalam (select portion from Srimadbhagavadgita)]		05 [1 × 5 = 5]
Unit - III	दृश्याकाव्य (Drama) : 'दुर्वाससः अभिषापः' (महाकवि कालिदास रचित 'अभिज्ञान-शकुन्तलम्' नाटकेर अंशविशेष) [Durvasasah Avisapah (select portion from Mahakabi Kalidasa's Abhijnana- Śākuntalam)]		05 [1 × 5 = 5]
Part II	संस्कृत व्याकरण ओ संस्कृत साहित्येर इतिहास (Sanskrit Grammar & History of Sanskrit Literature)	60 (40 + 20)	25
Unit - IV	व्याकरण (Grammar) : १. प्रत्यय [Suffix] – ➤ तद्धित – अण्, मतुप, तरप्, झयसुन्, तमप्, इष्ठन् (taddhita: an, matup, tarap, iyasun, tampap, isthan) ➤ नामधातु-प्रत्यय : काम्याच्, क्यच्, क्यङ् (suffix for nāmadhāni: kāmyac, kyac, kyan)	40	15 [1 × 5 = 5]

	১. কারক-বিভক্তি ও সমাস [Case-ending and Compound]- ➤ কারক-বিভক্তি - কর্তৃ, কর্ম, করণ (Case-endings: <i>kartr</i>, <i>karma</i>, <i>karaṇa</i>) ➤ সমাস - অব্যয়ীভাব, তৎপুরুষ (Compounds: <i>avyayībhāva</i>, <i>tatpuruṣa</i>)		[1 × 5=5] [1 × 5=5]
Unit – V	পৌরাণিক ও লৌকিক সংস্কৃত সাহিত্যের ইতিহাস (History of Purāṇic and Classical Sanskrit Literature):	20	10
	১. পুরাণের সংক্ষিপ্ত পরিচয় [Brief Introduction to the <i>Purāṇas</i>]		[1 × 3=3]
	২. ভাস, কালিদাস ও ভবভূতির সাহিত্যকৃতির সংক্ষিপ্ত পরিচয় [Brief Introduction to the literary works of Bhāsa, Kālidāsa and Bhavabhūti]		[1 × 4=4]
	৩. আর্যভট্ট ও ব্রহ্মগুপ্ত [গ্রন্থকার ও গ্রন্থের সংক্ষিপ্ত পরিচয়] [Brief Introduction to Āryabhaṭa and Brahmagupta and their works]		[1 × 3=3]

CLASS - XII

SUBJECT : SANSKRIT (SNSK)

SEMESTER - IV

FULL MARKS : 40

CONTACT HOURS : 80 Hours

COURSE CODE: THEORY

UNIT NO.	TOPICS	CONTACT HOURS	MARKS [QUESTION TYPE: SAQ INCLUDING DQ]
Part I	संस्कृत साहित्य (Sanskrit Literature)	40	20
Unit – I	गद्य (Prose): हासविद्याकथा (कवि-विद्यापति-कृत 'पुरुषपरिक्षा'র অংশবিশেষ) [Hāsavidyākathā (select portion from Puruṣaparīkṣā of Vidyāpati)]		20 - SAQ: 06 Marks 3 questions out of 4 (covering all the given texts of this part) each containing 2 marks to be answered in Sanskrit [সংক্ষিপ্ত উত্তরের প্রশ্ন: চারটি (এই অংশে প্রদত্ত সকল গ্রন্থ অবলম্বনে) ২ নং-এর প্রশ্নের মধ্যে যেকোনো তিনটির সংস্কৃত-তে উত্তর করতে হবে]. [2×3=6] - SAQ: 04 Marks 2 questions out of 3 (1+1+1) each containing 2 marks to be answered in Sanskrit/Bengali/ English/ Hindi [সংক্ষিপ্ত উত্তরের প্রশ্ন: তিনটি (১+১+১) ২ নং-এর প্রশ্নের মধ্যে যেকোনো দুটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [2×2=4] - DQ: 10 Marks 2 questions out of 3 (1+1+1) each containing 5 marks to be answered in Sanskrit/Bengali/ English/ Hindi [ব্যাখ্যামূলক উত্তরের প্রশ্ন: তিনটি (১+১+১) ৫ নং-এর প্রশ্নের মধ্যে যেকোনো দুটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [5×2=10]
Unit – II	পদ্য (Verse): বনেচরভাষণম্ (ভারবি-রচিত 'কিরাতার্জুনীয়ম্' মহাকাব্যের প্রথম সর্গের নির্বাচিত অংশ) [Vanecarabhāṣaṇam (select portion from the 1 st canto of Kirātārjunīyam of Bhāravi)]		
Unit – III	দৃশ্যকাব্য (Drama): আত্রেয়ী-বনদেবতা-সংবাদঃ (মহাকবি-ভবভূতি- রচিত 'উত্তররামচরিতম্' এর দ্বিতীয় অঙ্কের প্রারম্ভে বিষ্ণুস্তকের নির্বাচিত অংশ) [Ātreyī-vanadevatā-saṁvādaḥ (select portion from the Viṣkambhaka part of 2 nd act of Uttararāmacarita of Bhavabhūti)]		
Part II	संस्कृत व्याकरण ও সংস্কৃত সাহিত্যের ইতিহাস (Sanskrit Grammar & History of Sanskrit Literature)	40 (20+20)	20

Unit – IV	ব্যাকরণ (Grammar): ১. প্রত্যয় [Suffix]- ➤ সনাদি - সন, যঙ, গিচ্ (<i>san, yañ, ñic</i>) ➤ স্ত্রী - টাপ্, ঙ্গীপ্ (<i>strī: tāp, ṅīp</i>) ২. কারক-বিভক্তি ও সমাস [Case-ending and Compound] - ➤ কারক-বিভক্তি - সম্প্রদান, অপাদান, অধিকরণ ও সম্বন্ধ পদ (Case-endings: <i>sampradāna, apādāna</i> and <i>sambandha pada</i>) ➤ সমাস - সাধারণ-কর্মধারয়, দ্বিগু (সমাহার), দ্বন্দ্ব, বহুব্রীহি (সমানাধিকরণ, ব্যধিকরণ, নঞ) (Compounds: <i>sādhāraṇa-karmadhāraya, samāhāra-dvigu, dvandva, bahuvrihi - samānādhikaraṇa, vyadhikaraṇa, nañ</i>)	20	10 SAQ: 10 Marks 5 questions out of 6/7 (covering all the topics of this unit) each containing 2 marks to be answered as per given instructions [সংক্ষিপ্ত উত্তরের প্রশ্ন: (এই অংশের সকল বিষয় অবলম্বনে প্রদত্ত) ছয়/সাতটি ২ নং-এর প্রশ্নের মধ্যে যেকোনো পাঁচটির প্রদত্ত নির্দেশানুসারে উত্তর করতে হবে]. [2×5=10]
Unit – V	লৌকিক ও আধুনিক সংস্কৃত সাহিত্যের ইতিহাস (History of Classical and Modern Sanskrit Literature): ১. গদ্য - দণ্ডী ও বাণভট্টের সাহিত্যকৃতি (সংক্ষিপ্ত পরিচয়) [Brief Introduction to Daṇḍī and Bāṇabhaṭṭa and their literary works] ২. চম্পু - নলচম্পু ও ভারতচম্পু (সংক্ষিপ্ত ধারণা) [Brief Introduction to <i>Nalacampū</i> and <i>Bhāratacampū</i>] ৩. আয়ুর্বেদ - চরক ও সুশ্রুত (গ্রন্থকার ও গ্রন্থের সংক্ষিপ্ত পরিচয়) [Brief Introduction to Caraka and Suśruta and their works] ৪. আধুনিক বাঙালি সংস্কৃত সাধক ও সাহিত্যকৃতি (সংক্ষিপ্ত পরিচয়) [Bengalee authors of Modern Sanskrit literary works: a brief Introduction] - ➤ সিদ্ধেশ্বর চট্টোপাধ্যায়, সীতানাথ আচার্য, তারাপদ ভট্টাচার্য, বীরেন্দ্রকুমার ভট্টাচার্য ও শ্রীজীব ন্যায়তীর্থ (Siddheswar Chattopadhyay, Sitanath Acharya, Tarapada	20	10 • SAQ: 6 Marks 3 questions out of 4 (covering all the topics of this unit) each containing 2 marks to be answered in Sanskrit/Bengali/ English/ Hindi [সংক্ষিপ্ত উত্তরের প্রশ্ন: (এই অংশের সকল বিষয় অবলম্বনে প্রদত্ত) চারটি ২ নং-এর প্রশ্নের মধ্যে যেকোনো তিনটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [2×3=6] • DQ: 4 Marks 1 question out of 2 each containing 4 marks to be answered in Sanskrit/Bengali/ English/ Hindi [ব্যাখ্যামূলক উত্তরের প্রশ্ন: দুটি ৪ নং-এর প্রশ্নের মধ্যে যেকোনো একটির সংস্কৃত/বাংলা/ইংরাজি/ হিন্দি-তে উত্তর করতে হবে]. [4×1=4]
	Bhattacharya, Birendrakumar Bhattacharya, Srijeeb Nyayatirtha)		

CLASS: XII**SUBJECT: SANSKRIT (SNSK)****COURSE CODE : PRACTICAL/PROJECT****FULL MARKS : 20****CONTACT HOURS : 20 HOURS**

	TOPICS (PROJECT)	CONTACT HOURS	MARKS
১.	সংস্কৃত মনীষায় বিজ্ঞানচেতনা [Scientific thoughts in Sanskrit intellectual tradition]	20	20 1. Pictorial and Informative Project work in Sanskrit language with Devanāgarī script on any one from first two topics & 2. Dialogue/Debate in Sanskrit in any relevant topic [১. ছবি ও তথ্য সহ সংস্কৃত ভাষায় দেবনাগরীলিপিতে প্রথম দুটি বিষয়ের মধ্যে যে কোনো একটি বিষয় অবলম্বনে প্রকল্প-কর্ম এবং ২. সংস্কৃত বার্তালাপ/বিতর্ক – যেকোনো প্রাসঙ্গিক বিষয়ে]
২.	তৃতীয় ও চতুর্থ সেমেস্টার-এর পাঠ্যান্তর্গত বিষয়সমূহ (সংস্কৃত সাহিত্য ও সাহিত্যের ইতিহাস) থেকে একটি বিষয় [Any one topic from the syllabus of Semester III or IV (Sanskrit Literature and History of Sanskrit Literature)]		
৩.	বার্তালাপ/বিতর্ক – যেকোনো প্রাসঙ্গিক বিষয়ে [Dialogue/Debate in any relevant topic]		
			[10 + 10 = 20]

SUGGESTED READING:

1.	<i>Helps to the Study of Sanskrit</i> , Janakinatha Sastri, Kolkata: Sanskrit Book Depot.
2.	সংস্কৃত সাহিত্যের ইতিহাস ও সংস্কৃত ত্রিধারা, শ্রীকৃষ্ণগোপাল গোস্বামী ও আলপনা গোস্বামী। কলকাতা: গোস্বামী প্রকাশনী।
3.	সমগ্র ব্যাকরণ কৌমুদী, ঈশ্বরচন্দ্র বিদ্যাসাগর, হেমচন্দ্র ভট্টাচার্য্য বিদ্যাবিনোদ সম্পাদিত। কলকাতা: চলন্তিকা প্রকাশক।
4.	সমগ্র ব্যাকরণ কৌমুদী, ঈশ্বরচন্দ্র বিদ্যাসাগর, দুর্গাচরণ সাংখ্য-বেদান্ততীর্থ সম্পাদিত। কলকাতা: দেব সাহিত্য কুটীর প্রাইভেট লিমিটেড।
5.	নবরূপে ব্যাকরণ কৌমুদী, শ্রীকৃষ্ণগোপাল গোস্বামী ও আলপনা গোস্বামী। কলকাতা: গোস্বামী প্রকাশনী।
6.	সংস্কৃত সাহিত্যের ইতিহাস, যুধিষ্ঠির গোপ। কলকাতা: সংস্কৃত বুক ডিপো।
7.	বেদের পরিচয়। যোগীরাজ বসু। কলকাতা: ফার্মা কেএলএম প্রাইভেট লিমিটেড।
8.	বৈদিক সাহিত্যের রূপরেখা। শান্তি বন্দ্যোপাধ্যায়। কলকাতা: সংস্কৃত পুস্তক ভাণ্ডার।
9.	সংস্কৃত সাহিত্যের ইতিহাস (বৈদিক ও লৌকিক), জাহ্নবীচরণ ভৌমিক। কলকাতা: সংস্কৃত পুস্তক ভাণ্ডার।

COURSE OVERVIEW :

Sociology is a wide-ranging modern social science discipline which investigates the structure of society and the diversity of social activity in theory and practice. Sociology evolves along side society, with its focus areas ranging from globalization and societal structures to cultural practices and customs as well as diverse social relationships.

Sociology is introduced as an elective subject at the senior secondary stage. The syllabus is designed to help learners to reflect on what they hear and see in the course of everyday life and develop a constructive attitude towards society in change; to equip the learners to understand the dynamics of human behavior in all its complexities and manifestations. The learner's today need answers and explanations to satisfy the questions that arise in their minds while trying to understand social world. Therefore, there is a need to develop an analytical approach towards the social structure so that they can meaningfully participate in the process of social change.

There is scope in the syllabus not only for interactive learning, based on exercises and project work but also for the teachers and students to jointly innovate new ways of learning and divergent thinking.

OBJECTIVES :

- Sociological outlook makes it possible to challenge everyday assumptions and consciously distance the learners from notions that are considered self-evident.
- It would help the learners to relate the theoretical knowledge with the actual social practices.
- It enables the learner's to appreciate socio-cultural differences and develop empathy and inclusive attitude towards 'others'.
- It equips the learners to unearth the underlying meaning of reality and suggest new alternatives to bring about changes in desired direction.

CLASS – XI

SUBJECT : SOCIOLOGY(SOCG)

SEMESTER – I (MCQ)

FULL MARKS : 40

CONTACT HOURS : 100 Hours

COURSE CODE : THEORY

UNIT NO.	TOPICS	SUBTOPICS	CONTACT HOURS	MARKS
Unit 1	Introducing Sociology	- What is sociology- 'Sociological Imagination' – C. W. Mills ; Sociology and its relationship with other social sciences (Anthropology, History, Political Science, Economics, Philosophy and Psychology) - Emergence of sociology: French Revolution, Industrial Revolution, Enlightenment (a very brief outline) - Founding fathers of sociology: a) Auguste Comte-Law of Three Stages b) Emile Durkheim-Division of Labor c) Max Weber- The Protestant Ethic and the Spirit of Capitalism - Is Sociology a science-the debate	30	06

UNIT NO.	TOPICS	SUBTOPICS	CONTACT HOURS	MARKS
Unit 2	Basic Sociological Concepts(only conceptual outlines to be discussed)	- Society- structure, process - Community and Association - Social Institutions - Social Organization - Social Groups - Social Stratification - Social Change - Social Investigations- a) Definition and Stages b) Methods: i) Field Studies with special reference to Participant Observation ii) Social Survey c) Tools: Questionnaire and Interview Schedule	25	15
Unit 3	Culture and Society	- Meaning and Importance - Material and Non-material Culture (e.g. Technology and Language) - Culture as a Normative System- Institutions, Values, Norms, Beliefs, Folkways, Mores and Laws - Cultural Lag, Sub-culture, Counter Culture, Mass Culture, Folk Culture, Consumer Culture, Multiculturalism, Ethnocentrism, Xenon centrism (only conceptual understanding of each topic) - Culture ,Personality and Society	25	15

UNIT NO.	TOPICS	SUBTOPICS	CONTACT HOURS	MARKS
Unit 4	Socialization	• Concept and Importance • Socialization and Social Interaction (verbal and non verbal) • Agencies- Family, Peer, Educational Institutions, Work Place and Media • Formation of 'Self'-stages • Social Reproduction and Social Transformation	20	04

CLASS -XI

SUBJECT : SOCIOLOGY (SOCG)

SEMESTER – II (SAQ & LAQ)

FULL MARKS : 40

CONTACT HOURS : 80 HOURS

COURSE CODE : THEORY

UNIT NO.	TOPICS	SUB TOPICS	CONTACT HOURS	MARKS
Unit 5	Social Institutions	• Kinship, Family and Marriage- forms and functions; changing trends like single parent family, same sex marriage etc. • Economic Institutions- Work, Market, Online Transaction • Political Institutions- Power and Authority (concept), State (concept); Government (Democratic, Dictatorship, Parliamentary, Presidential); Non-State Institution (UN- in brief) • Cultural Institution- Religion, Education, Fashion (cultural variation) • Health and Illness – concept(physical and mental health); Health Scare-Pandemic (e.g.COVID19)	60	30

Unit6	Social Control and Crime	• Concept and meaning • Conformity and Non conformity • Deviance and Delinquency • Types of Crime (White-Collar Crime, Cyber Crime, Organized Crime- Terrorism)	20	10
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PROJECT WORK : Full marks - 20	
A. Written Internal Assessment- Any topic of social relevance can be investigated applying social science research methods and tools • 1.Introduction- 1 Mark • 2.Research Question- 2 Marks • 3.Research Objectives- 2 Marks • 4.Method of Data Collection and its Rationale- 3 Marks • 5.Data Analysis- 3 Marks • 6.Conclusion-2 Marks • 7.Reference and Bibliography- 2Marks	15 Marks
B. Viva-voce-on the basis of the assigned Project Work	05 Marks

- Project to be submitted during the second semester or within the guideline as directed by the WBCHSE.
- For Remedial Classes, Tutorials and Home Assignments no. of periods allotted-15

CLASS – XII
SUBJECT : SOCIOLOGY (SOCG)
SEMESTER – III (MCQ)

FULL MARKS : 40

CONTACT HOURS : 100 HOURS

COURSE CODE : THEORY

UNIT NO.	TOPICS	SUB TOPICS	CONTACT HOURS	MARKS
Unit1	Sociology in India	- Pioneers of Indian Sociology G. S. Ghurye – Caste and Race R. K. Mukherjee -Social Ecology M. N. Srinivas –Sanskritization And Westernization	30	8
Unit2	Social Change in India: Continuity And Change	- Colonialism, Nationalism - Post-Colonial India- Liberalization and Globalization	20	7
Unit3	Change in Social Structure	- Family: Changes in structure and functions, factors of change, recent trends (family disorganization) - Rural and Urban Communities: Difference between City and Village, emerging Agrarian Society in the 21st century, Panchayat, Municipality, Smart City - Caste and Tribe: Pollution and Purity, Caste Association – SC, ST, OBC, Tribal Absorption, Tribal movements - Religion: Communalism, Secularism - Education: Education Policy perspective in Independent India, Sarva Shiksha Abhiyan (SSA), Right To Education (RTE), State Education Policy of West Bengal (2023) in the context of National Education Policy (NEP-2020 from foundation to senior Secondary level, in brief)	50	25

CLASS – XII
SUBJECT : SOCIOLOGY (SOCG)
SEMESTER – IV (SAQ & LAQ)

FULL MARKS : 40

CONTACTHOURS : 80 HOURS

COURSE CODE : THEORY

UNITNO.	TOPICS	SUBTOPICS	CONTACT HOURS	MARKS
Unit4	Contemporary Social Issues	- Population Explosion, Poverty, Illiteracy and Unemployment, Mobility and Migration, Consequences and Remedies - Corruption: Concept, Recent Legislations - Gender and Sexuality: Concept of Sex and Gender, Women and Different forms of Sexuality (LGBTQ), Different forms of gender violence - Environment: Crisis and Responses (State and Non- State Initiatives) - Mass Media: Print and Tele Media, Social Media- Face book, Instagram, Reddit, LinkedIn, Vloggs, Reels and their impact in social life - Social Movements: Women's movement and Environment movement	80	40

PROJECT WORK : Full marks – 20	
A. Written Internal Assessment- Any topic of social relevance can be investigated applying social science research methods and tools • 1.Introduction-1 Mark • 2.Objectives-2 Marks • 3.Research Question-2 Marks • 4.Methodology and its Rationale-3 Marks • 5.Data Analysis-3 Marks • 6.Conclusion-2 Marks • 7.Reference and Bibliography-2 Marks	15 Marks
B. Viva-voce-on the basis of the assigned work	05 Marks

- Project to be submitted during the second semester or within the guideline as directed by the WBCHSE.
- For Remedial Classes, Tutorials and Home Assignments no. of periods allotted-15